

step ahead

Combined Science

FORM 2

NEW
Curriculum

2017

Approved by MOPSE

MADHANGOVE

PEARSON

Learner's Book

 **Pearson**

K Chavunduka

step ahead

Combined Science

FORM 2

Learner's Book



K Chavunduka

C₂ 01/18

 **Pearson**

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Cells and levels of organisation

Objectives

- Point out differences between living organisms.
- Compare continuous and discontinuous variation.
- Draw bar graphs to show variations in living organisms.

Introduction

In studying the world about you, you will notice that there is a difference between living things (such as goats, ferns, Msasa trees, lizards and butterflies) and non-living things (such as iron ore, tables, clay pots, cars and houses). Living things, or organisms, are similar in that they all display the same basic activities we call life processes. All living organisms:

- need nutrition
- get energy from their food (they **respire**)
- get rid of waste products (they **excrete**)
- control their internal conditions (maintain **homeostasis**)
- respond to changes in their environment
- produce offspring (they **reproduce**)
- show growth and development.

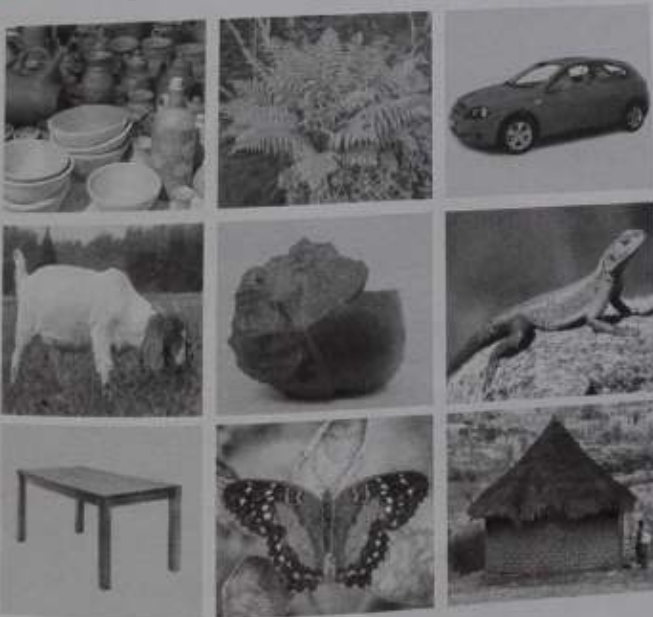


Figure 1.1 Some living organisms and non-living things

Similarities and differences between living things

Because there are so many different organisms, it is complicated to study life scientifically. For this reason, Biologists classify living things, their subject matter, by looking at what they have in common. Characteristics, such as how organisms get nutrition, what their cells are like, their body structure and how they reproduce, are considered.

The modern classification system of organisms



Figure 1.2 A portrait of Carolus Linnaeus, the pioneer in classifying organisms

The first classification work was done by a Swedish doctor and botanist, Carolus Linnaeus, in the eighteenth century. From there, the modern classification system developed, in which the first broad division of living organisms places them into one of five **kingdoms**:

- Animalia (animals)
- Plantae (plants)
- Fungi
- Protista
- Monera (bacteria).

Table 1.1 gives an overview of the five kingdoms, together with some of the shared characteristics of members in each of the kingdoms. You will notice that the organisms in a kingdom are only very broadly similar.

Table 1.1 An overview of the five kingdoms

Kingdom	Examples of organisms	Characteristics
Animalia  Example: <i>Hippotragus niger</i> (sable antelope)	All multicellular animals, including jellyfish, worms, arthropods, molluscs, echinoderms, fish, amphibians, reptiles, birds and mammals	• Multicellular • Have no cell walls • Feed on living organisms (ingest food) • Have no chlorophyll • Can move
Plantae  Example: <i>Chlorophytum comosum</i> (hen-and-chicken plant)	All green plants, including green algae, ferns and mosses (plants that do not produce seeds) conifers and flowering plants (plants that do produce seeds)	• Multicellular • Make their own food by photosynthesis • Have cell walls of cellulose • Usually have vascular tissue • Have distinct parts, such as roots, stems and leaves
Fungi  Example: <i>Rhizopus stolonifer</i> (bread mould)	Mushrooms, yeast, <i>Rhizopus</i> (bread mould)	• Absorb food directly from other organisms (or from dead and decaying material) • Cell walls made of chitin • Have more than one nucleus • No chlorophyll
Protista  Example: <i>Amoeba proteus</i>	Red and brown algae, protozoa (<i>Plasmodium</i> , <i>Amoeba</i>)	• Usually single-celled organisms • Have a distinct nucleus
Monera  Example: <i>Escherichia coli</i>	Bacteria, blue-green algae	• Single-celled organisms • Have no nucleus

The next levels of classification group fewer and fewer organisms together by focusing on more specific and specialised shared characteristics than each of the previous levels. In this way, kingdoms are subdivided into phyla. Each phylum is subdivided into classes, each class into orders, each order into families, each family into genera, and each genus into **species**. The species is thus the smallest unit of classification of living organisms.

A species is defined as a group of organisms (such as humans, zebras, horses, mangoes and oranges) that have many features in common and that can freely interbreed to produce fertile offspring. Organisms from different species usually cannot interbreed to produce offspring that can also reproduce.

Scientific nomenclature

Carolus Linnaeus is also known for developing the two-part system for scientific names we use today. The **binomial** system combines the last two levels of classification. The genus name appears first and the species name last. For example, humans are known as *Homo sapiens* and leopards as *Panthera pardus*. You can see that the genus name always starts with a capital letter and both the genus and species name should be in italics if you type them. If you write out the names, you must underline both parts separately.

The use of these names instead of common names helps to avoid confusion, especially when scientists are from different language groups. Common names are also sometimes used for more than one species or do not accurately describe the organism they refer to. Think of the silverfish, which is actually an insect!

Activity 1.1

1. You studied the differences between plant and animal cells in Form 1. Revise this on your own. Then draw and label a typical plant cell and a typical animal cell. Remember to give your drawings headings.
2. In pairs or small groups, choose one of the five kingdoms and do some more research on the organisms belonging to the kingdom you chose. You can guide your research by asking yourself questions such as:

- Are these organisms useful as food sources?
- Are they useful in the production of manufactured goods?
- Do they cause diseases?
- Are they in any other way useful or harmful?

Make a poster of your chosen kingdom in order to summarise your findings visually. Your teacher may assess your poster or ask your classmates to peer-assess it.

3. Scientists do not all agree about whether viruses are living or non-living. Your teacher will divide your class into two groups. One group will research and defend the view that viruses are alive, and the other group will defend the opposite view during a debate in class.

Variation in a species or population

We have no trouble picking out our friends from among a crowd of other people. People vary in how they look and behave. They also function quite differently and there is even **variation** between people at the molecular level. For example, people have different blood groups, but we all belong to one species, *Homo sapiens*.



Figure 1.3 Variation in humans, peas and Nguni cattle

What we see in humans is also true in other species. Even in a specific **population**, that is, in a group of individuals from one species living in one place at the same time, we will find differences in their characteristics that we can observe, that is, in their **phenotype**.

We can ask ourselves a lot of questions about variation in species and populations. For example, we can ask what kinds of variation there are. We can also ask what causes variation and we can ask ourselves if our knowledge about variation can be put to any use.

Types of variation

There are two types of variation in the features of individuals in a species or a population: continuous and discontinuous variation.

Continuous variation

When a phenotype shows continuous variation, the values it can take on change gradually over a range of possible values. An example is the height of children from a certain age group. They are not

either short *or* tall, as any height in between the shortest and the tallest child is possible. Other examples of this kind of variation are:

- weight
- foot length
- the number of seeds in a pod
- leaf length.

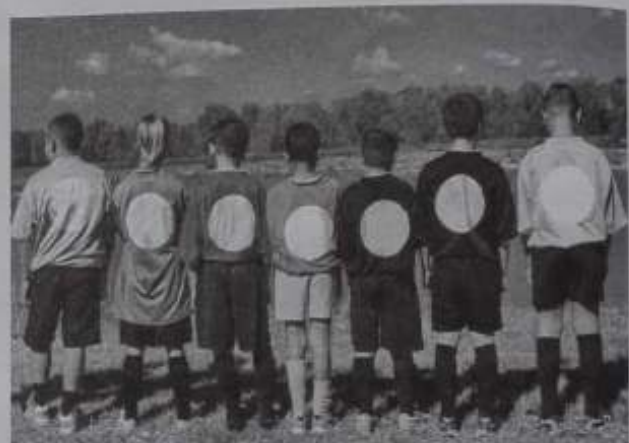


Figure 1.4 Height is an example of continuous variation.

Experiment 1.1

Aim: To investigate continuous variation

Materials: tape measure, graph paper, pencil

Procedure

1. Attach a tape measure to the wall, making sure that '0' is at floor level. Your tape measure should be at least 2 metres long.
2. Measure the height of each learner in your class and record these measurements.
3. Arrange the heights of learners in your class, in order from the shortest to the tallest learner, to determine the range of heights in your class. (For example, a range may be 145 to 183 cm.)
4. Calculate the **mean** height for your class by adding all the values and dividing your answer by the number of learners you have measured.
5. Draw a **frequency** table for your results by making use of the height intervals in Table 1.2.

Table 1.2 Frequency table for results of Experiment 1.1

Height interval	Number of learners
131–140 cm	
141–150 cm	
151–160 cm	
161–170 cm	
171–180 cm	
181–190 cm	

6. Draw a histogram, with the height intervals on the x-axis and the number of learners for each interval on the y-axis. Give your histogram a suitable heading.

Results

Compare your histogram with the example shown below in Figure 1.5. Does your histogram look similar and show a similar curve? If not, can you explain why?

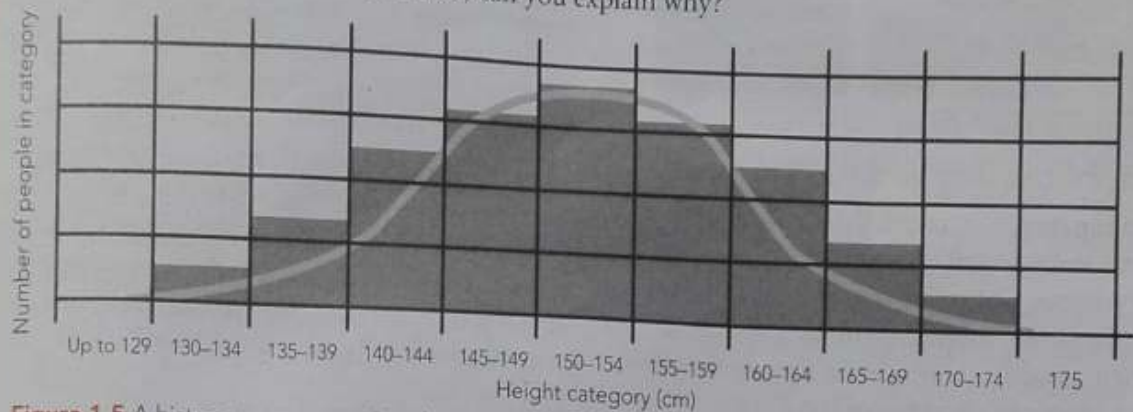


Figure 1.5 A histogram representing the heights of people in a given population

Discontinuous variation

When there are only a limited number of possibilities for a phenotype in a species, it is discontinuous. Think of peas that produce either tall or dwarf plants, or either green or yellow seeds. There are no in-between values.

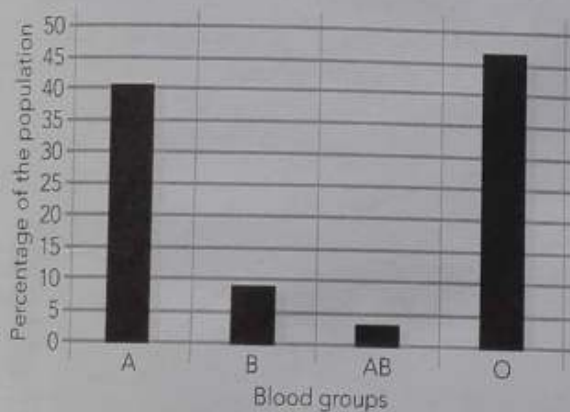


Figure 1.6 The blood groups of a population, given in percentages

The same is true of human blood groups. There are only four possibilities: A, B, AB and O. Data related to this kind of variation are represented in a bar graph, such as Figure 1.6. Note that the bars do not touch one another, because the groups are not continuous.

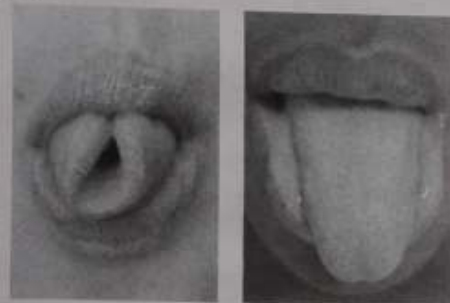
Other human characteristics also show discontinuous variation:

- There are two genders (male and female).
- We are either left- or right-handed.
- We can either roll our tongues or cannot roll our tongues.
- We have earlobes that are either attached or unattached.
- We can either have a gap or no gap between our front teeth.

Activity 1.2

1. Make an inventory of your classmates, showing the types of discontinuous variation indicated in the pictures below, and calculate the percentages in each group. Draw a bar graph to show your results.

a)





Factors causing variation in species

Some variation in a species is due to individuals' genetic make-up and is **inherited**. Think about how you look a little like your mother and a little like your father. Some kinds of variation are due to environmental influences on the genetic make-up, or **genotype**, of an individual. For example, hydrangea flowers of the same genetic variety change colour from blue to pink, depending on the acidity of the soil in which they grow. A single tree with its specific inherited genotype has leaves with differences in size, shape and greenness, depending on their exposure to sun and wind. The number of seeds in a bean pod depends on factors such as water supply, pollination and richness of soil.

For humans, nutrition will influence an individual's height, exercise can change your build, time spent in the sun can influence your skin colour and experience will improve your

performance in intelligence tests. However, you would not be able to pass on these changes to your offspring.

Other factors resulting in variation in a species are age, injury, disease and **mutation**. Mutation, that is, the change of genetic material during fertilisation and cell division, can introduce new **heritable** traits into a species.

Changes in species over time

In the nineteenth century, the scientist Charles Darwin proposed the theory of evolution to explain the origin of species. He noticed how species adapted to their environment and this lies at the basis of his view that species change over time and are not fixed.

Natural selection

Together with Alfred Wallace, Darwin proposed **natural selection** as the mechanism for evolutionary change. This mechanism is based on the following observations the two scientists made:

- Some variations in a population can be passed on from one generation to the next.
- Individuals with variations best suited to their environment survive longer than others.
- More individuals with favourable features will survive to reproduce than other individuals with less favourable features.

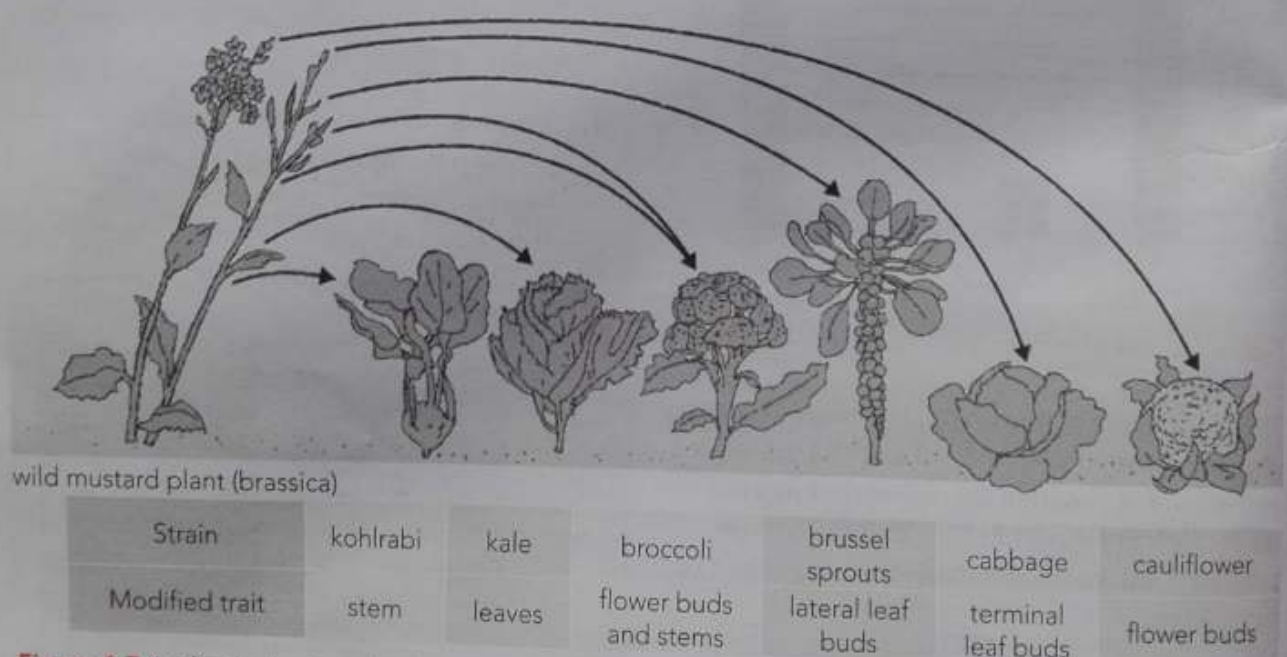


Figure 1.7 Artificial selection of wild mustard

Over time, the population will become more adapted to its environment.

You will notice that variation is a key element in this theory.

Artificial selection

Over the ages, humans have benefited from the principles of natural selection by artificially selecting animals and plants with specific traits for breeding.

For example, the plants we grow for food are very different from their ancestors. Cabbage, Brussels sprouts, kohlrabi, kale, broccoli and cauliflower are all descendants from wild mustard. They are the result of people noticing variations and artificially encouraging them to suit their needs.

In the Mashona cattle, we see the results of both natural and artificial selection. The cattle are reared for their meat, and they are hardy and very docile work animals. Their traits are believed to be linked with their long history with humans. Cattle were kept in kraals and taken out by herders to feed for a couple of hours only, before being taken back to the safety of the kraal. The cattle had to be able to eat a lot in a very short time. During droughts, cattle had to walk far to get to grazing and weak animals died along the way. Less hardy traits thus disappeared. Similarly, very lively cattle jumping kraal fences at night got eaten by predators. Only the genes of the gentler animals were passed on.



Figure 1.8 The Mashona cattle breed is distinctive of Zimbabwe.

Genetic engineering and health sciences

The realisation that genes are responsible for certain variations is put to use in genetic engineering. Plant and animal organisms, and even bacteria can be **genetically modified**. This is done by isolating the genes responsible for certain variations in a species or an organism and introducing these genes into other plant, animal or bacterial material that people want to change. Strains of cotton, corn potatoes and even bananas have been successfully made resistant to certain pests, because their cells now produce a chemical that kills fungi or insects.

Health sciences have also benefited from studying variations that may be harmful or beneficial to human health. Doctors can test for **congenital** diseases or disorders before a baby is born. Research is also done on how to benefit from individuals' natural resistance to some diseases. For example, the genetic variation that is linked to sickle cell anaemia makes humans immune to malaria.

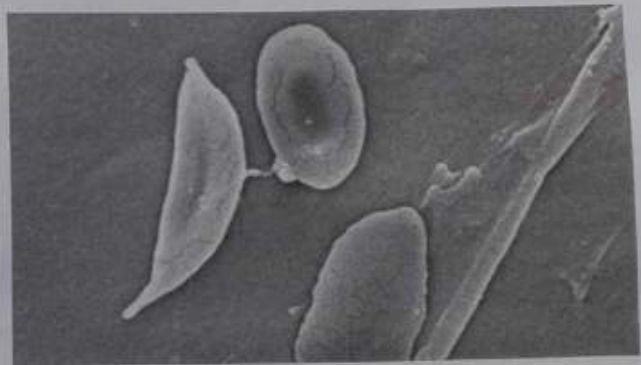


Figure 1.9 A normal red blood cell (right) and sickle cell (left)

Exercise 1.1

1. Define the following terms:
 - a) Continuous variation
 - b) Phenotype
 - c) Genotype
2. List the observations Darwin made on which he based his theory of evolution.
3. Explain what natural selection is and illustrate it by giving an example.
4. Give two examples of how humans make use of variation to their benefit.
5. Explain the difference between artificial selection and genetic modification.

Summary

- Despite their variety, all living organisms display the same basic activities, referred to as life processes.
- Biologists classify living things according to shared characteristics, into groups.
- In the five-kingdom system, organisms are grouped into categories that become more and more specific: kingdom-phylum-class-order-family-genus-species. The kingdom to which an organism belongs is always the broadest category, containing the most organisms, and species is the most specific category, containing the fewest organisms.
- The scientific name for organisms consist of the genus name (capitalised) and species name, both given in italics.
- Continuous variation for a characteristic in a species or population refers to characteristics for which values can take on a range of values between two extremes.
- Variation is discontinuous when there is only a limited number of values it can have.
- Natural selection occurs when a population becomes more adapted to its environment because certain individuals' traits are more suited to their environment.
- Artificial selection occurs when humans choose to interbreed individuals from a species with specific characteristics.
- Genetic modification occurs when genes that are responsible for certain characteristics are isolated in one species or organism, and then introduced into a plant, animal or bacteria one wants to change.

Glossary

binomial – with two names

congenital – present before or at birth

excrete – to get rid of waste products

fertile – able to reproduce

frequency – how many times something happens

genetically modified – genetically modified organisms (GMOs) whose genes have been changed by introducing foreign genes into their gene pool

genotype – the genetic make-up of an individual organism

heritable – characteristics that can be inherited

histogram – a type of bar graph in which data are grouped into successive intervals of the same size

homeostasis – a state of internal balance in an organism

inherited – characteristics received from one's parents

kingdom – the biggest grouping of organisms with similar characteristics

mean – the value you get by adding all values and dividing the total by the number of values

mutation – a change in genetic material

natural selection – process in which organisms that are better suited to their environment tend to survive and produce more offspring

phenotype – the observable characteristics of individuals in a species or population

population – a group of individuals of the same species living and breeding in the same place at the same time

reproduce – to produce offspring

respire – to release energy together with carbon dioxide from food

species – a group of organisms with common characteristics and qualities (features); can freely interbreed to produce fertile offspring; the smallest grouping of organisms showing similarities

variation – the differences in members of the same species or population

Revision questions

1. Name the five kingdoms of organisms. (5)
2. Arrange the following levels of classification in the correct order. Start with the level in which groups contain the smallest number of organisms: family; species; order; genus; phylum; kingdom; class (7)
3. What is the difference between continuous variation and discontinuous variation?

Give an example of each type of variation in humans.

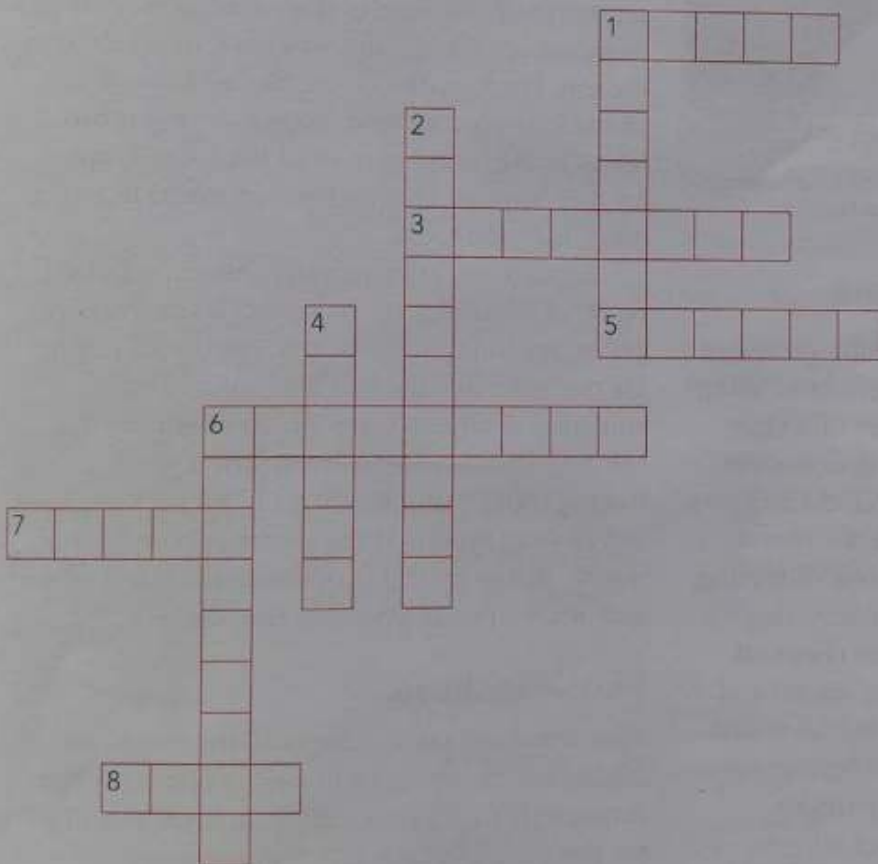
4. What is meant by the term 'genotype'? (2)
5. What is a congenital disease? (2)
6. Use the following clues to complete the crossword puzzle. (10)

Across

1. The plural of 'fungus'
3. Red and brown algae belong to this kingdom
5. Complete: All living things get _____ from their food.
6. To describe something that can be inherited
7. They determine the genotype of an individual
8. The value that you get when you divide the sum of values in a set by the number of values in that set

Down

1. Able to reproduce
2. A group of individuals from the same species living and breeding in the same place at the same time
4. The surname of the father of evolution
6. A type of bar graph that groups data into intervals of the same size



TOTAL: 30

Objectives

- Explain the importance of plants as producers of food.
- Give the word equation for the process of photosynthesis.
- Draw and label the human digestive system.
- Outline the route followed by food in the human digestive system.



Figure 2.1 The plants of the woodlands at Mana Pools produce food for all living things in that ecosystem. The photo shows warthogs feeding.

Introduction

All living organisms need nutrition and get energy from their food. This is one characteristic of living things that distinguishes them from non-living things. Like animals, plants are living organisms. But we never see them taking in food, chasing prey or chewing food in the way animals do. This is because green plants have a special way of feeding.

Plants make their own food by converting the energy contained in sunlight into chemical energy contained in **glucose** (carbon sugars) and later into other organic substances such as **starch**. The process by which plants convert light energy to chemical energy is called **photosynthesis**. Humans and other organisms depend, directly or indirectly, on the process of photosynthesis for food. This unit focuses on plant nutrition, the feeding relationship between plants and other

organisms and on human nutrition as an example of nutrition in **mammals**.

Plants as producers of food

Green plants are 'self-feeders' or **autotrophs**. They are organisms that make all their own **organic matter** from nutrients that are **inorganic**: carbon dioxide from the air, and water and minerals from the soil. Photosynthesis, the process by which plants manufacture food, takes place mainly in leaves as they contain most of the tissue adapted for this purpose. The food is then stored in leaves, roots, fruit and seeds.

By contrast, humans and other animals feed on other organisms or their products and cannot get energy from inorganic nutrients to fuel their life processes. We say that humans and other animals are 'other-feeders' or **heterotrophs**. You will find that, in most **ecosystems**, if you trace any feeding relationship, you usually get back to plants and photosynthesis as the starting point. For this reason, plants are called **producers**, while humans and other animals are called **consumers**.

Photosynthesis

Photosynthesis takes place in all the green parts of plants, but the leaves of flowering plants contain **mesophyll tissue** with cells, which are specialised for this process. The leaf is thus the main photosynthetic organ and the leaf system is the main photosynthetic system in flowering plants (see Figure 2.2).

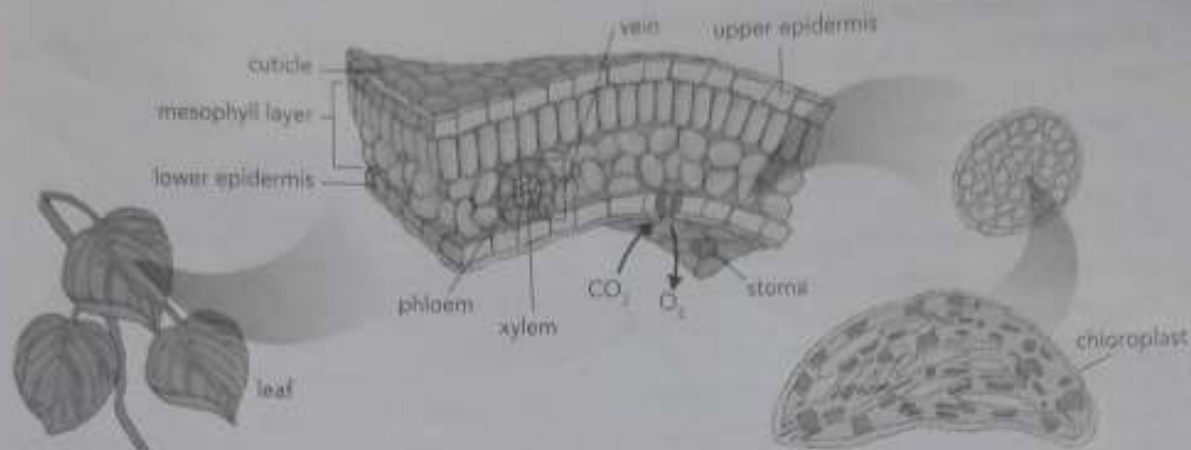


Figure 2.2 The location of photosynthesis in a plant, from the leaf system to the inside of a cell

The raw materials (or reactants) for photosynthesis are water and carbon dioxide. The roots of a plant absorb water from the soil, and this water travels up the stems to the veins of the leaves in special tubes called **xylem**. Carbon dioxide diffuses into the leaves through small leaf openings called stomata (singular: **stoma**). Once inside the leaf, water and carbon dioxide diffuse into **chloroplasts**.

Chloroplasts are the organelles inside the mesophyll cells that carry out photosynthesis. Chloroplasts contain the green chlorophyll pigment that absorbs light energy to drive the process of photosynthesis. Without the help of chlorophyll and light energy, plants cannot photosynthesise.

The end-products of photosynthesis are glucose and oxygen. The glucose that has been produced can be used immediately by the plant or stored in the form of starch, a more complex carbohydrate than glucose. The special tissue transporting the food produced in the leaves to where it is needed in the plant, is called **phloem**. The second product, oxygen, diffuses out of the leaf through the stomata, and into the atmosphere.

The overall process of photosynthesis can be represented as follows:

The raw materials of photosynthesis are the same as the waste products of **cellular respiration**: carbon dioxide and water. Photosynthesis also produces what respiration uses to provide energy for the cell: glucose and oxygen. Besides transforming light energy into chemical energy in the form of glucose, photosynthesis thus has two other advantages for any ecosystem: it recycles the waste products of respiration and it makes oxygen, which is essential to all living organisms, a renewable resource. This is a very good reason to protect vegetation and to plant trees around us. You will learn more about respiration in the next unit.

The suitability of leaves for photosynthesis

The typical leaf is suited for its photosynthetic function in a number of ways:

- There are usually many leaves on plants so that they provide a large area for photosynthesis to take place.
- Leaves are well-spaced and exposed to the sunlight due to the way that they are positioned on the leaf stems.

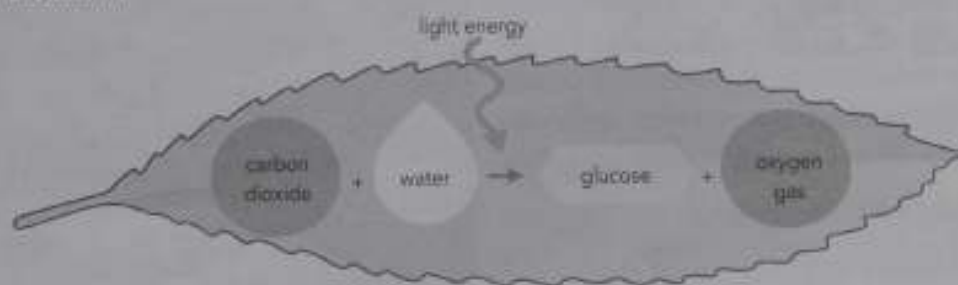


Figure 2.3 The word equation for photosynthesis

- The leaf blade is usually broad in order to present a surface area as large as possible to the atmosphere.
- A thin, waxy, transparent layer called the cuticle protects the leaf while it allows light to enter.
- The stomata on both the upper and lower epidermis allow gases to diffuse in and out of the leaf quickly.
- Because the leaf blade is thin, the substances involved in photosynthesis can move quickly to places they are needed.
- There are air spaces between the mesophyll cells so that gases can be exchanged freely between the atmosphere and these cells.
- The mesophyll tissue in the leaf contains many chloroplasts with the green pigment chlorophyll that is necessary to absorb light energy.
- The leaf veins allow water to pass into the xylem. Phloem carries away the sugars manufactured in the leaf to where they are needed in the plant.

Exercise 2.1

Look at Figure 2.4 and answer the questions that follow.

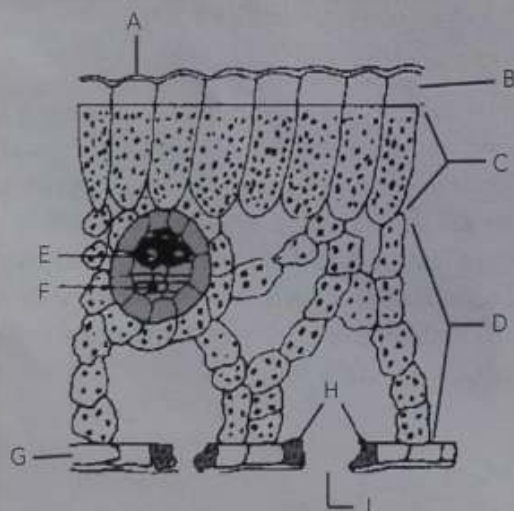


Figure 2.4 Cross-section through a leaf

1. Provide labels for structures A to H.

2. What are the main functions of structures E and F with regard to photosynthesis?
3. How is structure C adapted for photosynthesis?
4. Give two characteristics of structure D that makes it well-adapted for photosynthesis.
5. Complete: The leaf is the main photosynthetic _____ in the plant.

The interdependence between plants and other organisms

You have seen that plants are autotrophs, making their own food (glucose) from inorganic nutrients (carbon dioxide and water) in the presence of an outside energy source (light energy). You have also learnt that plants can store food in their leaves, roots, fruits and seeds or grains. The food that plants produce sustains whole ecosystems, either directly or indirectly.

Consumers

Figure 2.5 shows the interdependence between plants as producers of food (autotrophs) and other organisms that are consumers (heterotrophs).



Figure 2.5 Animals are consumers.

Activity 2.1

1. Study Figure 2.5 and explain the relationship between the different animals and the plants shown.
2. Evaluate each of the relationships. Are these relationships direct or indirect?
3. How can you show these relationships using symbols and words?
4. What happens in the soil and what relationships are represented in Figure 2.5?
5. Point out the differences in feeding habits of these organisms.

You will notice that there are different types of consumers and that they feed in different ways. Biologists make a distinction between the following types of feeders:

- herbivores
- carnivores
- omnivores
- detritivores and decomposers.

Herbivores

Herbivores are consumers that feed only on plants. Examples of this type of consumer are grazers feeding on grass and shorter vegetation, and browsers feeding on tree leaves. The teeth of grazers and browsers enable them to pluck and chew leaves, and they have a specialised digestive system to digest plant material. Some insects like caterpillars and bees are also small herbivores, while some fish are herbivores feeding on algae in water.

Carnivores

Carnivores eat meat only. In the wild, carnivores have to hunt and kill their prey. You will notice that **predators**, such as wild cats, are equipped with sharp senses, quick speed, and sharp teeth and nails. Birds that feed on insects are also carnivores. Vultures are scavengers. They feed only on dead animals and are also carnivorous.

Omnivores

Omnivores feed on both plants and animals. Humans, dogs, chickens and pigs are omnivores.

Detritivores and decomposers

Detritivores feed on **detritus**, that is, organic waste and decomposing organic matter. Marine fan worms filter detritus from water. On land, earthworms, some beetles and ants are all detritivores. Bacteria and fungi are **decomposers** that get nutrients by absorbing them from dead organic matter. This group of feeders is very important, because they release inorganic substances back into the feeding cycle for plant use.

Food chains

When we follow a feeding relationship from one organism to another, we make a **food chain**. The first organism in the food chain is usually a plant. An example from Figure 2.5 is as follows:

grass → antelope → lion

This is a herbivore-predator food chain. Along a river bed, one could find a food chain that is composed as follows:

water lily → water snail → fish → human

In both these food chains, the arrows between the organisms indicate energy flow to the next feeding level. With each transfer of energy to the following level, energy is lost.

Remember that a plant uses some of the energy it produces for its own cellular respiration, for growth and reproduction, and heat energy is lost during transpiration. Only the energy stored in the plant can pass on to the herbivore that eats it. In the same way, the herbivore will use some of the energy in the plant it eats to build and repair body cells, to protect itself against disease, to move, to keep body temperature constant, to excrete and to reproduce. Only the energy stored in the body of the herbivore can be passed on to the next feeding level when a carnivore eats it.

This means that the longer a food chain gets, the less energy is passed on to the next level. Shorter food chains are more advantageous for the organisms involved, because larger amounts of energy will be available.

Food webs and ecological pyramids

In nature, single, isolated food chains do not occur. There is always more than one type of organism feeding on one food source. It also happens sometimes that organisms die before they can pass energy on to a next feeding level. The nutrients in the dead organisms are then recycled into the ecosystem by the decomposers, making them available for plant use. Figure 2.6 represents a food web. Try to isolate a few food chains in this food web, and note how more than one of these can pass through one food source.

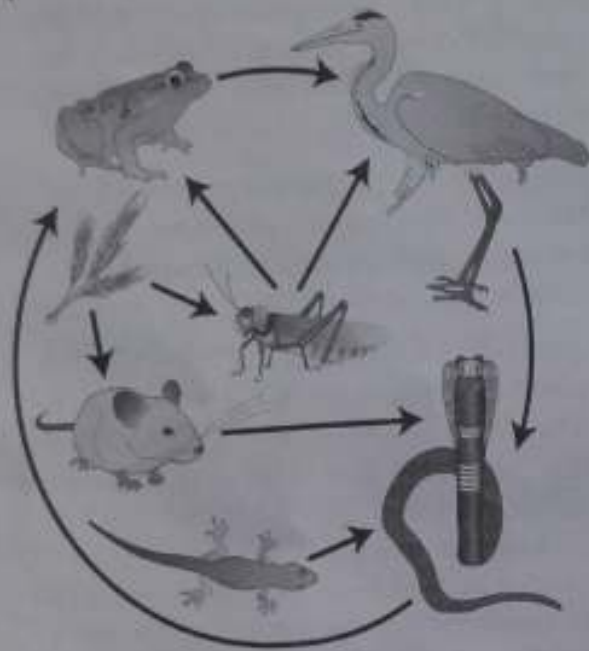


Figure 2.6 A food web

Sometimes food webs are represented in the form of an ecological pyramid. The number of organisms on each feeding level, the total biomass of the organisms on each level or the amount of energy available on the different levels is used to construct a pyramid as shown in Figure 2.7.



Figure 2.7 An ecological pyramid

Project

Work in small groups. Choose an ecosystem, for example, the ecosystem in the open ocean, a forest, a savannah plain, or in a garden. Your teacher will help you choose. Find out which organisms live in your ecosystem and find out what feeding relationships exist there. Draw a food web with at least two interconnected food chains.

You can find information in books, or on the Internet, or you can visit such an ecosystem or talk to people.

Your food web will be evaluated as follows:

Food web

1. Suitable heading (2)
2. Correctness of feeding relationships (8)
3. Two interconnected relationships shown (4)
4. Neatness and display (4)
5. Acknowledgement of sources (2)

TOTAL: 20

Human nutrition

In Form 1, you learnt about the nutrients humans need and why it is necessary to have a balanced diet. The food we eat comes in forms that cannot be used directly by the cells in our bodies. This means that it must first be processed to make the nutrients available to the cells in our bodies. We will now turn to the system that processes food in humans.

Processes involved in nutrition

Nutrition in humans occurs in five stages:

1. **Ingestion:** Food is put into the mouth and enters the **alimentary canal**.
2. **Digestion:** Mechanical processing and digestive juices (chemical processing) break food down into smaller pieces and simple, soluble substances.
3. **Absorption:** Soluble food substances move from the alimentary canal into the bloodstream.
4. **Assimilation:** From the bloodstream, soluble food substances move into the cell and they are incorporated into the cellular structures.
5. **Egestion:** Undigested food substances pass out of the body through the anus in the form of faeces.

The human digestive system

Like most animals, the human organism has a specialised system in which food is processed. This system is made up of two groups of organs. The first group of organs makes up the alimentary canal, while the second group consists of accessory organs.

The alimentary canal is a long, hollow tube with two openings, the mouth at one end and the anus at the other end. The organs making up the alimentary canal are the mouth, the oesophagus, the stomach, the small intestine and the large intestine. The accessory organs include the teeth and the tongue as well as glands that secrete digestive juices through ducts into the alimentary canal. These glands are three sets of salivary glands, the pancreas, the liver and the gallbladder. As food moves down the alimentary canal, each of the organs making up the digestive system performs a specialised task.

Mouth and tongue

Ingestion and the first steps in processing food occur in the mouth. The mouth opening is enclosed by the lips in the front and the cheeks at the sides. The upper jaw is fused with the skull and cannot be moved. Strong muscles move the lower

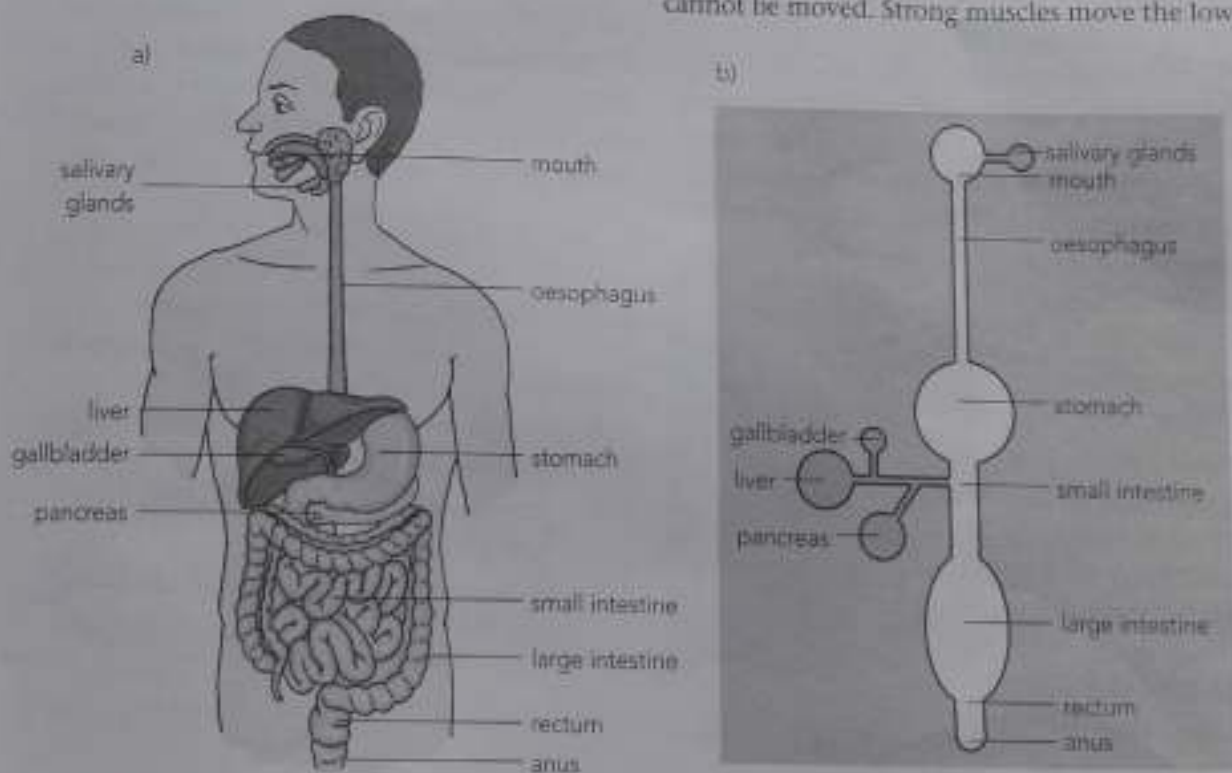


Figure 2.8 a) and b) The components of the human digestive system

jaw up, down and sideways, while the teeth tear, chew and grind food.

The tongue is a muscular organ that is attached to the floor of the mouth at the back. The function of the tongue is to:

- keep food between the teeth while chewing
- make swallowing food easier by mixing food with saliva from the salivary glands
- help push the food to the back of the mouth for swallowing
- taste food to send messages to the brain about the food. (Taste makes you salivate or it makes you realise food is unsafe for eating.)

Saliva from the salivary glands starts the process of chemical breakdown of food.

Oesophagus

When food is swallowed, it enters the oesophagus, a tube connecting the mouth and the stomach.

The oesophagus makes sure that the food moves to the stomach, even when a person is lying down. This is done through its muscles contracting and relaxing in a wave-like movement along the tube. These muscular contractions are called **peristalsis**.

At the end of the oesophagus a **sphincter** allows the swallowed food to enter the stomach, but prevents it from entering the oesophagus again.

Stomach

The stomach is a bag-like organ with a thick muscular wall. Glands in the **gastric pits** secrete gastric juice containing hydrochloric acid. This is mixed with the food by the stomach wall. The wall is elastic so that the size of the stomach can become three times bigger, or even more, during a large meal. While some absorption of nutrients do take place in the stomach, its main function is to store food while continuing digestion. A second sphincter allows only small squirts of the food and gastric juice mixture, called **chyme**, into the



Figure 2.9 Glands in the gastric pits inside the stomach secrete gastric juices.

duodenum. This slows down the digestive process to make it more effective.

Small intestine

Most of the chemical processing and nutrient absorption takes place in the small intestine. It is the longest section in the alimentary canal (about 5 m), but it is called the *small* intestine because its diameter is smaller than that of the large intestine. Digestive juices from the pancreas, the liver, the gallbladder and the walls of the small intestine itself are mixed with the half-digested food. The surface area for absorption of nutrients into the bloodstream across the lining of the small intestine is enlarged by finger-like structures called villi. Each villus has many microvilli, making the surface area for absorption even bigger.

Large intestine

The main part of the large intestine is called the **colon**. The colon extends up the right-hand side of the **abdominal cavity**, passes just under the stomach to the left-hand side and then descends. No digestion takes place in the large intestine. The large intestine absorbs water from the undigested food. As the undigested food passes through the colon, it becomes more and more solid. This undigested food is called **faeces**. Faeces are stored in the **rectum**, the final section of the large intestine. There are two sphincters between the rectum and the **anus**, through which faeces are finally passed out of the alimentary canal.

Activity 2.2

1. As a class, collect recyclable waste materials and natural products to make a model of the alimentary canal.
2. Design your model so that it will be a view from the front.
3. Think of how you are going to label the different parts.
4. Make your model and display it in the classroom.

Summary

- All living organisms get energy from their food.
- Some organisms, like green plants, produce their own food from inorganic material.
- Plants convert light energy into chemical energy. They do this by producing glucose from carbon dioxide and water in the presence of sunlight and chlorophyll.
- The main photosynthetic organ of the plant is the leaf.
- Because plants make their own food, they are called autotrophs. They are also the producers in an ecosystem as they provide food for all the other organisms in that ecosystem.
- Organisms that depend on the producers in their ecosystem for food are called heterotrophs. They are the consumers in their ecosystem.
- Examples of consumers are herbivores, carnivores, omnivores, detritivores and decomposers.
- Humans are consumers. When they ingest their food, it enters the alimentary canal, in which the food is processed before the body cells can release the energy in the food.
- The processes involved in human nutrition are ingestion, digestion, absorption, assimilation and egestion.

Glossary

abdominal cavity – the space containing the digestive system, the reproductive system, and so on

accessory organ – an organ that does not form a primary part of the digestive system, but plays a role in digestion

alimentary canal – the long, hollow tube that stretches from the mouth to the anus, forming the digestive tract

anus – the opening at the end of the alimentary canal

autotroph – an organism, for example a green plant, that makes its own food from inorganic materials

cellular respiration – the process by which cells release energy, together with carbon dioxide, from food

chloroplast – the chlorophyll-containing organelle in plant cells; location at which photosynthesis occurs

chyme – the mixture of gastric juice and food in the stomach

colon – the main part of the large intestine where absorption of water takes place

consumer – any organism that cannot produce its own food, but gets food by eating other organisms

cuticle – a waxy layer that is found on the outside of the epidermis of stems and leaves to prevent water loss

detritus – dead organic matter

ecological pyramid – a diagram showing the numbers or biomass of the organisms on each feeding level, or the energy available on the feeding levels

ecosystem – all the organisms in a given area, together with their environment

epidermis – outer layer of cells of leaves and stems

faeces – the food wastes discharged from the body

food chain – the pathway along which food energy is passed from producers to consumers; consists of a series of organisms that feed on the previous group and which are eaten by the next group

food web – the interconnected feeding relationships in an ecosystem

gastric pit – one of the folds inside the stomach wall containing the cells that secrete gastric juice

glucose – a simple carbohydrate or sugar produced during photosynthesis

heterotroph – an organism that feeds on other organisms and cannot produce its own food

inorganic – not organic, usually of mineral origin

mammal – any animal species grouped under the class Mammalia, named after the mammary glands; females have to produce milk for their offspring

mesophyll tissue – the chlorophyll-containing tissue in leaves that is specialised for photosynthesis

organic matter – matter found in living or once-living things; matter containing carbon and carbon-hydrogen bonds

peristalsis – the involuntary, muscular contractions and relaxations that push food along the alimentary canal

phloem – vascular tissue in plants that transports the food produced in the leaves to the rest of the plant

photosynthesis – the process by which green plants manufacture glucose from water and carbon dioxide in the presence of sunlight

predator – an animal that hunts and feeds on other animals

producer – one of the organisms that makes its

own food from inorganic materials, and thus supplies other organisms with food
rectum – the end of the colon, terminating in the anus

sphincter – a ring-like valve of muscle fibre found between certain sections of the alimentary canal

starch – a complex carbohydrate consisting of a chain of many simple sugar molecules

stoma – an opening in the epidermis of a leaf permitting gases to be exchanged

villus – a small, finger-like structure on the lining of a surface such as is found in the small intestine

xylem – vascular tissue that supports the plant and transports water and dissolved nutrients

Revision questions

1. Explain what role producers play in their ecosystem. (2)
2. Pick out the organisms that are NOT producers from the following list: humans; horses; bean plants; snails; earthworms. (2)
3. Give the word equation for photosynthesis. Indicate which substances are the reactants (raw materials), which are the catalysts (helpers) and which are the products. (7)
4. Define the term 'cellular respiration'. (3)
5. Outline the five processes involved in human nutrition. (10)
6. Look at the schematic diagram of the organs in the digestive system below.

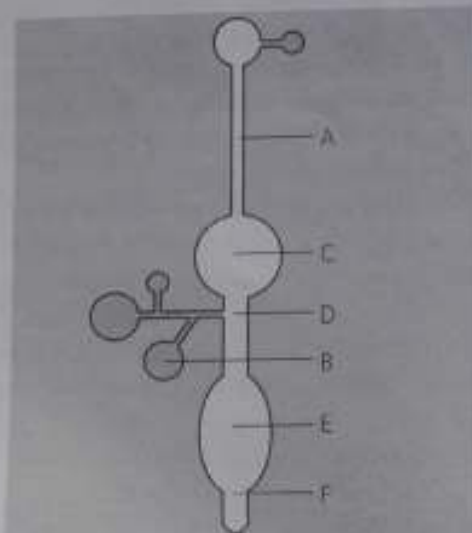


Figure 2.10 Digestive system organs

- a) Use these clues to fill in the missing labels (A to F). Write down only the letter and the correct label.
rectum; pancreas; stomach; large intestine; small intestine; oesophagus (6)
- b) Explain what aspect of the large and small intestines gave them their names. (1)
- c) Is the following statement true or false? Explain your answer.
The shaded organs (in Figure 2.10) are the main organs in the human digestive system. (2)
- d) Which letter represents the structure in which gastric pits are found? (1)
- e) Give the letter labelling the structure that is mainly responsible for the absorption of water in the alimentary canal. (1)

TOTAL: 35

Respiratory systems

Objectives

- State the word equation for cellular respiration.
- Label parts of the human respiratory system.

Introduction

The processes in living organisms (such as growth, cell repair and reproduction) require a lot of energy. Energy is stored in the food you eat, but your body cannot use it directly. Food is first digested and the nutrients are taken up in the bloodstream. Before your cells can make use of the energy in the nutrients, this energy must first be released. The process of setting energy free from nutrients to be used in the body's cells is called cellular respiration.

Cellular respiration requires a cell to exchange two gases with its surrounding environment. The cell takes in oxygen gas (O_2) and it gets rid of carbon dioxide gas (CO_2) as a waste product of cellular activity. You will remember from Form 1 that breathing involves the exchange of these same two gases between your blood and the outside air. Your lungs are the organs that make this exchange possible. The proportions of gases in the air you breathe in and the air you breathe out are very different.

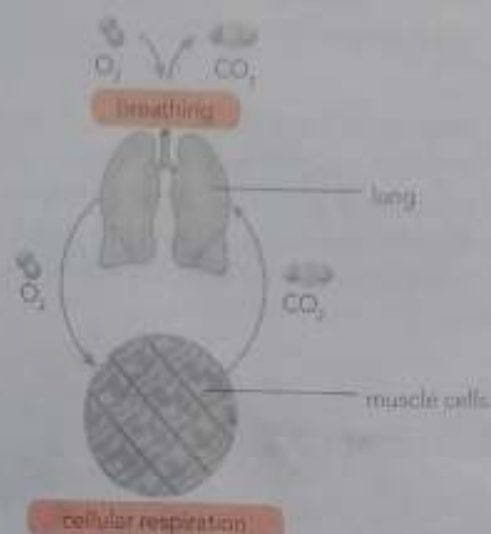


Figure 3.1 How breathing is related to cellular respiration

Cellular respiration

When seeds **germinate**, they make use of the food stored in the cotyledons (seed leaves) to provide energy to start growing. This means cellular respiration takes place. You get energy from food in much the same way. The next experiments explore aspects of cellular respiration.

Experiment 3.1

Aim: To demonstrate that carbon dioxide is a product of cellular respiration

Materials: two similar flasks with lids, seeds (beans or mealies), two beakers with lime water

Figure 3.2 Investigating if carbon dioxide is produced during cellular respiration



Carbon dioxide is released during respiration.

Procedure

1. Take half of the seeds the day before the experiment and soak them to start the process of germination. Keep these seeds in a damp cloth until the next day.
2. On the day of the experiment, place a beaker of lime water inside one of the flasks and place the germinating seeds around it. Seal the flask.
3. Also place a beaker of lime water in the second flask and place the dry seeds around it. Seal the flask. (Make sure this flask is completely dry so that the seeds will not start to germinate.)
4. Place both flasks in a dark place in the classroom. Observe what happens over the next two or so days.

Results

1. What were the results of the experiment after about two days?
2. What can you conclude?

Questions

1. What is the purpose of using germinating seeds in this experiment?
2. Why did you prepare a second flask containing dry instead of germinating seeds?
3. What is the purpose of placing a beaker of lime water inside the two flasks?
4. Why did you keep the flasks in a dark place?
5. What can go wrong with this experiment?

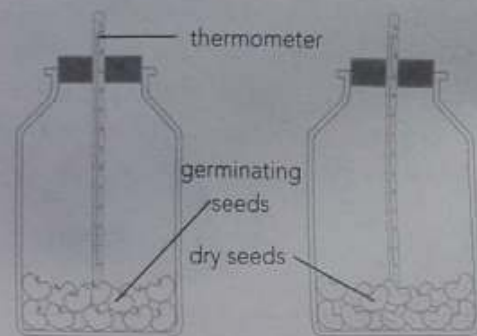
Experiment 3.2

Aim: To demonstrate that energy (heat) is a product of respiration

Materials: two similar flasks that can be fitted with a cork stopper, seeds (beans or mealies), two thermometers

Procedure

1. Take half of the seeds the day before the experiment and soak them to start the process of germination. Keep these seeds in a damp cloth until the next day.
2. On the day of the experiment, place the germinating seeds inside one of the flasks. Seal the flask with a cork stopper fitted with a thermometer as shown. The tip of the thermometer must be surrounded by seeds.
3. Do the same with the dry seeds. (Again, make sure this flask is completely dry so that the seeds will not start to germinate.)
4. Place both flasks in a dark place where the temperature remains fairly constant. Note the start temperature for both flasks after a lapse of about fifteen minutes.
5. Record the temperature readings on both thermometers every two hours for 24 hours. (Make arrangements with the school to do this in a safe way.)



heat is evolved in respiration

Figure 3.3 Investigating if energy is released during cellular respiration

Results

1. Plot temperature against time for both flasks.
2. What can you conclude?

Questions

1. Why did you use two flasks, one with germinating seeds and one with dry seeds in this experiment?
2. What is the purpose of measuring the temperature of the seeds inside the two flasks?
3. Why did you not start your temperature readings immediately after setting up the apparatus?
4. Why did you keep the flasks in a dark place with a fairly constant temperature?
5. How can you make this experiment more reliable and fair?

Experiment 3.3

Aim: To investigate energy release when burning sugars (food)

Materials: retort stand and clamp, Bunsen burner, beaker with water, thermometer, evaporating dish, glucose powder or mealie meal, gauze mat, tripod

Procedure

1. Set up the apparatus as shown in Figure 3.4, using a small amount of glucose powder (or mealie meal). (Use Figure 3.4, without the food burning, to help you set up the experiment.)
2. Record the temperature in the beaker of water before you start the experiment.
3. Heat the evaporating dish with a Bunsen burner until the glucose catches fire.
4. Immediately remove the Bunsen burner.
5. Measure the increase in temperature of the water once burning has stopped.

Questions

1. Calculate the increase in temperature of the water.
2. What does the increase in water temperature suggest about the energy in the glucose powder or mealie meal?
3. What does a substance react with when it is burnt?
4. What other observations did you make?
5. What precautions did you take in this experiment?

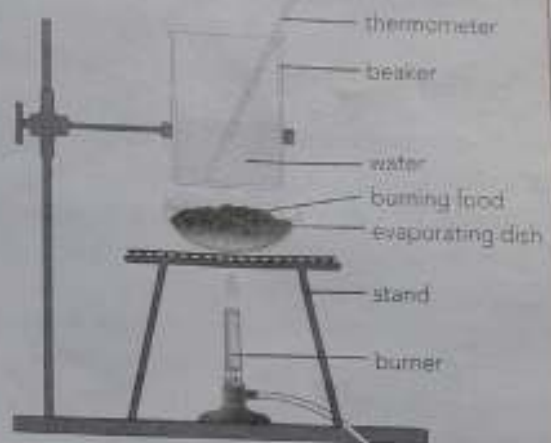
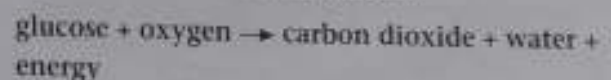


Figure 3.4 Investigating energy release during the burning of food

The experiments above show us that two of the products of cellular respiration are carbon dioxide and energy. If you breathe on a cold surface, such as a mirror or a glass pane, you will see water vapour condensing on the surface. On a cold winter's morning, you can also see a cloud of water vapour forming in the air as you breathe

out. This is the third product.

The reactants (raw materials) of cellular respiration are glucose and oxygen. The word equation for cellular respiration is:



The energy produced during cellular respiration is used by the cells to perform their functions. Some energy is lost as heat energy. Carbon dioxide and water are the waste products of cellular respiration. You have already learnt that these two substances are the reactants in photosynthesis. We say that there is **chemical cycling** between photosynthesis and cellular respiration.

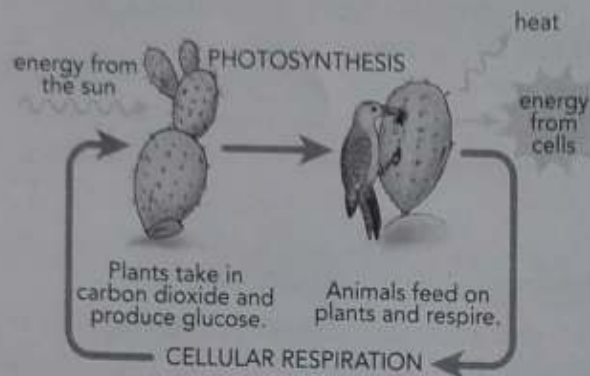


Figure 3.5 Chemical cycling between photosynthesis and cellular respiration

The human respiratory system

The human respiratory system is made up of the following:

- the air passages
- two lungs
- the respiratory muscles
- the respiratory control centre in the brain.

The human transport system makes **external gaseous exchange** and **internal gaseous exchange** possible. This will be discussed in the next unit.

The air passages

The air passages allow free movement of air into and out of the lungs. They include:

- the nasal cavities
- the pharynx (or throat)
- the glottis (opening to the voice box)
- the larynx (or voice box)
- the trachea (or windpipe)
- the bronchi (singular: bronchus)
- the bronchioli (singular: bronchiolus).

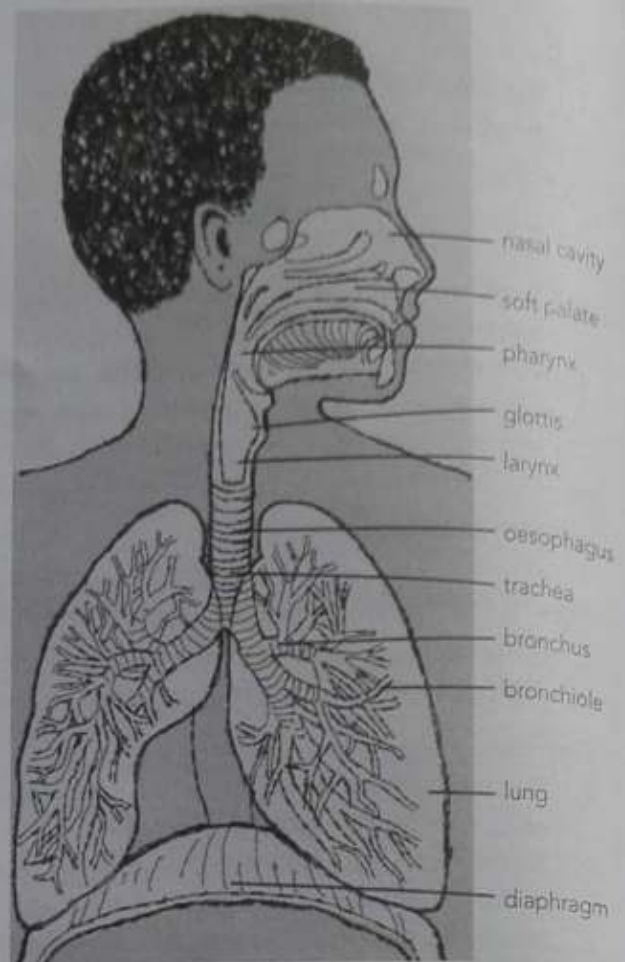


Figure 3.6 The air passages

The nasal cavities

When you breathe in through your nostrils, the hairs and **cilia** (singular: cilium) inside act as a screening device, trapping dust and other small particles. Irritation of the nasal lining can make you sneeze, which also helps to keep the air passages clean. The lining is supplied with blood **capillaries**, which warm the air you **inhale**. There are also mucous glands to moisten the air before it reaches the lungs. The **mucus** is antiseptic and helps to prevent the spread of germs.

The pharynx, the glottis and the voice box

The air and food passages meet in the pharynx. This means that food can enter the air passages to the lungs, but it also allows you to breathe through your mouth when you exercise very hard or when your nose is blocked.

You are prevented from choking when you swallow due to the upward movement of the trachea and the larynx, making the epiglottis close over the glottis. At the same time, the soft palate moves backwards, covering the entrance to the nose, and the food passes down the oesophagus.

When you breathe, the air passes freely through the glottis into the larynx, where your vocal cords are and then into the trachea.

The trachea

The wall of the trachea is strengthened and held open at all times by a series of C-shaped cartilage rings. You can feel these rings very clearly when you tilt your head backward and run your fingers up and down your neck. The Adam's apple is one of these rings and is clearly visible in adult men. The trachea forks into the left and right bronchus.

The bronchi and bronchioli

The left bronchus enters the left lung and the right bronchus enters the right lung. In the lungs, both left and right bronchi branch out into a network of fine tubes called the bronchioli. The system of air tubes looks like a tree turned upside down.

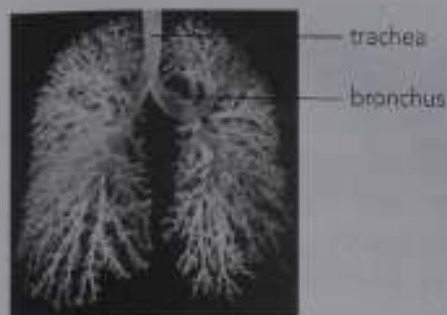


Figure 3.7 The bronchial tree.

The lining of the major branches of the bronchial tree is covered by cilia and a thin layer of mucus. The mucus traps particles, and the cilia beat the mucus upward to the throat, where it can be swallowed into the oesophagus. This is sometimes referred to as the 'mucus escalator' and it keeps the respiratory system clean.

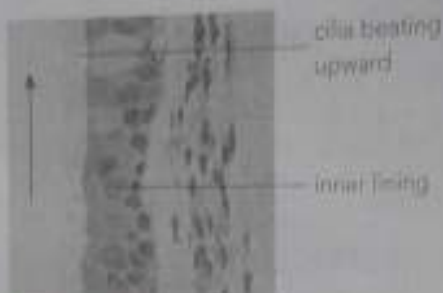


Figure 3.8 The cilia covering the lining of the major bronchial branches clean the system of harmful particles.

The lungs

External structure

The lungs are located inside the chest (thoracic) cavity, where they are well protected by the ribcage. They consist of spongy tissue, are very elastic and have a rich supply of blood capillaries. They are shaped like cones and are pinkish-grey in colour. The heart is situated in the space in between the lungs.

Internal structure

The bronchioli branch into smaller and smaller tubes deep into the lungs. Each bronchiolus ends in a cluster of tiny air sacs called alveoli. Each alveolus has an extremely thin wall and is surrounded by a network of tiny blood capillaries. Gaseous exchange takes place between the air in the alveoli and the blood in the capillaries. The capillaries also ensure the health of the alveoli. There are millions of alveoli in each lung and together they have a surface area about fifty times of that of the skin.

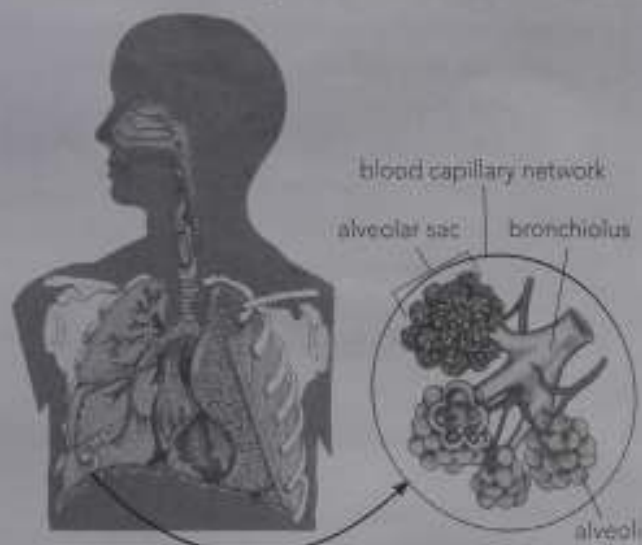


Figure 3.9 The external and internal structure of the lungs

The respiratory muscles

The muscles used for breathing are the **diaphragm** at the base of the chest cavity and the muscles between the ribs. These muscles are used when you inhale and exhale.

The mechanism of breathing

It is impossible to see what happens inside your chest cavity, but you can do activities to help you describe what happens when you breathe.

Activity 3.1a

Aim: To investigate the mechanism of breathing using your own body

Materials: your own body, hands and mind

Procedure



Figure 3.10 Breathing in and out with your hands placed a) on the sides of the ribcage b) on your diaphragm c) at the top of your ribcage

1. Stand up straight with your hands in different positions on your chest as shown in Figure 3.10.
2. Breathe in deeply and concentrate on what you feel as you do this. Repeat this until you feel you can describe the movement of your chest.
3. Now repeat the exercise while you breathe out deeply. Try to describe what you feel.

Results

Write down a description of the movements you felt while breathing in (inhaling) and breathing out (exhaling).

You will have felt how your ribcage moves up and out when you inhale. Your ribcage moves down and in when you exhale. You will also have felt how your diaphragm moves down (contracts) when you inhale and moves up (relaxes) when you

exhale. In order to imagine what happens to your lungs as you breathe in and out, you can make use of a breathing model.

Activity 3.1b

Aim: To investigate the mechanism of breathing using a breathing model

Materials: a plastic 2-litre cold drink bottle, rubber bands or string, rubber sheet big enough to be fitted around the bottom of the bottle, two balloons, Y-shaped tube, rubber bung or cork

Procedure

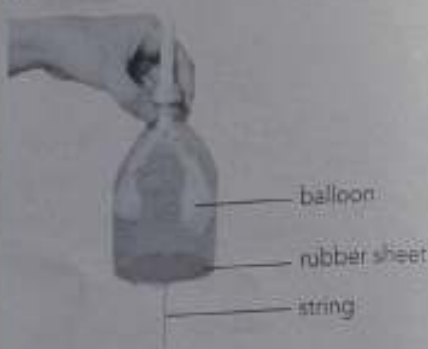


Figure 3.11 Assembling a breathing model

1. Cut off the bottom of the bottle.
2. Make a hole in the rubber bung for the Y-shaped tube to pass through. Put the bung in position at the top of the bottle.
3. Tie a balloon around each short arm of the Y-shaped tube. Push the long arm through the rubber bung so that your 'lungs' are inside the 'chest cavity'.
4. Fit the rubber sheet securely around the bottom of the bottle with rubber bands. This represents the diaphragm for your breathing model. Make sure that it is not too tight to pull down or push up.
5. Observe what happens when you pull the rubber sheet downwards and push the sheet upwards.

Results

1. Copy the breathing model in your exercise book and draw in the shape of the balloons:
 - a) before the start of the activity
 - b) while pulling down the rubber sheet
 - c) while pushing up the rubber sheet
2. Give your drawing a suitable heading.

Questions

Use your drawings to help you answer the questions below.

1. Compare the pressure inside the balloons that are inside the bell jar to the air pressure outside the bell jar before the start of the activity.
2. How did this change when you pulled the rubber sheet down?
3. What movement in your chest is represented by pulling the rubber sheet down?
4. What changed when you pushed the rubber sheet up and what did it represent?
5. What is the limitation of this model? In other words, what cannot be shown by this breathing model?

Inhalation is the active phase of breathing. The diaphragm contracts and moves down. The external muscles between the ribs contract so that the ribcage moves out and up. The chest cavity increases in volume. The pressure in the lungs decreases so that air from outside flows in.

Activity 3.1b illustrates the pressure change in the lungs and how air flows into the lungs, but it cannot show the movement of the chest cavity accurately.

The respiratory control centre

Although you can decide to hold your breath or to breathe faster or slower, you are, most of the time, unaware of the fact that you are breathing. This is because you do not have to concentrate to do it.

Breathing is controlled automatically by the respiratory control centre at the base of the brain. We say it is an **involuntary** action. Breathing control is only effective in keeping the correct levels of oxygen and carbon dioxide in the body if there is coordination with blood flow.

In the next activity, you are going to observe the relationship between your breathing rate and your heart rate. Both rates are normally measured at rest, that is, when a person is sitting or lying quietly. For the breathing rate, count one for each

inhalation-exhalation cycle. For your pulse, count one for each beat you feel.

Both rates are given per minute. You can take the counts over 30 seconds and multiply them by two. A normal breathing rate at rest is between 16 and 20 per minute. A normal pulse rate at rest is between 60 and 100 per minute.

Activity 3.2

Aim: To investigate the relationship between heart rate and breathing rate

Materials: stopwatch

Procedure



Figure 3.12 Taking a pulse

1. Work in pairs and measure each other's pulse rate at the wrist as shown in Figure 3.12. Do not use your thumb, because it has its own pulse that you may feel.
2. Record each other's breathing rate.
3. Let your partner jump up and down twenty times and take the pulse rate again.
4. Jump up and down twenty times and let your partner take your pulse rate.
5. After a rest, repeat steps 3 and 4 in order to take your breathing rates.

Results

Draft a table with the measurements you have taken:

	At rest	After 20 jumps
Heart rate (per minute)		
Breathing rate (per minute)		

Your results should show that both rates increased after exercise.

Glossary

capillary – any of the smallest blood vessels
chemical cycling – process during which chemical substances are used and produced in a continuous cycle
cilia – short, hair-like projections in the respiratory tract which sweep away fluids and particles in waves
diaphragm – the muscle sheet at the bottom of the chest that pulls air into the lungs when it contracts
epiglottis – a cartilaginous flap that closes over the opening to the voice box, when swallowing
exhale – to breathe out
external gaseous exchange – the exchange of gases across the lining of the lungs to and from the atmosphere

germinate – to start growing a root and shoot from a seed or grain

internal gaseous exchange – the exchange of gases across the cell membrane to and from a cell

inhale – to breathe in

involuntary – done without intention, not controllable

mucus – a sticky and slippery mixture secreted by a mucous membrane to moisten and protect the membrane

water vapour – water in the form of a gas

Summary

- Cellular respiration is the process of setting energy free from nutrients to be used in the cells of an organism.
- The word equation for cellular respiration is: glucose + oxygen $\longrightarrow$ carbon dioxide + water + energy.
- Cellular respiration and photosynthesis are two processes that are related through chemical cycling.
- The human respiratory system is made up of the air passages, two lungs, the respiratory muscles and the respiratory control centre in the brain.
- The air passages allow free movement of air into and out of the lungs. They include the nasal cavities, the pharynx (or throat), the glottis (opening to the voice box), the larynx (or voice box), the trachea (or windpipe), the bronchi and the bronchioli.
- External gaseous exchange takes place in the alveoli when gases diffuse into and out of the bloodstream.
- Internal gaseous exchange happens in the cells and makes cellular respiration possible.
- The respiratory control centre in your brain controls your breathing rate. The breathing rate and heart rate are closely linked to maintain the correct levels of gases in the body.
- Breathing is mostly involuntary and the mechanism makes use of the decrease of air pressure in the lungs when the chest cavity expands.

Revision questions

1. State the word equation for cellular respiration. (5)
2. Complete: Cellular respiration and _____ are related through chemical cycling. (1)
3. A learner made the model shown in Figure 3.13 using the following materials: a plastic cooldrink bottle, a straw fitted through a cork, a balloon, a piece of thick plastic sheeting, glue, an elastic band and masking tape.
 - a) State the aim of the experiment. (2)
 - b) What parts are represented by the straw, the plastic bottle, the balloon and the plastic sheeting? (4)
 - c) State two limitations of this model. (2)
 - d) Describe the process in the human body that is shown by pulling the elastic band. (8)
4. What is the function of the respiratory control centre in the brain? (2)
5. Choose the correct option: There is a direct/indirect relationship between breathing rate and pulse rate. (1)

TOTAL: 25



Figure 3.13 Experimental set-up for the learner's experiment in question 3

Unit
4

Transport systems

Objectives

- Give an overview of the internal structures of plant roots and stems.
- Describe the uptake of water and ions by plants.
- Describe the human circulatory system.
- Draw and label the structures of the heart.
- Name the main blood vessels transporting blood to and from the human heart.
- Identify the functions of the human heart.

Introduction

Flowering plants and animals need transport systems because they are multicellular. In single-celled organisms, all life processes take place within a single cell, and can be controlled by one cell that is in direct contact with its environment. But multicellular organisms have to overcome the problem of distance between cells. For example, a cell producing or processing food may be very far away from where the food is needed. For this reason, flowering plants and humans have specialised transport systems to get substances from one part of the organism to another.

Processes involved in transport

There are three main processes involved in transporting substances across cell membranes:

- diffusion
- osmosis
- active transport.

Diffusion

The molecules in **fluids** (gases and liquids) are in constant motion. They move in all directions, without any pattern. When there are more molecules in one region and fewer of them in another region, it creates a **concentration gradient** between the two regions. Molecules then tend to show an overall movement towards the region where they are in a lower concentration.

Diffusion is the **net movement** of molecules down a concentration gradient, that is, the overall movement of molecules from a place where they are at a higher concentration to one in which they are at a lower concentration. This net movement continues until the concentrations are the same in both regions and the molecules have reached a state of **dynamic equilibrium**. The molecules do not stop moving, but are spread out evenly. This is illustrated in Figure 4.1.

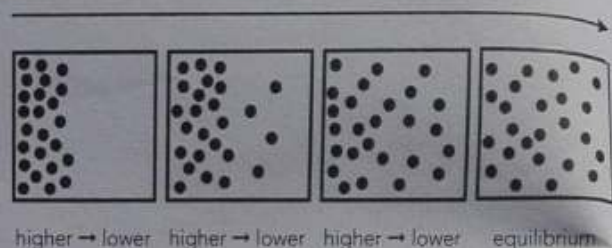


Figure 4.1 Molecules move down a concentration gradient to reach equilibrium.

Diffusion plays an important role in the life processes of plants and animals. For example, nutrients, gases and waste products move from one cell to another by diffusion. Several factors can influence how fast diffusion takes place. These factors include:

- the size of a particle (smaller particles diffuse faster)
- temperature (particles move and spread faster at higher temperatures)
- concentration gradient (a big difference in concentration increases the rate of diffusion)

Experiment 4.1

Aim: To demonstrate diffusion in a liquid

Materials: a glass beaker, hot water, a tea bag

Procedure

1. Fill the beaker with the hot water and wait for the water to become still.
2. Carefully lower a tea bag into the beaker.
3. Observe the changes in the beaker until you cannot see any more changes.

Questions

1. Describe your observations.
2. Why did you wait for the water to become still and why did you lower the tea bag gently into the water?
3. Did you have to use hot water for the experiment?
4. Make drawings to illustrate your observations at the beginning of the experiment and at the end.
5. Label areas of high concentration and low concentration.



Figure 4.2 Diffusion in a beaker of tea.

- distance (the further particles have to travel, the longer it takes to reach equilibrium)
- surface area (the larger the surface area of a membrane to cross, the greater the rate of diffusion)
- type of membrane (a membrane can be highly permeable or selectively permeable).

Osmosis

The movement of water across selectively permeable membranes is an example of a special type of diffusion called osmosis. When substances are dissolved in water, two solutions can have the same concentration of water, or one of the solutions can have a higher concentration of water than the other. Suppose these two solutions are separated by a selectively permeable membrane that prevents the *dissolved substance* from crossing. In order to reach a state of equilibrium, the *water molecules* will move across the membrane from where water is at a higher concentration to where it is at a lower concentration. The water will not stop crossing the membrane when equilibrium is reached, but the same number of water molecules will move in both directions across the membrane. This is shown in Figure 4.3.

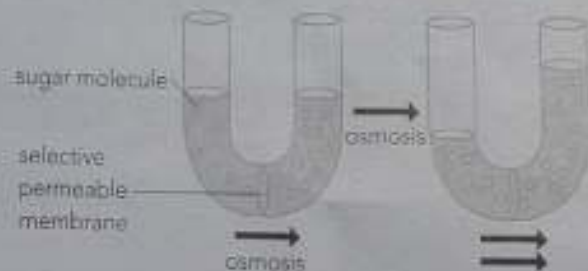


Figure 4.3 Osmosis is a special kind of diffusion.

Active transport

Both diffusion and osmosis depend on the **spontaneous** movement of molecules down their concentration gradient. No additional energy is necessary to make these processes happen. We say that they are **passive processes**.

When molecules or ions are transported **against** their concentration gradient, across a selectively permeable membrane, that is, when they move across a selectively permeable membrane from where they are in a lower concentration to where they are more concentrated, active transport happens. Active transport needs extra energy to drive **transport proteins** in the cell membrane to pump particles

in a direction opposite to one in which the particles would normally flow. Transport proteins recognise only specific particles and they will help only those particles across the membrane. This is shown in Figure 4.4 where only the "round particles" are recognised by the transport proteins. The advantage of this type of transport is that more of a certain particle can be taken up into a cell that already has more of it than in the surroundings of the cell.

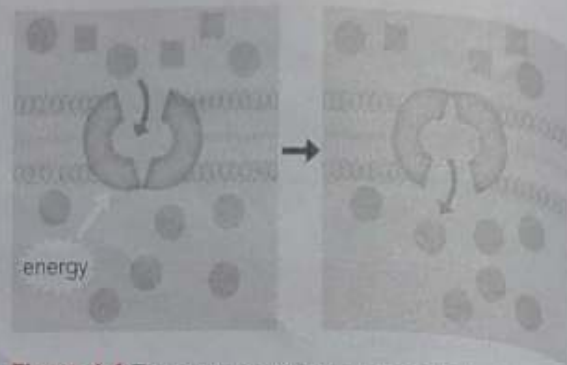


Figure 4.4 Transport proteins pump particles they recognise against their concentration gradient across a membrane.

Table 4.1 gives examples of these three types of transport in plants and humans.

Table 4.1 Examples of diffusion, osmosis and active transport in plants and humans

	Plant	Human
Diffusion	Oxygen and carbon dioxide move through the stomata of leaves Water vapour moves from the stomata to the atmosphere	Oxygen and carbon dioxide move between the alveoli and the bloodstream Oxygen and carbon dioxide move between cells and the bloodstream Glucose and other nutrients move from the small intestine into the bloodstream
Osmosis	Water movement from the soil into the roots of plants Water movement from plant cell to plant cell	Water reabsorption in the colon Water is reabsorbed from the kidneys into the bloodstream
Active transport	Nitrates and other ions from the soil are absorbed into the roots of plants Nutrients dissolved in xylem sap into cells Manufactured carbohydrates move from phloem to plant cells	Useful substances, such as glucose, amino acids and other minerals, are reabsorbed in the kidneys into the bloodstream

Experiment 4.2

Aim: To track the movement of water and minerals in the vascular tissue of plants.

Materials: a white flower (chrysanthemum or carnation), two measuring cylinders or test tubes, a scalpel, a tile to cut on, two solutions prepared by adding different colours of food colouring to water, sticky tape, paper

Safety precautions: Be careful when working with a scalpel. Always cut away from your body on a cutting surface. Put your scalpel in a safe place after using it.

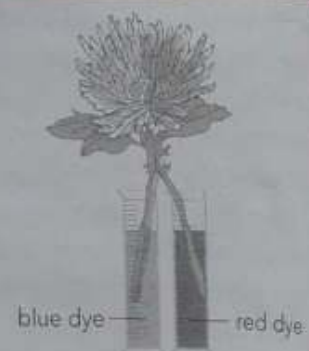


Figure 4.4 Tracking the movement of water and minerals in the vascular tissue of plants

Procedure

1. Use a white flower with a long stalk. A white chrysanthemum or a white carnation will work well.
2. Fill your measuring cylinders (or test tubes) with the different coloured water solutions.
3. Cut the stalk of the flower carefully along its length. Do not cut it right up to the base of the flower.
4. Arrange the flower with each half of the stalk in one of the measuring cylinders (or test tubes).
5. You can fold wedges of paper and place these under the measuring cylinders to make them tilt towards each other. This makes the arrangement of the flower easier.
6. Tape the measuring cylinders down so that they will not fall over.
7. Leave the flower for a few hours like this and observe what happens.
8. Leave the flower in the measuring cylinders for about a week so that you can observe it. You will have to top up the water solutions in the measuring cylinders.

Questions

1. Describe what you observed after a few hours and as the days passed.
2. What was the role of the dye in this experiment?
3. What happened to the water in the solutions? And to the dye? Why?
4. Why did you use two colours instead of only one?
5. Do water and nutrients move only upward in a plant? Explain your answer.
6. What conclusions can you draw from this experiment?

Exercise 4.1

1. Fill in the missing words:

- a) Diffusion occurs when molecules have a net movement from a region of _____ concentration to a region of _____ concentration. They move down a concentration _____ until _____ is established.
- b) An example of osmosis is when _____ molecules show an overall movement across a _____ permeable membrane from a point at which water molecules are present in a _____ concentration to where they are in a _____ concentration, until _____ is reached.
- c) _____ transport happens when molecules are moved across a cell _____ against their _____ gradient.
- d) Both _____ and _____ are examples of passive _____.

2. Choose the correct option to complete the sentences below:

- a) When equilibrium is reached through the process of diffusion, the particles *(continue to move/stop moving)*.
- b) Active transport *(needs/does not need)* extra energy.
- c) Active transport of a nutrient will happen when a cell has a *(higher/lower)* concentration of that nutrient than the environment of the cell.

Transport in plants

The body of a flowering plant consists of different plant organs, namely, roots, stems, leaves and flowers. Each of these has their own specialised functions. In this section, you are going to learn how the roots and the stem of a dicotyledonous plant are adapted for their function as transport organs.

Vascular tissue in plants

Plants have **vascular** or conducting systems like animals do. But they are made up differently and function differently. The two types of vascular tissue in plants are xylem and phloem. You have already learnt a little about them. Let us look at them in more detail.

Xylem

Water and dissolved minerals are taken up by the plant through the roots. But the water is needed much further away in the leaves, stem and other green parts of the plant, where photosynthesis takes place. The cells can store some of the water until they need it, but water is also lost from the leaves during **transpiration**. This means that a constant supply of water is needed. Xylem tissue allows for this.

Xylem tissue is made up of long cells with **perforated** ends. The cells are lined up to form tubes through the entire length of the plant. These cells are called vessels and tracheids. Their walls are thickened by a build-up of a woody, strengthening substance called lignin. The thickening structures can be arranged in different ways, such as rings or spirals. The thickening is necessary to help the plant resist the pressure of the water moving through the tubes. It supports the whole plant.

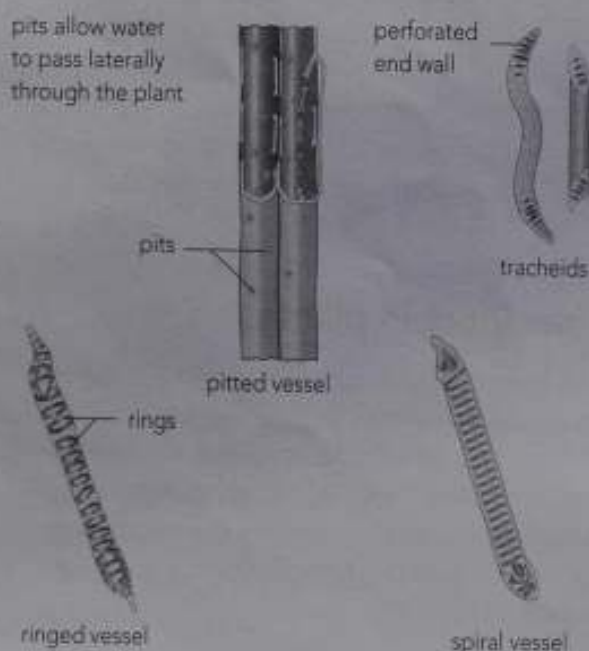


Figure 4.5 Xylem tracheids and vessels showing different kinds of thickening

The older xylem cells are dead, their cell content has disappeared and the end walls often **disintegrate** completely, leaving only a continuous hollow tube for xylem sap (the **cellular solution** of water and nutrients inside xylem) to move through. The tracheids are special cells with pits in the side walls, making it possible for water to flow sideways too. This is useful if parts of the plant have been damaged. You observed the upward and sideways movement in Experiment 4.1.

Some examples of xylem segments are shown in Figure 4.5.

Phloem

Phloem tissue carries food made in the leaves to all the other parts of the plant. This transport of simple sugars through the cell sap of the phloem is called **translocation**. Translocation mostly happens in a direction opposite to the transport direction in the xylem tubes, and is associated with active transport.

The main components of phloem are sieve tubes and companion cells. The sieve tubes have perforated end walls and they transport the products of photosynthesis. Sieve tube cells have very little **cytoplasm**, no nucleus and very few organelles. In order to function, they depend on the companion cells, which have large **nuclei**. Scientists still do not know exactly how companion cells work.

Unlike xylem, phloem has no important strengthening function in the plant. Figure 4.6 shows some of the important aspects of phloem.

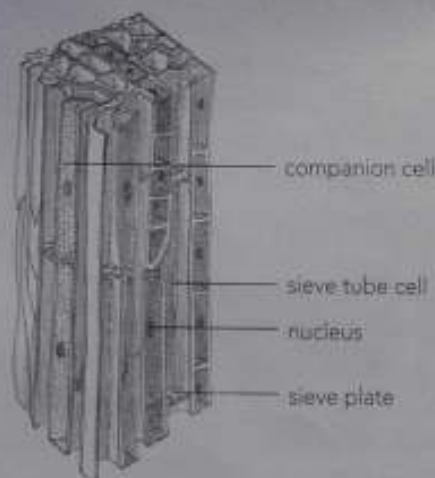


Figure 4.6 The characteristics of phloem tissue

The root of a dicotyledonous plant

The function of the roots of plants is to absorb water and dissolved nutrients, such as nitrates, phosphates, potassium ions, and so on, from the soil. This causes the root system of plants to branch out widely in order to ensure a large surface for absorption. Roots can, in fact, branch out wider than the part of the plant you see above the soil.

A dicotyledonous plant has a **tap root** or main root, **lateral roots** and **root hairs**. The roots anchor the plant firmly in the soil and allow water and dissolved nutrients to enter the plant. Roots can store water and starch.

Internal structures of the root of a dicotyledonous plant

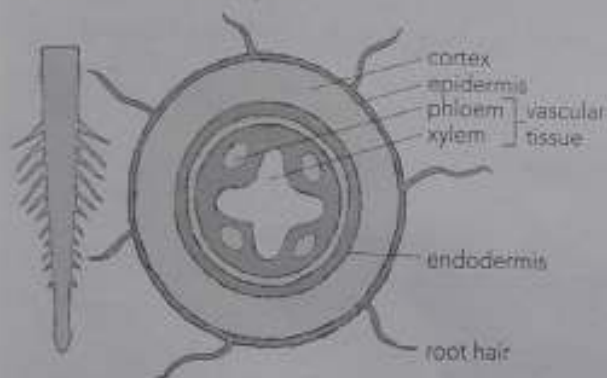


Figure 4.7 A line diagram of a transverse section through the young root of a dicotyledonous plant

The root of a dicotyledonous plant consists of three main regions:

- the outer epidermis together with root hairs
- the cortex together with the endodermis
- the central part containing vascular tissue.

The outer epidermis

The root epidermis is a single layer of cells on the outside of the roots with the function of protecting the inner root tissues. Root epidermis does not have a waxy layer covering it, as this would prevent the absorption of water. The cells of the root hairs have large central vacuoles and large surface areas for the uptake of water through osmosis.

The cortex

The cortex consists of round cells with thin walls and air spaces between them. They provide a

pathway for water to the xylem, but can also store water and food. The air spaces between them make the exchange of gases by diffusion possible in the root area.

The endodermis is a single layer of cells with the function of regulating the flow of water and other substances into the plant. They allow the root to select which substances reach the vascular tissue in the central area of the root.

The central part

The central part of the root contains the vascular tissue grouped in a single vascular cylinder.

In a dicotyledonous root, the xylem tissue is arranged in the form of a cross with rounds of phloem between the arms of the cross.

The arrangement of xylem in a cross makes it easy to identify a transverse section of a dicotyledonous plant.

The stem of a dicotyledonous plant

The stems of plants are adapted to perform certain functions:

- They connect the roots and leaves with the rest of the plant so that there is a steady supply of water, nutrients and food.
- Stems produce new growth and they can also store food and water.
- Green parts of stems can photosynthesise.

Stems must be strong, because they carry leaves, flowers and fruits. In a dicotyledonous plant, the arrangement of leaves, flowers and fruits is different from what we see in monocotyledonous plants.

Internal structure of the stem of a dicotyledonous plant

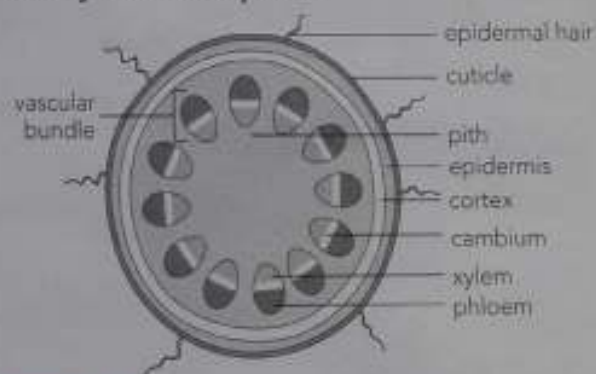


Figure 4.8 A line diagram of a transverse section through the stem of a young dicotyledonous plant

A young dicotyledonous stem has three regions: the epidermis, cortex and the central area containing the vascular tissue grouped together in vascular bundles. This arrangement of vascular tissue is characteristic of the stem of a dicotyledonous plant.

The epidermis

In young green plants, this is a single layer of closely packed cells. A waxy layer (the **cuticle**) covers the epidermis to reduce water loss. There may be epidermal hair. In woody stems, the epidermis is made up of several layers and forms the bark of the plant.

The cortex

The cells of the cortex that are just below the epidermis are tightly packed and the contain chloroplasts. The stems of young plants can thus photosynthesise. The tightly packed cells strengthen the stems. The cortex cells that are found below this layer have thinner walls with air spaces between them. Organic food such as starch is stored here.

The central part

In young stems of dicotyledonous plants, the vascular tissue is arranged a ring of vascular bundles.

In the vascular bundles, the xylem is found on the inside and the phloem on the outside. A strip of cambium cells separate the xylem and the phloem. Secondary growth of vascular tissue can occur here.

The pith consists of large thin-walled cells with air spaces in between so that gases can be exchanged.

Activity 4.1

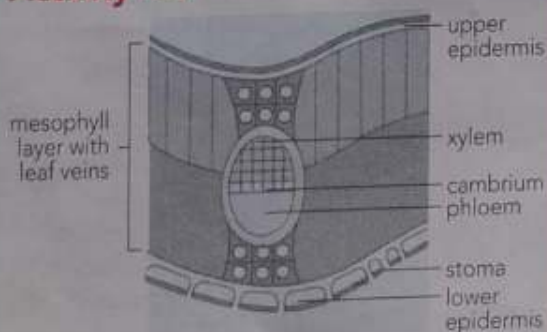


Figure 4.9 A line diagram of part of a dicotyledonous leaf (transverse section)

In Unit 2, you learnt how the structure of the leaf is adapted for photosynthesis. Study the line diagram of a transverse section of a dicotyledonous leaf in Figure 4.9 as well as the transverse sections of a dicotyledonous root and stem in Figures 4.7 and 4.8, respectively. Now do the following.

1. Use these diagrams to identify the micrographs in Figure 4.10 A, B and C as a stem, a root or a leaf of a dicotyledonous plant. (Your teacher may also provide you with prepared biology slides from which to work instead.)
2. Make a drawing of each, clearly showing the vascular tissue and the arrangement of tissues specific to the part of the plant concerned. (You may choose to draw a detail of the whole only.)
3. Label your drawings and give them suitable headings.

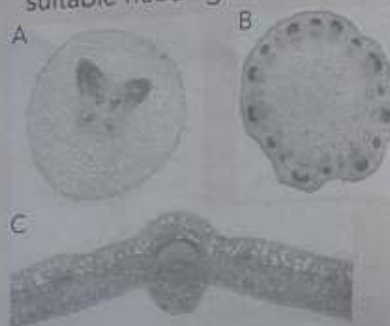


Figure 4.10 Micrographs to identify

Transport in humans

You have seen that the transport system of flowering plants is very specialised. It is a response to the fact that plants are multicellular and need to get nutrients, water and gases to cells that are far away from the surroundings, or from where food is produced.

Animals have a different way of responding to the challenge of getting nutrients and getting rid of waste materials across many cells. Because animals are active, their needs may require an immediate response and quick transport of some substances. Think of how you feel when you have held your breath for a long time, or if you have not had breakfast and become very hungry.

The substances that need rapid transport in the human body are:

- oxygen from the lungs to all body cells
- nutrients from the alimentary canal to all body cells
- carbon dioxide from all body cells to the lungs
- waste products from the cells to the kidneys for excretion.

Activity 4.2

1. In pairs, look at the substances that need rapid transport, listed above. Discuss and give examples of what could happen if transport is not quick enough.
2. Report back to the class and compare your answers.

Blood transport organs

The human blood transport system can be divided into three parts:

- The heart serves as a pump, ensuring that blood and its contents are circulated throughout the whole body.
- The blood vessels are made up of arteries, veins and capillaries, and they provide the pathways for blood.
- The blood transports substances by following the pathways of the blood vessels.

The heart

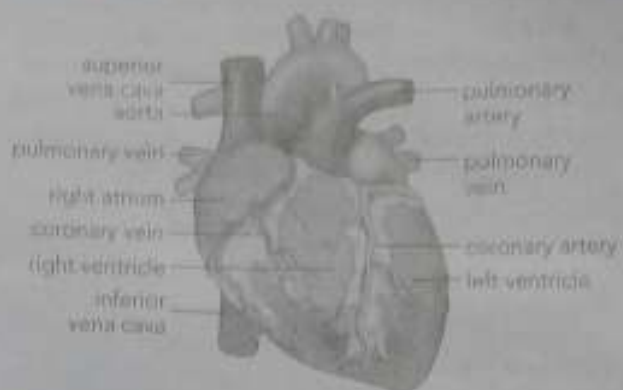


Figure 4.11 The external structure of the heart.

The heart is about the size of a person's fist. It is a hollow, muscular sac, broad at the top and pointed at the bottom. It hangs between the two lungs inside the thoracic (chest) cavity, slightly off centre and pointing to the left. The heart is surrounded by the **pericardium**, a thin sac with two walls. The inside layer fits snugly around the heart, but the outside layer fits more loosely. The function of the pericardium is to reduce friction between the movement of the beating heart and the other organs. The big blood vessels entering and leaving the heart help to keep it in place. On the outside, the blood vessels carrying blood to and from the heart are also clearly visible.

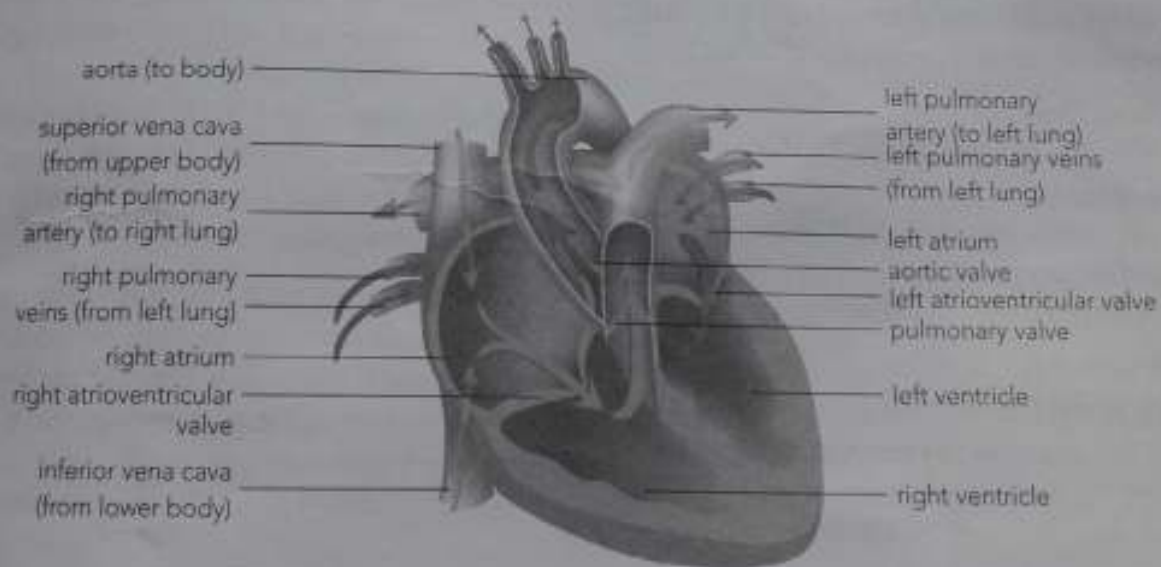


Figure 4.12 The interior structure of the heart.

There are two grooves running along the outside surface of the heart. They show where the inside walls of the heart divide it into **chambers**. The heart is divided into two halves by an internal wall called the **septum**. Each half is further divided into two chambers that are connected, but can be closed off from each other, by **valves**. There are thus four chambers in the human heart. The two chambers at the top of the heart are called **atria** (singular: **atrium**) and are smaller than the two bottom chambers, the **ventricles**.

The right atrium

The right atrium receives **deoxygenated** blood from the whole body. Because the right atrium only has to pump blood to the right ventricle, it does not have a very thick wall.

The following blood vessels enter the right atrium:

- The *superior vena cava* receives blood from the different veins transporting oxygen-poor blood from the head and upper body.
- The *inferior vena cava* brings oxygen-poor blood from lower parts of the body into the right atrium.
- The *coronary vein* drains oxygen-poor blood from the heart itself.

The right ventricle

The walls of the right ventricle are thicker than those of the right atrium. The right ventricle receives blood from the right atrium through the right **atrioventricular** opening. The blood is pumped into the main **pulmonary artery** (pulmonary trunk) that divides into the *left and right pulmonary arteries* to take blood to the lungs.

The left atrium

The left atrium receives **oxygenated** blood from the two lungs from *four pulmonary veins*. The walls of the left atrium are relatively thin as they have to pump this oxygen-rich blood only into the left ventricle through the left atrioventricular opening.

The left ventricle

The walls of the left ventricle are very thick. This chamber has to pump oxygen-rich blood to almost all parts of the body through the **aorta**.

The valves in the heart

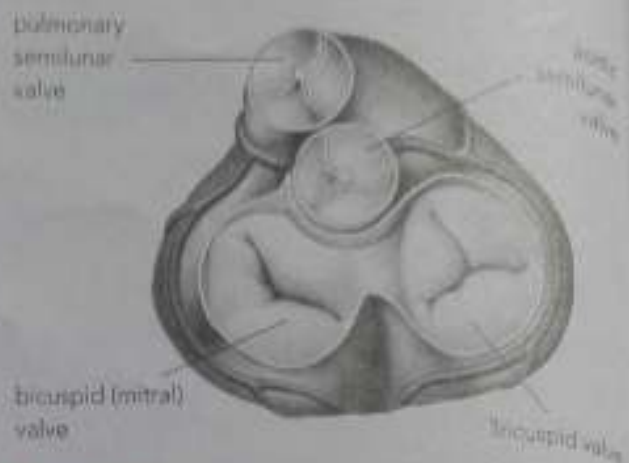


Figure 4.13 The four valves of the heart

The heart has four valves:

- The tricuspid valve has three folds (cusps). It separates the right atrium from the right ventricle and prevents blood flowing back into the right atrium when the right ventricle contracts.
- The bicuspid valve with its two folds separates the left atrium from the left ventricle. It prevents blood from flowing back into the left atrium when the left ventricle contracts.
- The semilunar valve of the pulmonary artery has three flaps that look like half-moons. From the inside of the artery they look like little bags. When the ventricle pumps blood into the artery, the blood pushes the valves open. Once inside the artery, the pressure in the artery causes blood to flow back towards the ventricle that is now relaxing. The blood flows into the little bags and pushes them tightly shut against one another.
- The semilunar valve of the aorta has a similar structure and function as the semilunar valve in the pulmonary artery. It prevents blood from flowing back into the right ventricle.

There are no valves where the two *venae cavae* (singular: *vena cava*) and the four pulmonary veins enter the heart. The muscle tissue of the heart forms a circle around these openings and they are closed when the heart contracts.

The pathway of blood through the heart

From the description above, you will already have some idea of how blood flows through the heart. The heart of a healthy adult beats about 72 times a minute. That means one heartbeat takes about 0.8 seconds. A **cardiac cycle** of one heartbeat completes with the simultaneous contraction of first the atria and then of the ventricles. The blood flows as follows:

- The superior and inferior venae cavae with oxygen-poor blood (with high carbon dioxide concentration) empty into the right atrium.
- The right atrium sends blood through the tricuspid valve to the right ventricle.
- The right ventricle sends blood through the pulmonary semilunar valve into the main pulmonary artery, branching into the two pulmonary arteries to the lungs.
- The four pulmonary veins empty oxygen-rich blood into the left atrium.
- The left atrium sends blood through the bicuspid valve to the left ventricle.
- The left ventricle sends blood through the aortic semilunar valve into the aorta to the rest of the body.

You will notice that oxygenated blood never mixes with deoxygenated blood. You will also notice that blood has to go through the lungs in order to pass from the right side to the left side of the heart. This means there is a double loop in the blood transport

system. And it means *not all* oxygen-rich blood links with arteries and *not all* oxygen-poor blood links with veins. Figure 4.14 summarises this. Let us look at the two loops in the pathway blood follows through the human body in detail.

The pulmonary circuit

In the pulmonary circuit, the path along which blood travels is as follows:

- Blood that is poor in oxygen and rich in carbon dioxide from all parts of the body collects in the right atrium and is sent to the right ventricle.
- The right ventricle pumps the blood into the main pulmonary artery.
- The main pulmonary artery divides into the right and the left pulmonary arteries, which carry the blood to the lungs.
- This blood now passes through the capillaries into the lungs.
- Carbon dioxide diffuses into the alveoli from the bloodstream and oxygen diffuses from the alveoli into the bloodstream.
- Oxygen-rich blood returns to the heart and passes into the left atrium through the pulmonary venules that join to form pulmonary veins.

The systemic circuit

In the systemic circuit, blood is transported to and from all parts of the body, excluding the

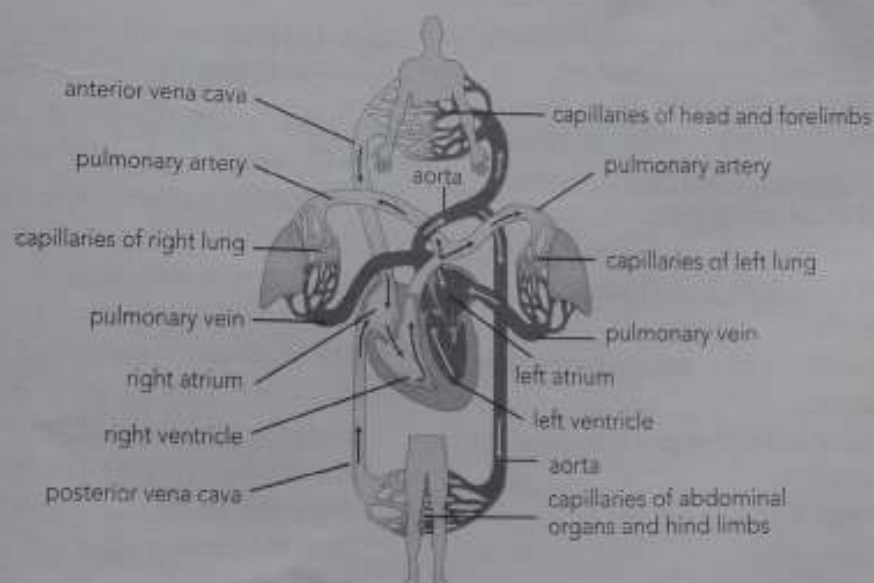


Figure 4.14 The two circuits of the human blood transport system

lungs. The aorta and the venae cavae are the main blood vessels in this circuit. You can trace the path of the blood to any organ by starting from the left ventricle of the heart; you will mention the aorta, then the particular branch of the aorta to the organ. You will then mention the capillary network that supplies the organ. As the blood moves through the organ's tissue, oxygen diffuses into the organ's tissue and carbon dioxide diffuses into the bloodstream. You will see that the capillaries join to form the vein that returns the now oxygen-poor blood to the correct venae cava (superior or inferior). Finally, the venae cava enters the right atrium of the heart.

Exercise 4.2

Study the diagram of the heart below and answer the questions that follow.

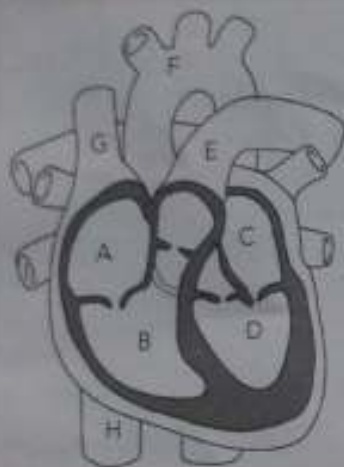


Figure 4.15 A diagram of the heart

1. Provide labels for structures A to I.
2. Say if the following statements are true or false.
 - a) Blood flows into the heart from structure F.
 - b) Blood flows from C to D inside the heart.
 - c) Blood flowing in structures G and H is oxygen-rich.
 - d) The valve between A and B is called the tricuspid valve.
 - e) Structures A and C contract at the same time.
 - f) Structures C and D relax at the same time.

Activity 4.3

1. Practise redrawing Figure 4.15.
2. Practise labelling the diagram:
 - a) Label all the structures.
 - d) Indicate the blood flow inside the heart, and to and from the heart.
3. Once you are confident doing this, your teacher may give you a model of the heart to draw.

The blood vessels

Humans, like other mammals, have a closed blood system. This means that blood flows only in the blood vessels, even though some of the blood components leave the blood vessels. There are three types of blood vessels: arteries, capillaries and veins. They have different features and functions.

Arteries

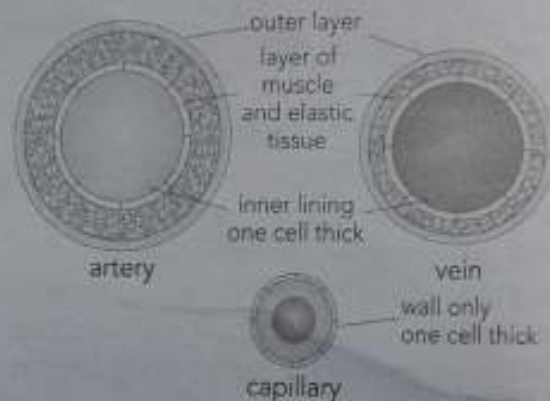


Figure 4.16 A comparison of the walls of an artery, a capillary and a vein

These blood vessels carry blood away from the heart. When the heart contracts, blood is pumped into the arteries, placing them under huge pressure. Their thick elastic walls make it possible for them to stretch without being damaged. The blood vessels comprise three layers:

- The outer layer of connective tissue with elastic fibre gives the artery elasticity and strength.
- The middle layer consists of smooth muscle fibres that are circular and have more elastic fibre. The nerves connected to them control how widely the artery tube opens or closes, depending on the blood flow.

- The layer on the inside is made of flat epithelial cells which are tightly packed together. This layer runs from the inside surface of the heart and makes the artery tubes very smooth inside.

As arteries branch out, they become finer vessels. The finer branches are called arterioles. Arteries do not have valves, except where they start in the heart.

Capillaries

As the arterioles branch out, they gradually lose the two outer layers until only the inner layer of **epithelium** remains. The vessels are now microscopically thin and are called capillaries. The capillaries branch around the tissue cells throughout the body. Every cell in your body is very close to a capillary: less than 0.1 mm. The blood flow in the capillaries is very slow and their walls are made up of only one layer of cells.

Blood plasma, the liquid part of blood, passes out through the capillary walls into the spaces between cells. It is now known as tissue fluid. Although red blood cells are very small, they are too big to pass through the walls. They go along the capillaries in single file.

Substances such as oxygen and nutrients enter the bloodstream. Carbon dioxide and other waste products leave the cells to be taken up into the bloodstream: carbon dioxide is **expelled** from the lungs and other waste products are excreted appropriately.

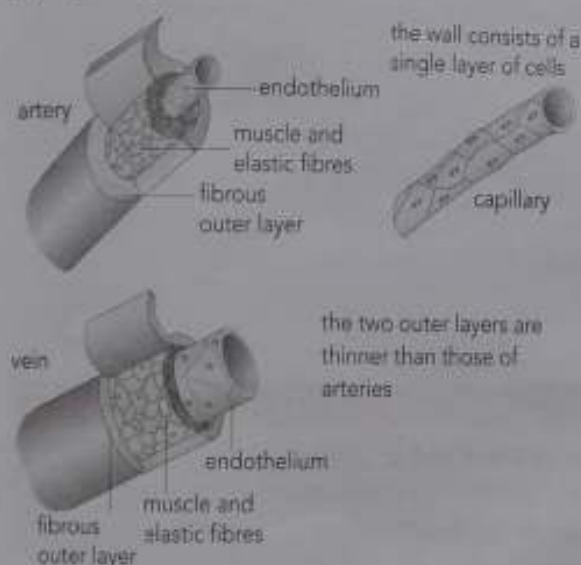


Figure 4.17 Diagram of transverse sections through blood vessels

Veins

The capillaries join up again to form bigger vessels called **venules**. The venules, in turn, join to form veins. Venules and veins carry blood *towards the heart*. The pressure inside veins is not as high as in the arteries so that they will not easily stretch open too widely. This is why the walls of veins are thinner, even though they also have three layers of different tissues. Their thinner walls make it easier for them to change shape when muscles exert pressure on them. This helps blood move along. But, because there are semilunar valves inside veins, blood can only move in one direction. The loose flaps of the valves are on the side nearest to the heart.

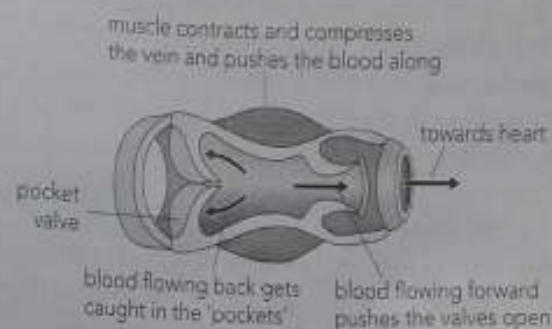


Figure 4.18 Valves in veins make blood flow possible in one direction only

Blood

Blood is a fluid tissue and its functions include:

- transporting oxygen and carbon dioxide between the cells and the lungs
- transporting waste substances (such as urea) in the bloodstream from the liver to the kidneys
- transporting hormones to locations in the body at which they are required
- controlling infection because white blood cells attack and destroy harmful bacteria and proteins
- clotting to protect wounds
- regulating the body's water balance and pH levels
- regulating body temperature

Blood consists of plasma, red and white blood cells and platelets.

Blood plasma

About 55% of the volume of blood is made up of a clear, yellowish liquid, called plasma. It is mostly water, but some of the substances blood carries dissolve in the plasma. Examples of these substances are oxygen and carbon dioxide, glucose, some nutrients, mineral salts, urea and hormones.

Various plasma proteins, such as **antibodies**, fulfil important functions in the body.

Red blood cells

The red blood cells are some of the smallest cells in the body. There are about 5 to 6 million of them in a cubic millimetre of blood. They are shaped like round discs with thicker edges; we say they are **biconcave**. Their shape makes it easy for them to bend and stretch without tearing, as they move through the capillaries. They contain the red pigment **haemoglobin**, which is necessary to carry oxygen to the cells and carbon dioxide from the cells. There are about 250 million haemoglobin molecules in each red blood cell. Red blood cells are formed in red bone marrow and broken down in the liver after about 120 days. They have no nucleus.

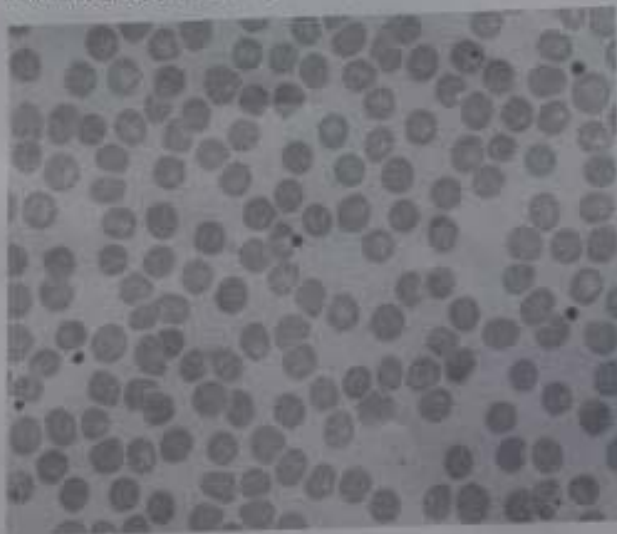


Figure 4.19 Red blood cells viewed through a microscope

Exercise 4.3

1. Explain what the difference is between the structure of a vein and an artery.
2. What are the functions of veins and arteries?
3. What is the name given to the smallest blood vessels in the body?
4. List four of the functions of blood.
5. List the components of blood.

White blood cells

There are about 5 000 to 7 000 white blood cells per cubic millimetre of blood. They are colourless and almost transparent. They have no fixed shape. White blood cells can pass through the walls of capillaries to move between tissue cells. There are five types of white blood cells and their main function is to protect the body. For example, some white blood cells ingest and destroy bacteria, while others manufacture antibodies and others form a barrier to an infected area in order to prevent germs from spreading. White blood cells are also formed in the red bone marrow.

Platelets

Blood platelets are very small. They form when larger cells in the red bone marrow break down. Their numbers vary, but there are about 250 000 per cubic millimetre of blood. They play a role in blood clotting. Blood platelets live for about four days only.

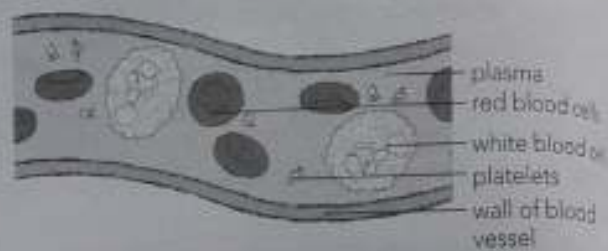


Figure 4.20 Blood components

Summary

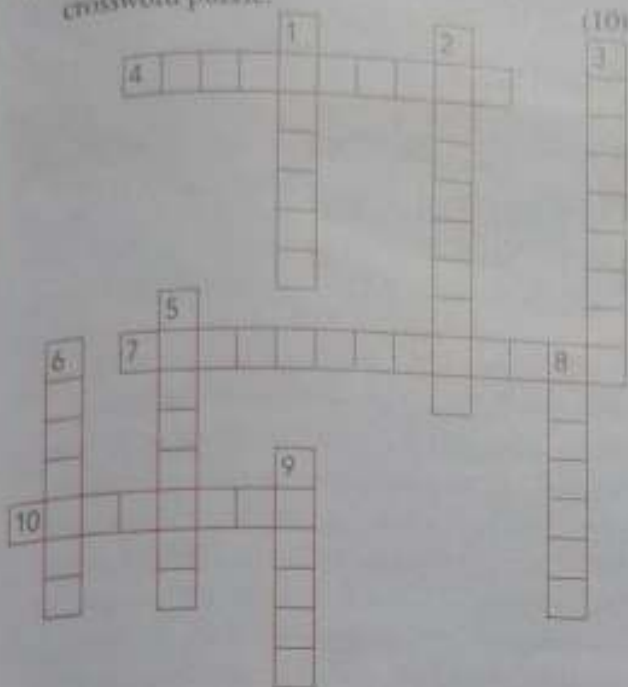
- The three main processes involved in transport across cell membranes are diffusion, osmosis and active transport.
- Diffusion occurs when particles show an overall movement from a place where they are present at higher concentration to one in which they are present at lower concentration, and so spread out evenly. We say that they move down their concentration gradient.
- The movement of water across selectively permeable membranes is an example of a special type of diffusion called osmosis. Water will move from a solution where it is at higher concentration (a weaker solution) to a solution where it is at lower concentration (a stronger solution) in order to reach equilibrium.
- When substances move against their concentration gradient across a membrane, additional energy is necessary to drive special transport proteins. The transport proteins use energy to bind and pump the particles in a direction opposite to one in which they would normally flow. This is called active transport.
- Flowering plants have special vascular tissue to transport substances throughout the plant.
- Xylem transports water and dissolved nutrients from the roots to the leaves.
- Phloem transports the food produced in the leaves to where they are needed in the plant or to where they are stored.
- The human blood transport system is made up of the heart that serves as a pump, the blood vessels that are the pathways and the blood that transports substances to where they are needed.
- The heart is divided into four chambers (the atria and the ventricles), each with a special function in the cardiac cycle.
- The right side of the heart receives deoxygenated blood from all parts of the body and pumps this to the lungs to become oxygenated.
- The left side of the heart receives oxygenated blood from the lungs and pumps this to almost all parts of the body.
- The three main types of blood vessels are arteries, veins and capillaries. Arteries carry blood away from the heart and veins carry blood to the heart. Capillaries make the exchange of gases and substances at cell level possible.
- Blood consists of plasma, red and white blood cells, and platelets. These components all fulfil specialised functions.

Glossary

- antibody** – any protein secreted by the plasma cells in building immunity against a foreign substance
- arteriole** – the smaller branches of arteries
- atrioventricular** – used to describe aspects of the heart or heart function dealing with both the atria and the ventricles
- atrium** – any of the two upper chambers of the heart
- biconcave** – indented on both sides
- cardiac cycle** – one full cycle of contraction and relaxation for all four heart chambers
- chamber** – the compartments into which the heart is divided
- closed blood system** – a system in which blood remains in the blood vessels
- concentration gradient** – difference in concentration of a dissolved substance in a solution; molecules will move from an area with a higher number of particles to an area with a lower number of particles; the areas are typically separated by a membrane.
- cuticle** – the waxy layer covering some epidermal cells in plants in order to prevent water loss
- cytoplasm** – jelly-like substance making up the inside of a cell; surrounds organelles
- deoxygenated** – does not contain oxygen any more
- dicotyledonous** – describes plants with two initial cotyledons
- disintegrate** – to break up
- dynamic equilibrium** – a state of balance between continuing processes
- epithelium** – tightly packed cells, like the cells lining the blood vessels
- expel** – to force out
- fluid** – liquids and gases that do not have a definite shape and which can flow
- haemoglobin** – the red iron-containing pigment in red blood cells facilitating the transport of oxygen
- highly permeable** – allowing almost any substance to pass
- lateral root** – a root that grows sideways from an existing root
- net movement** – overall movement
- nuclei** – the plural form of nucleus
- oxygenated** – contains oxygen
- passive process** – a process that does not require energy to be spent
- perforated** – a hole, or a series of holes
- pericardium** – the sac enclosing the heart
- pulmonary** – of the lungs or to do with the lungs
- root hairs** – small, tube-like extension of a root cell which absorbs water and minerals from the soil
- selectively permeable** – allowing only certain substances to pass
- septum** – the wall inside the heart, which divides it into two halves
- spontaneous** – happening by itself, without any external cause
- tap root** – the main root that tapers down as it grows vertically downwards; develops lateral roots
- translocation** – the transport in phloem of food produced in the leaves
- transpiration** – the release of water vapour from plant leaves through the stoma
- transport protein** – a protein in a membrane that helps the passage of a certain substance across a membrane
- valve** – a structure allowing the flow of, for example blood, only in one direction
- vena cava** – any of the two veins, the *venae cavae*, transporting oxygen-poor blood to the heart and opening into the right atrium
- vascular** – made up of or containing vessels for transporting sap (in plants) or blood (in humans)
- ventricle** – any of the two bottom chambers of the heart
- venule** – the smaller branches of veins are called venules
- xylem sap** – the dilute solution of water and nutrients in xylem

Revision questions

1. Use the following clues to complete the crossword puzzle.



Across

4. Rich in oxygen
7. The transport associated with the phloem
10. Plays a role in blood clotting

Down

1. Smaller veins
2. The plural form of vena cava (two words)
3. Describes the cells that help phloem sieve cells to function
5. The name of a xylem segment
6. In the stem of a dicotyledonous plant, vascular tissues are arranged in _____
2. a) Explain the concept 'active transport'. (4)
b) Give one example of active uptake in flowering plants and one example in the human body. (2)
3. What structural element in xylem makes it possible to transport water sideways? (2)
4. Study Figure 4.20, which shows a blood vessel of a person who has both normal red blood cells and sickle red blood cells due to a disorder. Use this to explain how the structure of red blood cells makes it easier for them to carry oxygen to body cells through smaller



Figure 4.20 Normal and sickle red blood cells

5. Study Figure 4.21, which shows a cross-section through a dicotyledonous plant organ.
a) Identify the plant organ. (1)
b) Give the correct labels for the structures A to E. (5)

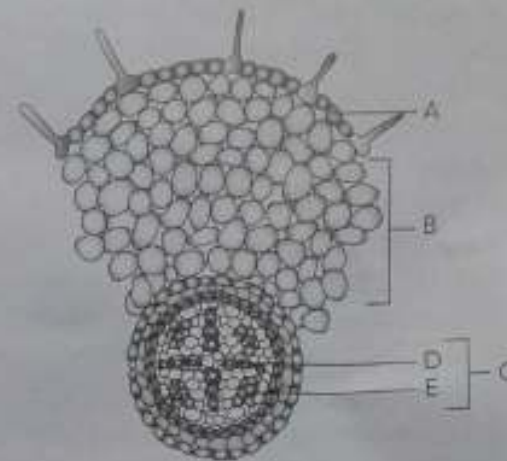


Figure 4.21 Dicotyledonous plant organ

- c) Give one function of structure A. (1)
6. In humans, blood returning to the heart from the lungs enters through:
A the right ventricle
B the right atrium
C the left atrium
D both the left and right sides of the heart. (1)
7. Which of these associations is false?
A White blood cells fight infections.
B Red blood cells clot blood.
C Plasma contains water and nutrients.
D Plasma contains wastes. (1)
8. All arteries in the body contain oxygen-rich blood, except the:
A aorta
B pulmonary arteries
C renal arteries
D coronary arteries. (1)

TOTAL: 32

Unit
5

Reproductive systems

Objectives

- Distinguish between monocotyledonous and dicotyledonous plant seeds.
- State the functions of the cotyledon and the endosperm.
- State the functions of the female and male reproductive systems in humans.

Introduction

Reproduction is one of the life processes of all organisms. Plants and animals reproduce either sexually or asexually. Sexual reproduction has many advantages to a species. For example, variation due to cell division after fertilisation may produce offspring that are better adapted to the environment than their parents.

Flowering plants are the most common plants in the world. Flowers are the structures, or organs, of reproduction of flowering plants. They play a role in producing gametes (sex cells), in providing a safe environment for fertilisation and they develop into fruits that will protect and help spread the embryo-containing seeds.

Humans reproduce sexually. The reproductive organs of males and females produce sex cells, protect the sex cells before and during fertilisation, and protect the developing embryo and foetus until the baby is born. Humans also give parental care to their offspring until they can take care of themselves.

Reproduction of flowering plants

In Form 1, you saw that pollen is produced in the male organs of the flower (inside the stamen, by the anther). Each pollen grain contains two male gametes. The female gamete (egg cell) is found in the flower's female organs (pistil), in the **ovule**, which is inside the **ovary**.

Some flowering plants, such as petunias, aloes and bean flowers, carry both male and female organs in the same flower. We call these flowers **bisexual**. Some plants, such as oaks, pumpkins, paw-paw, date palms and mealies, bear male and female organs in separate flowers. These flowers are **unisexual**. Unisex flowers of some plants (for example, oak, pumpkin, mealies) are found on the same plant (oak, pumpkin, mealies), but some plant species (for example, paw-paw and date palms) carry them on different plants.

Pollination and fertilisation

Pollination takes place when a pollen grain lands on the **stigma** of a flower of the same species. The pollen grain bursts open and a pollen tube grows down towards the ovule, enters it and releases its two male gametes. One of the male gametes joins with the female gamete (egg cell) inside the embryo sac and fertilises it. The other gamete fuses with two other cells in the centre of the embryo sac. This will develop into the **endosperm**, a food-storing tissue in the seed.

The process whereby two male gametes separately fuse with different cells in the embryo sac is called **double fertilisation**. Flowering plants are thus characterised by double fertilisation, with the development of an embryo and endosperm as the result.

Some flowers have more than one ovule, so that more than one pollen grain can start growing. This is shown in Figure 5.1. Only the ovules with fertilised egg cells will develop further into seeds.

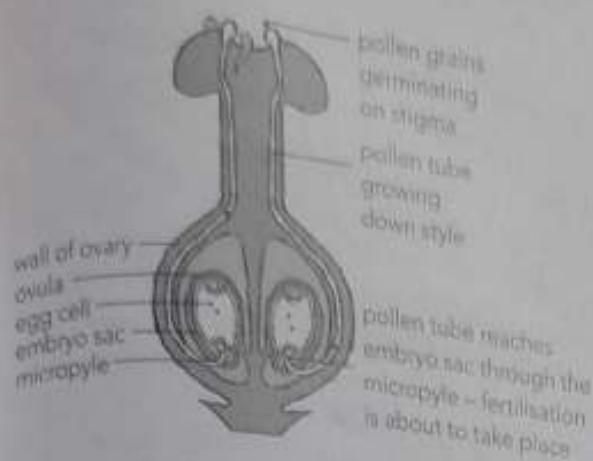


Figure 5.1 Fertilisation of more than one female gamete

Seed and fruit formation

After double fertilisation, the seed (or seeds) and fruit develop. The fertilised egg cell develops into an **embryo**. The embryo consists of a tiny root (**radicle**) and a tiny shoot (**plumule**). The ovule develops into a seed that contains the embryo and the endosperm, or food store.

The seed develops one or two seed leaves, called **cotyledons**. Seeds are called **monocotyledonous** if they have one seed leaf and **dicotyledonous** if they have two seed leaves. In general, the endosperm is gradually absorbed into the cotyledons in dicotyledonous seeds, while it remains unabsorbed in monocotyledonous seeds. The outer wall of the ovule forms a hard, protective seed coat or **testa**.

The ovary wall develops into a fruit wall (**pericarp**) enclosing the seed or seeds. There are many types of fruit, depending on how they develop and on their characteristics. As the fruit develops, some parts of the flower dry up and fall off. The receptacle (base) of the flower now carries the fruit.

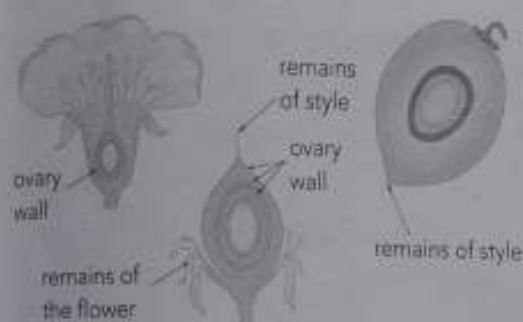


Figure 5.2 The formation of a plum fruit

The fruit and the seed of the bean plant and the mealie plant

Both the bean plant and the mealie plant have dry fruits that develop from the ovary of a single flower. Bean seeds are found in a pod, while mealie seeds are found in grains, where the fruit wall has fused with the seed coat. Bean seeds are dicotyledonous and mealie seeds are monocotyledonous. These seeds look very different from each other and they protect the embryo inside them in different ways.

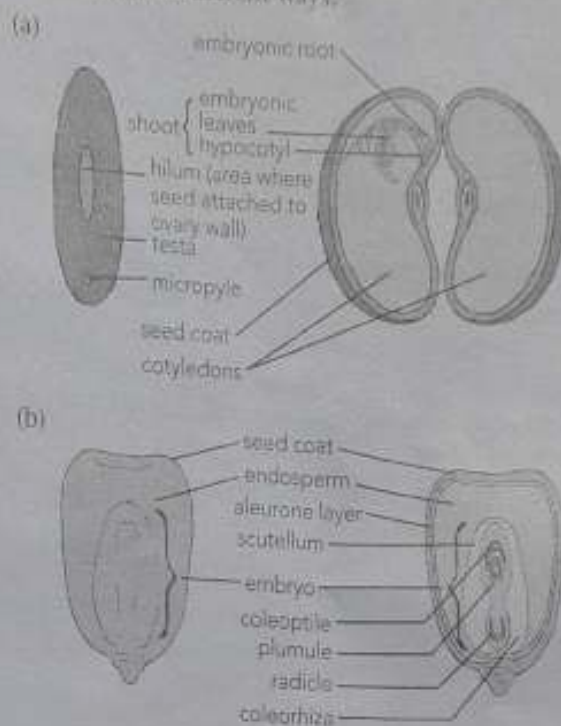


Figure 5.3 Exterior and interior diagrams of a a) bean seed and b) mealie grain

The bean seed

The bean seed is kidney-shaped. The hilum, the scar where the seed was attached to the pod, is clearly visible on the inside curve. If you remove the seed coat and open the seed leaves carefully, you will find the embryo. The embryo of the bean seed is situated between the two cotyledons to which it is attached on the side of the hilum. Above the attachment, the embryo consists of the **epicotyl** and the plumule. The plumule will develop into the plant's first leaves. Below the attachment, the embryo consists of the **hypocotyl**, which terminates in the radicle.

When the seed germinates, the root of the seedling appears first. The shoot grows up above the soil so that a little hook appears. This protects the young, developing leaves that are then pulled out of the soil. The cotyledons contain enough food to give the energy needed for the stem to straighten and the leaves to grow. When the leaves are mature and start to produce food for the plant by photosynthesis, the cotyledons shrivel up.

The mealie grain

The mealie grain is actually a combination of the mealie fruit and seed, as the fruit wall and seed coat are fused. You will see the woody fruit stalk at the bottom and the remains of the flower's style at the top. There is only one cotyledon in the form of a shield. This is characteristic of members of the grass family.

A longitudinal section of the grain shows the endosperm still clearly present and separate from the shield-like cotyledon. Both the root (radicle) and shoot (plumule) of the embryo are covered with protective sheaths, known as the **coleoptile** and the **coleorhiza**, respectively. When the seed germinates, the root grows directly downwards into the soil and the shoot grows directly upwards. The sheaths are shed as the roots and shoots grow and develop.

Activity 5.1

Aim: To compare the external and internal structures of bean and mealie seeds.

Materials: bean and mealie seeds that have been soaked for 24 hours, hand lens

Procedure

1. Study the diagrams of the external and internal structures of the bean seed and the mealie grain in Figure 5.3.
2. Examine the external structure of your soaked seeds first. Use Figure 5.3 to help you identify the external characteristics of both types of seed. Can you see anything of the internal structures through the soaked outside covers?
3. Draw labelled diagrams of the external structure of your seeds. (Do not copy the drawings from your textbook).
4. Remove the outside covers of the seeds carefully so that you do not damage the inside structures. Open the two cotyledons of the bean seed carefully. Identify the inside structures inside both seeds.
5. Draw labelled diagrams showing the internal structure of your seeds. Do not forget to give all your diagrams suitable headings.

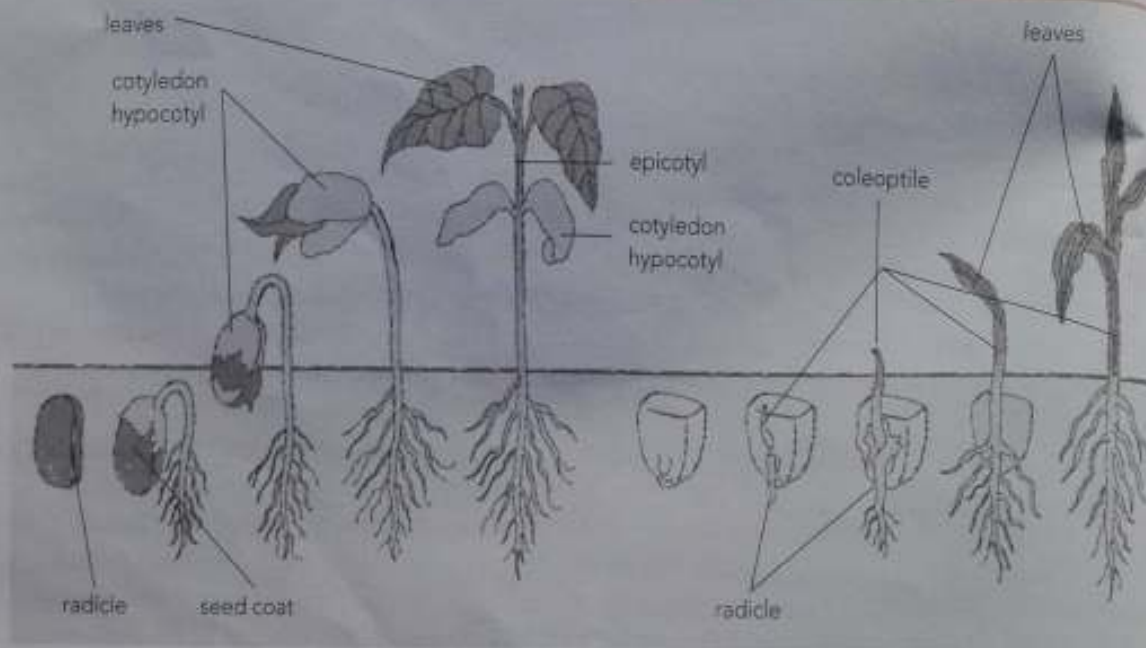


Figure 5.4 A seedling of a) a bean seed and b) a mealie grain

The human reproductive system

In humans, there are two separate sexes: male and female. Humans are born with primary sex characteristics. Secondary sex characteristics start to develop in both sexes during puberty. In boys, this development occurs due to an increased amount of testosterone in the testes. In girls, it is the female sex hormones, oestrogen (often referred to in the singular) and progesterone that stimulate the development of secondary sex characteristics.

The male reproductive system

The main components of the male reproductive system are:

- the paired testes (singular: testis) inside the scrotum
- the epididymis
- the sperm ducts
- the glands producing fluids
- the penis.

The penis and the scrotum are the external male genitals.

The testes

The testes are the organs that produce sperm and the male sex hormones. Sperm production occurs at a lower temperature than body temperature. For this reason, the testes, which start their development inside the abdominal cavity, descend into the scrotum during the last two months of foetal development in males. The function of the scrotum is to help regulate the temperature of the testis by holding them closer to or further away from the body.

The epididymis and sperm ducts

The sperm cells that have been produced in the

testes mature in the epididymis. From there, they enter the sperm ducts, where they may be stored for some time. The sperm ducts are cut or blocked in a vasectomy, a surgical form of birth control. Sperm are then unable to reach the urethra and are destroyed.



Figure 5.5 The male reproductive system

The male glands

As the sperm cells travel through the male reproductive tract, fluids from several male glands (seminal vesicles, prostate and bulbourethral glands) are added to them to make up the thick, whitish fluid called semen. Fluids in the semen provide nourishment for the sperm cells and enable them to move faster.

The penis

The penis is the male organ that is used during sexual intercourse. The function of the penis is to transfer the sperm cells from the male reproductive tract into the vagina of the female. During erection, the spongy erectile tissue in the penis fills with blood. This compresses the veins that normally take blood away from the penis, and the penis becomes erect. Semen is expelled from the opening of the urethra during ejaculation.

The female reproductive system



Figure 5.6 The female reproductive system

The main components of the female reproductive system are:

- the paired ovaries
- the paired Fallopian tubes
- the uterus
- the vagina
- the **vulva** (external female genitals).

The ovaries

The paired ovaries produce the female sex hormones, that is, oestrogen and progesterone, as well the female gametes (egg cells). They are situated in the abdominal cavity on both sides of the uterus. They are held in place by ligaments.

The Fallopian tubes

The Fallopian tubes stretch from the side of the uterus to the ovaries. They are not attached to the ovaries, but have finger-like projections that sweep over the ovaries. When an egg cell is released during **ovulation**, it is swept into the tube. Fertilisation usually takes place very high up in the Fallopian tube and the developing embryo is slowly moved down to the uterus by the movement of cilia and muscle contractions in the tube.

Some women choose to permanently prevent pregnancy by having the Fallopian tubes surgically closed off so that sperm cannot reach the egg cells. This is called **tubal ligation**.

The uterus

The uterus, or womb, is a thick-walled, muscular, pear-shaped organ. It is situated behind the bladder and is held in position by strong ligaments. The main functions of the uterus are to:

- shelter and protect the developing foetus
- provide nutrition for the foetus through the placenta.

The **cervix** is the narrow end of the uterus and it has a small opening into the vagina.

The vagina

The vagina is a muscular tube that connects the uterus with the outside world. The lining of the vagina is wrinkled so that it can stretch. The functions of the vagina are to:

- serve as birth canal
- receive the penis during sexual intercourse.

At birth and usually up until sexual intercourse tears it, the opening of the vagina at the vulva is partly covered by the **hymen**. Vigorous physical activity can also tear the hymen.

Exercise 5.1

1. List the main components of the human male reproductive system.
2. a) Where are sperm produced?
b) Explain how the male reproductive system is adapted to control the temperature of the organ producing sperm.
c) Why is it necessary to control the temperature of the sperm-producing organ separately from that of the rest of the body?
3. Name the external reproductive organs in the human male.
4. Name the main male reproductive hormone.
5. List the main components of the human female reproductive system.
6. Where are the female gametes produced?
7. Where are the female hormones about which you have learnt produced?

8. Which human female reproductive organ both receives sperm and acts as birth canal?
9. What are the external female reproductive organs called?
10. Name two surgical methods of birth control.

Fertilisation, pregnancy and birth

During human intercourse, about 2 to 5 ml of semen, with 70 to 130 million sperm cells in each millilitre, is transferred by the penis into the vagina. The sperm cells swim up through the uterus to the Fallopian tubes. Fertilisation,



Figure 5.7 Fertilisation occurs when a sperm cell enters an egg cell and fuses with it.

which is when a sperm fuses with an egg cell, usually happens in the top part of the Fallopian tube. Once a sperm enters the egg cell, a membrane forms to prevent any other sperm from entering.

As the fertilised egg is being moved down the Fallopian tube into the uterus by the movement of cilia and muscle contractions, it increases in size through successive cell divisions. The embryo then attaches itself to the inner lining of the uterus. This is called **implantation** and pregnancy begins here. At this stage, the embryo looks like a small hollow sphere. After two months, all the organs have started forming. The embryo is now known as a **foetus**.

Nutrition and protection during pregnancy

The **placenta** forms partly from tissue of the mother and partly from tissue of the embryo.

It is connected to the embryo by the **umbilical cord**. The embryo's blood flows through arteries in the umbilical cord to the maternal blood pool in the placenta, where exchange of substances occurs by diffusion, active transport and selective absorption. This happens across the membranes of the embryo's capillaries as the embryo's blood never leaves the blood vessels. Afterwards, veins passing through the umbilical cord transport the blood back to the embryo. There is thus no direct contact between the mother's and embryo's blood. The functions of the placenta include:

- providing the embryo (later foetus) with nutrients
- allowing for the excretion of waste products
- allowing for the exchange of oxygen and carbon dioxide through diffusion
- preventing harmful substances or bacteria from reaching the embryo
- secreting the hormone progesterone to sustain the pregnancy
- allowing antibodies to pass from the mother to the foetus with levels peaking just before birth.

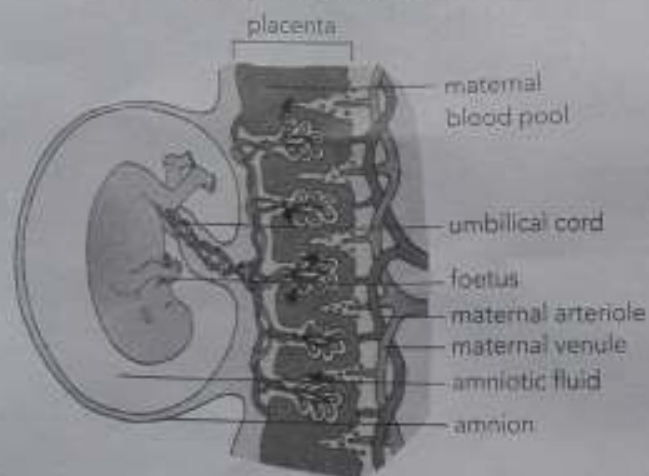


Figure 5.8 Transport in the placenta

Although the placenta protects the embryo (later foetus) from some harmful substances and bacteria, it is not a complete protection. Nicotine, drugs and alcohol, as well as sexually transmitted infections, may influence the baby's health before birth.

The embryo (later foetus) is surrounded by membranes that develop from the fertilised egg cell, but which are outside the embryo itself. The inner membrane is called the **amnion** and it is filled with the **amniotic fluid**. The functions of these membranes together with the amniotic fluid include:

- cushioning the embryo and protecting it from shocks
- allowing free movement
- allowing symmetrical development
- keeping the embryo at a constant temperature.

Childbirth (labour)

Pregnancy in humans lasts about nine months (280 days). During the last weeks of pregnancy, the foetus normally turns to lie upside down, with the head just above the vagina. Birth happens in three stages. During the first stage, the cervix widens so that the baby's head and body can pass through. The amnion around the baby breaks. The baby is born during the second stage when the cervix has widened enough. The umbilical cord is cut. The placenta is expelled during the last stage.

Parental care

Humans, like all mammals, have **mammary glands**. During pregnancy, hormones prepare the mother's body to produce milk after birth. Breastfeeding, as part of parental care after birth, is thought to be ideal for the baby's health. It provides a balanced diet and allows antibodies to move through the mother's milk to the baby. Unfortunately, the HI virus can also be passed on to the baby in this way. Mothers who live with the HI virus are thus advised to bottle-feed their babies. Humans generally take care of their young until they can take care of themselves. Education, mostly forms part of this parental care.

Exercise 5.2

1. What is the fertilised egg cell called during the first eight weeks?
2. State the functions of the following:
 - a) the placenta
 - b) the umbilical cord
 - c) the amnion.
3. What benefit, apart from nourishment, does breastmilk have for babies?

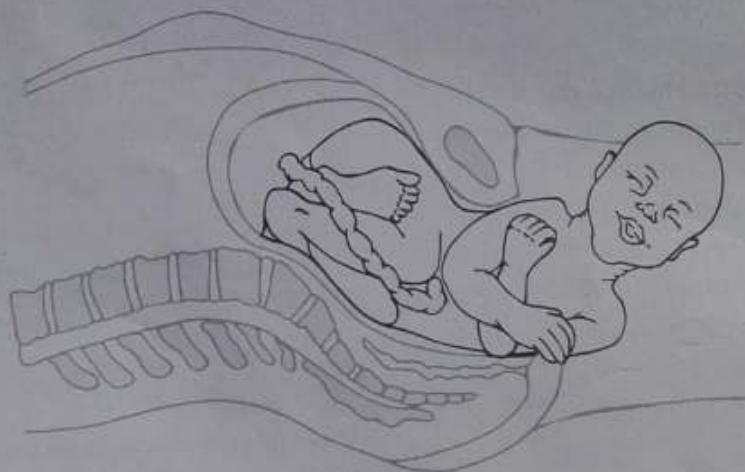


Figure 5.9 The second stage of the birthing process.

Summary

- Pollination takes place when a pollen grain lands on the stigma of a flower of the same species.
- The pollen grain grows down to the ovule to release its two male gametes.
- Flowering plants are characterised by double fertilisation where one male gamete fertilises the egg cell and the other fuses with two cells in the interior of the embryo sac.
- As a result, an embryo, together with endosperm, develops.
- Only ovules with fertilised egg cells develop into seeds.
- The ovary wall of the flower develops into the fruit wall enclosing the seed or seeds.
- The seed develops one or two seed leaves, called cotyledons.
- Seeds are called monocotyledonous if they have one seed leaf and dicotyledonous when they have two seed leaves.
- In dicotyledonous seeds, the endosperm is mostly absorbed into the cotyledons, while it remains unabsorbed in monocotyledonous seeds.
- The embryo of a seed already contains the elements of a plant. The seed contains enough nourishment for the young seedling from the moment it germinates until it starts photosynthesising.
- All humans are born with primary sex characteristics, while secondary sex characteristics start to develop in both sexes during puberty when they reach sexual maturity.
- The main components of the male reproductive system are the paired testes, the epididymis, the sperm ducts, glands producing fluids contributing to semen and the penis. The penis and the scrotum are the external male genitals.
- The main components of the female reproductive system are the paired ovaries, the paired Fallopian tubes, the uterus, the vagina and the vulva (external female genitals).
- The function of the penis is to transfer the sperm cells from the male reproductive tract into the vagina of the female.
- The transferred sperm cells swim up the vagina, into the uterus to the Fallopian tubes to fertilise the egg cell.
- The fertilised egg cell is moved down the Fallopian tubes into the uterus, where it implants itself in the lining.
- During the first eight weeks of pregnancy, the unborn human offspring is referred to as the embryo. It is called the foetus afterwards.
- The placenta that forms partly from the inner lining of the uterus and partly from the tissues of the embryo provides the embryo with nourishment.
- The embryo is attached to the placenta by the umbilical cord.
- The amnion is a membrane filled with fluid in which the embryo is cushioned and protected.
- Pregnancy in humans lasts about nine months.
- Birth happens in three stages: first, the cervix widens to let the baby's head and body pass as the amnion breaks, next, the baby is born and, last, the placenta is expelled.
- The mammary glands of female humans produce milk to feed the baby after birth. This is part of parental care.

Glossary

- amnion** – a membrane surrounding a fluid-filled cavity in which the embryo floats
- amniotic fluid** – the fluid in the amniotic sac, cushioning the embryo
- bisexual** – a flower that has both male and female structures
- cervix** – the opening at the base of the uterus into the vagina
- coleoptile** the sheath protecting the shoot of the embryo or seedling
- coleorhiza** – the sheath protecting the root of the embryo or seedling
- cotyledon** – a seed leaf
- double fertilisation** – the process where two male gametes separately fuse with different cells in the embryo sac in flowering plants
- ejaculation** – when semen is expelled from the opening of the urethra
- embryo** – the rudimentary plant in a seedling or the human offspring in the first eight weeks after conception
- endosperm** – the nutrient-rich tissue resulting from the fusion of a second male gamete with two cells in the centre of the embryo sac
- epicotyl** – the part of the embryo above the attachment to the seed leaves and below the plumule
- erection** – when the spongy erectile tissue in the penis fills with blood and the penis becomes erect
- foetus** – the unborn human from eight weeks after conception
- genitals** – sex organs
- grain** – the name for seeds when the seed wall and fruit wall are fused
- hymen** – the tissue that partially covers the opening of the vagina from birth up until it is torn by sexual intercourse or vigorous exercise
- hypocotyl** – the part of the embryo below the attachment to the seed leaves and above the radicle
- implantation** – when the embryo attaches itself to the inner lining of the uterus
- mammary gland** – a gland secreting milk in female mammals
- oestrogen** – the oestrogens are female sex hormones that develop and maintain female characteristics of the body (often referred to as the singular)
- ovary** – the large base of the pistil housing one or more ovule
- ovulation** – in humans, the monthly release of an egg cell from an ovary in the female
- ovule** – a structure inside the ovary of a flowering plant containing the egg cell
- pericarp** – a fruit wall
- placenta** – the organ forming partly from the uterus wall and partly from the tissue of the embryo serving as nourishment for the embryo and later the foetus
- plumule** – the small shoot of the embryo inside a seed
- primary sex characteristics** – the sexual organs, or genitals, a person is born with and that distinguish males from females
- progesterone** – the hormone that stimulates the development of the uterus in preparation of pregnancy
- puberty** – the period during which secondary sex characteristics become visible and sexual maturity is reached
- radicle** – the small root of the embryo inside a seed
- scrotum** – the sac outside the body containing the testes
- secondary sex characteristics** – the male and female characteristics of the body developing during puberty
- semen** – sperm cells together with fluids from several male glands
- sexual intercourse** – the insertion of the male's erect penis into the female's vagina
- sperm** – the male sex cell or gamete
- stigma** – the part of the pistil that receives pollen grains
- testa** – a seed coat
- testosterone** – male sex hormones that develop and maintain male characteristics of the body
- tubal ligation** – surgical procedure to permanently prevent pregnancy by having the female's Fallopian tubes surgically closed off

umbilical cord – the structure connecting the embryo and the placenta. It contains the umbilical arteries and veins.

unisexual – a flower that has only male or only female structures.

urethra – in males, the tube conducting both urine and semen to the outside of the body.

vagina – the tube that leads to the uterus and inside which the male's penis is inserted during sexual intercourse.

vasectomy – a surgical form of birth control where the sperm ducts in the male are cut or blocked.

vulva – the outside genitals of the female.

Revision questions

1. A seed contains:
 - A seed coat
 - stored food
 - an embryo.
 - All of these are correct.
2. The seed of a monocotyledonous plant:
 - is the result of single fertilisation.
 - does not have endosperm.
 - has one seed leaf.
 - is called a gamete.
3. Fruits:
 - nourish embryo development
 - help with seed dispersal
 - attract pollinators
 - develop from endosperm.
4. Bean seeds:
 - are called pods
 - are dicotyledonous
 - are actually fruits
 - develop from the ovary.
5. Say which pair is mismatched.
 - Dicotyledonous – coleorhiza
 - Monocotyledonous – one seed leaf
 - Coleorhiza – root
 - Coleoptile – shoot
6. Say which pair is mismatched.
 - Urethra – conduction of sperm
 - Testes – sperm production
 - Ovary – egg cell production
 - Plasma – semen
7. Draw the external and internal characteristics of a bean seed. Label each structure clearly and give your drawings labels. (5)

8. Study the diagram below and answer the questions that follow.

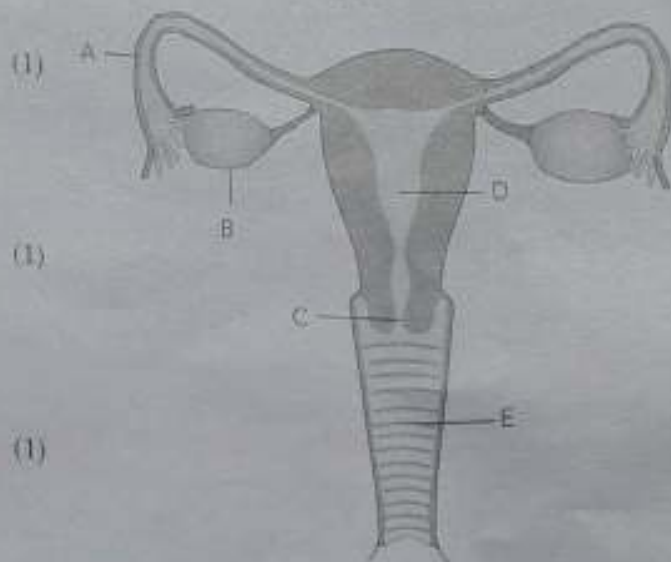


Figure 5.9 Diagram for question 8

- a) Give the diagram a suitable heading. (1)
 - b) Label structures A to E. (5)
 - c) State the two main functions of structure E. (2)
 - d) In which structure does fertilisation usually take place? (1)
9. List five functions of the placenta. (5)
 10. Give one function for each of the following:
 - a) testis
 - b) scrotum
 - c) semen
 - d) sperm
 - e) penis

TOTAL: 30

Unit
6

Health and disease

Objectives

- List causes of disease.
- Describe the cause of bilharzia in humans.
- Describe the life cycle of the bilharzia parasite.

Introduction

It is difficult to define the terms 'disease' and 'health'. The World Health Organisation (WHO) defines health as 'a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.' This means that, for humans to be healthy, one needs to be well physically, mentally and in one's social relations.

One way of looking at disease is to say that it is a disorder or malfunctioning of an organism, with particular signs and symptoms. When you are

ill, it may mean that you need to be treated by a health worker. Diseases do not always have clearly observable symptoms, so you may not always know that you are ill. Other diseases, however, show up with a clear set of signs and symptoms.

Classification of diseases

There are different ways of classifying diseases. It is possible to distinguish six categories of human diseases based on this view. These categories are summarised in Table 6.1.

Table 6.1 Six categories of human disease

Cause of disease	Examples of diseases
1. Diseases caused by living organisms Viruses, bacteria, fungi, protozoans, roundworms and flatworms are organisms that can cause disease. They are parasitic organisms interfering with the normal functioning of their hosts.	Viruses: HIV/AIDS, influenza, chickenpox, measles bacteria: cholera, tuberculosis, typhoid fever Fungi: thrush, athlete's foot, ringworm Protozoans: malaria, sleeping sickness Roundworms: Guinea worm disease Flatworms: bilharzia, tapeworms
2. Diseases humans bring upon themselves Humans bring these diseases upon themselves either as individuals (for example, choosing to smoke can cause lung cancer later in life) or as a society (for example, using CFCs caused many people to get skin cancer as a result of the thinner ozone layer). These diseases are sometimes referred to as social diseases or social disorders and are often associated with modern, industrialised societies.	Some coronary heart diseases, alcoholism, substance abuse, lung cancer, domestic and industrial accidents, industry- and pollution-related diseases, such as asbestosis, asthma, some cancers

Cause of disease

3. Deficiency disorders
These conditions are caused by the lack or absence of specific nutrients or food groups in your diet. If deficiencies in your diet are not addressed, they may develop into diet-related diseases.

4. Genetic or congenital disorders and diseases
These are related to diseases that are present at birth or exist from birth. Genetic screening can often predict the possibility of offspring inheriting certain conditions.

5. Ageing and degenerative diseases
Any disease that appears with age or that is due to cell damage as we grow older.

6. Mental illnesses
Any disease or condition that influences the way you think, feel, behave or relate to others and to your surroundings falls into this category. These conditions are often due to chemical imbalances in the brain and can be treated. The stigma people attach to these diseases makes it difficult for people to seek help.

Examples of diseases

Scurvy is caused by a lack of vitamin C; night blindness is linked with a lack of vitamin A; marasmus and kwashiorkor are both associated with a lack of protein; rickets is caused by a lack of vitamin D; goitre is linked with a lack of iodine.

Cystic fibrosis, Huntington's disease, Down's syndrome, sickle cell anaemia, albinism, congenital syphilis and some other venereal diseases can be transmitted from mother to baby.

Arthritis, dementia, cataracts, macular degeneration, osteoporosis, age-related diabetes, hypertension, loss of hearing, Alzheimer's disease.

Schizophrenia, bipolar mood disorder, depression, anxiety, posttraumatic stress disorder, Tourette's syndrome, autism, eating disorders.

When you look at the diseases listed above, you will see that they can also be divided into two broad categories: **infectious diseases** and **non-infectious diseases**.

Infectious diseases and non-infectious diseases

Non-infectious diseases or **non-communicable diseases** are those diseases that are not caused by a **pathogen** and cannot be passed on from one person to another through direct or indirect contact. Non-infectious diseases are caused by the environment, deficiencies in the diet and lifestyle choices, or are passed on genetically.

Infectious diseases are caused by pathogens, such as bacteria, viruses, parasites or fungi. These diseases spread directly or indirectly from person to person. Infectious diseases are also known as **communicable diseases**. Each infectious, or communicable, disease has a unique cycle of progression with chain elements that keep it going.

The infectious disease cycle

There are six chains in the infectious disease cycle: a pathogen or disease-causing organism, a reservoir, a means of escape, a mode of transmission, a means of entry and a susceptible host. This is represented in Figure 6.1. For an infection to be able to spread, these six chains or links must be intact. In order to stop an infectious disease, the cycle can be interrupted by breaking any of these links.

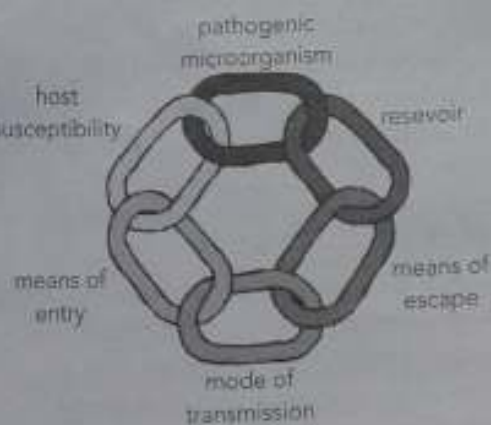


Figure 6.1 The chains in the infectious disease cycle

Pathogen (disease-causing agent)

Each type of pathogen is responsible for a specific type of disease associated with a specific set of symptoms and signs. Examples of pathogens include viruses, bacteria, fungi and parasites. Organisms cause infections when they are virulent (able to multiply and grow), invasive (enter tissue) and pathogenic (able to cause disease).

Reservoir (source)

The human body, animals, the environment (air, water, soil, medical equipment, food) can all act as a reservoir or source for the pathogen. This is the place where pathogens live, thrive and reproduce. For sexually transmissible diseases (STDs), the warm and moist mucous membranes of the mouth and the genital or anal regions are the perfect reservoir. Other examples of reservoirs for some common pathogens are standing water, a dog with rabies and a person with a common cold or syphilis.

Means of escape (portal of exit)

Means of escape, or portal of exit, refers to the route by which the pathogens escape or leave the reservoir. Organisms causing respiratory diseases usually leave the body through the nose and mouth during coughing, sneezing, talking or breathing. Pathogens of the gastrointestinal system usually leave the body in body fluids or faeces. Some ways of breaking this link are to:

- stay at home when you are ill
- practice good cough and sneeze etiquette by covering your mouth and nose
- practice good hand hygiene by washing your hands before you handle food
- handle sharp things that can cause injury carefully.

Means of entry (portal of entry)

Pathogens generally gain entry into the body of their host by a limited number of routes. These sites are referred to as portals of entry. This is the opposite of the means of escape or portal of exit. Pathogens can enter the human body by inhalation (through the respiratory tract), ingestion (the digestive tract) or absorption

(through the mucous membranes of the eyes, breaks in the skin and through the urogenital tract). Some ways to break this link are to:

- have good ventilation
- stay away from people with respiratory problems
- practice good hand hygiene and respiratory etiquette
- practice good food handling and housekeeping.

The means of entry and the means of escape are sometimes collectively referred to as the routes of disease transmission.

Mode of transmission

The means by which an organism is transmitted from the reservoir to an appropriate port of entry to the host is either through direct contact (kissing, shaking hands and sexual intercourse) or indirect contact (when handling medical equipment, clothing, bedding, money and cooking utensils).

Susceptible host

The person who is next exposed to the pathogen becomes the new host. Sometimes a person does not develop an infection because their immune system can fight it off. Such a person can, however, become a 'carrier' without symptoms, and able to be the next 'mode of transmission' to another 'susceptible host'. An infected host may therefore become a reservoir for future transmission of the disease. Some people will be more susceptible to become a host than others; for example, children who are very young and people who are very old, those with inadequate diets, those with chronic illnesses and people with open wounds.

Opportunities to break the chain of infection and to interrupt transmission occur when:

- the disease-causing agent is eradicated, inactivated or cannot exit the reservoir
- there are safe infection control practices with regard to the portals of exit
- there are barriers that can prevent transmission between objects or people
- the portals of entry are protected through safe infection control practices
- hosts are no longer susceptible.

Exercise 6.1

1. Discuss what is meant by the words 'health' and 'disease'.
2. State the six chains in the cycle of an infectious disease.
3. Distinguish between an infectious disease and a non-infectious disease.
4. Give another name for 'infectious diseases'.
5. Give two examples of non-infectious diseases.

Bilharzia (schistosomiasis)

Bilharzia, 'snail fever' or schistosomiasis, is a disease caused by one of several varieties of the parasite *Schistosoma*. Infected freshwater snails act as alternate or secondary hosts to humans, who are the primary hosts. The disease is found mainly in tropical and sub-tropical climates and it is estimated that about 258 million people needed

preventative treatment in 2014 in the 78 countries where cases are reported. The heaviest burden of the disease is in Africa. After malaria, it is the most common parasitic disease in tropical countries. Statistics also shows that it is the deadliest of the neglected tropical diseases (NTD) listed by the World Health Organisation (WHO). An estimated 280 000 people are killed by bilharzia annually.

Transmission and lifecycle of the parasite *Schistosoma*

Schistosomiasis is transmitted when humans come into contact with contaminated freshwater sources, such as lakes, dams, ponds and rivers, where there are parasite-carrying snails. The two species of the parasite found in Africa are *Schistosoma mansoni* and *Schistosoma haematobium*. The contamination of freshwater sources means that swimming, bathing, fishing and even domestic chores, such as washing clothes and herding livestock, put people at risk of acquiring the disease.

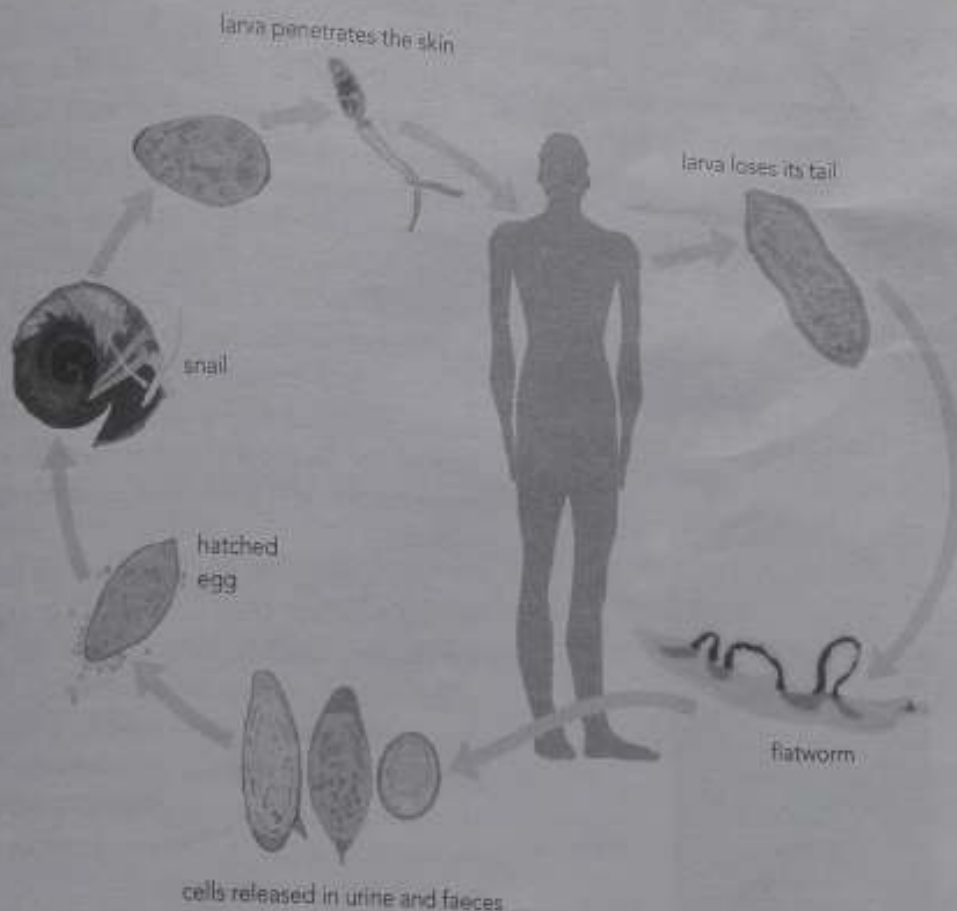


Figure 6.2 The lifecycle of the *Schistosoma* parasite

The larvae that developed and matured through asexual reproduction in the snails enter the water, where they can live for up to 48 hours. The larvae swim towards their future human host and penetrate the skin. Humans are their primary hosts and here they reproduce sexually. Once inside the human body, the larvae develop into adult male and female flatworms that pair up and live together in the blood vessels. Over years, the female flatworms will release thousands of eggs, of which some are passed out in urine and faeces. When infected people urinate or defecate in freshwater sources, snails, the secondary hosts, become infected and a new cycle starts.

Some *Schistosoma* eggs stay in the body and the larvae migrate to specific organs where they can cause major internal damage. There are two types of schistosomiasis: urogenital schistosomiasis and intestinal schistosomiasis. They attack different organs in the body.

Urogenital schistosomiasis generally causes blood in the urine. Scarring and tearing of the bladder and kidneys occur in advanced cases. Bladder cancer is also a complication in advanced stages of the disease. Women can develop genital lesions, nodules in the vulva and vaginal bleeding. Sexual intercourse can be painful. In men, the testes and other male reproductive organs can also be affected. Infertility can be one of the irreversible consequences.

Intestinal schistosomiasis develops slowly. It causes bleeding in the abdomen, while the liver, lungs and spleen become enlarged and the intestines are damaged. Blood in the faeces and diarrhoea can be indicators of the disease.



Figure 6.3 Schistosomiasis disfigures sufferers.

Children can develop anaemia, stunted growth, malnutrition and learning disabilities. Chronic infections in children can have major consequences for them, due to absences from school. Schistosomiasis can also disfigure sufferers severely as their organs enlarge.

Diagnosis

Diagnosis of schistosomiasis is mostly made by taking specimens of urine and stools and analysing these for parasite eggs. The presence of antibodies or antigens in blood or urine samples can also confirm infection. Other methods of testing are also possible. For example, for people who live in areas of lower infection rates, serological and immunological tests may be useful in showing whether there was exposure and if there is a need for thorough examination, treatment and follow-up.

Treatment and efforts to control the disease

The treatment of choice is a drug called praziquantel. There are no vaccines available for the disease and reinfections can occur. Some campaigns to contain the disease are based on targeted distribution of praziquantel in endemic areas. Schistosomiasis 'hot-spots' are mapped by looking at blood levels in the urine of school-going children and looking for eggs in their stools. Where the levels are high, praziquantel is given to the whole community. Annual treatment with praziquantel is recommended in high-risk areas where reinfection may occur. This also helps to manage the disease that is characterised by major organ damage and severe symptoms in chronic sufferers.

The strategy the WHO proposes on their website is similar. Their target group is school-age children, adults in endemic areas that are at risk and people with occupations involving contact with water, such as fishermen, farmers, irrigation workers and women whose domestic tasks bring them in contact with infected water. In communities living in highly endemic areas, they target the whole community.

Education about the risks of infection with bilharzia forms a major part of the effort to control

the disease. The aim is to break the life cycle of the disease by improved water and sanitation, as well as good hygienic practices.

Some worrying factors

In developing countries there is an increase in the number of water resource projects. While this is essential for industrial and agricultural expansion, man-made lakes and irrigation systems are possible breeding places for freshwater snails. They also encourage contact between humans and possibly infected water.

Climate change seems to have had a negative influence on efforts to contain the disease. In fact, the snails that are secondary hosts thrive in hot, wet conditions. Global warming thus contributes to endemic conditions.

Recent research has also shown that schistosomiasis is a key predisposing agent to Human Immunodeficiency Virus (HIV) transmission. For example, urogenital schistosomiasis in HIV-infected women increases the risk of transmitting HIV to male sexual partners. There also seems to be a higher prevalence of HIV infection in women who have eggs in their pap smear.

Some hopeful factors

Some African countries have contributed towards finding solutions for the challenges that schistosomiasis poses. Reintroducing edible freshwater prawns to lakes and rivers in Senegal may prove to be a sustainable way of controlling the disease, while providing an income to nearby communities.

Recent research also seems to indicate that some traditional herbal remedies against urinary schistosomiasis that have been used by traditional healers in Zimbabwe do indeed kill adult parasites.

Until schistosomiasis is eradicated, education seems to be the best way of opening up ways to contain the disease.



Figure 6.4 Map indicating the number of neglected tropical diseases in Africa

Project

In 2015, Health and Child Care minister David Parrenyatiwa said that 57 out of Zimbabwe's 63 districts were prone to bilharzia. This is one of the neglected tropical diseases (NTD) on our continent. The map in Figure 6.4 indicates the number of NTDs in African countries.

Find out which of the other NTDs can be found in Zimbabwe. Research ways to break the cycle of these infections. In groups, design a leaflet in which you inform people about one of these infections. Your leaflet will be evaluated as follows:

1. Suitable heading (2)
2. Correctness of information (8)
3. Neatness and organisation of information (3)
4. Acknowledgement of sources (2)

Total: 15

Summary

- The WHO defines health as 'a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.'
- Disease is a disorder or malfunctioning of an organism, and it has particular signs and symptoms.
- Infectious diseases are caused by pathogens such as bacteria, viruses, parasites or fungi. These diseases spread directly or indirectly from person to person. Infectious diseases are also known as communicable diseases.
- Non-infectious diseases are caused by the environment, deficiencies in the diet, lifestyle choices or are passed on genetically.
- Each infectious, or communicable, disease has a unique cycle of progression with chain elements to keep it going.
- There are six chains in the infectious disease cycle: a pathogen or disease-causing organism, a reservoir, a means of escape, a mode of transmission, a means of entry and a susceptible host.
- Bilharzia, 'snail fever' or schistosomiasis is a disease caused by one of several varieties of the parasite *Schistosoma*.
- Bilharzia is one of the neglected tropical diseases that places a heavy burden on the African continent.
- Schistosomiasis is transmitted when humans come into contact with contaminated freshwater sources, such as lakes, dams, ponds and rivers where there are parasite-carrying snails.
- Humans are the primary hosts in the life cycle of the bilharzia parasite while freshwater snails are the secondary or alternate hosts.
- The adult bilharzia worms are killed by Praziquantel. There is no vaccine available, but Praziquantel can be administered as a preventative measure in high-incidence areas.

Glossary

communicable disease – infectious disease

endemic disease – a disease that is characteristic of a certain population

infectious disease – a disease that can be spread from human to human indirectly or directly

neglected tropical disease a disease found mainly in poor communities in tropical and sub-tropical areas

non-communicable disease – non-infectious disease

non-infectious disease – a disease that cannot be spread from human to human by contact although it can be inherited

parasitic disease – a disease caused by a parasite

pathogen – a disease-causing organism

Revision questions

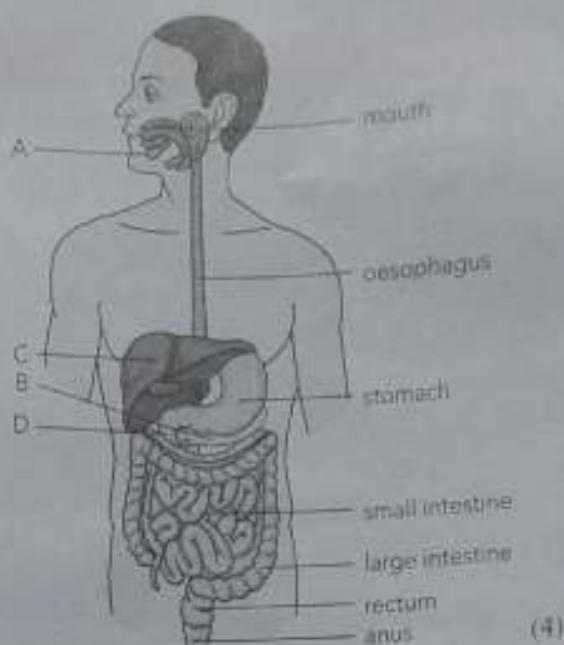
1. What is meant by the term 'health'? (2)
2. What is meant by the term 'disease'? (2)
3. Which of the following pairs is mismatched?
A NTD – non-communicable
B Parasitic – infectious
C Sickle cell disease – non-communicable
D Sickle cell disease – hereditary (2)
4. Name the six chains that must be present for an infectious disease to spread. (6)
5. Give the scientific name of the two organisms that cause schistosomiasis in Africa. (4)
6. Name four activities that can put people at risk of getting bilharzia. (4)

TOTAL: 20

End-of-topic revision test

- List five of the life processes that distinguish living organisms from non-living things. (5)
- Which characteristic listed below is NOT a characteristic of animals? (1)
 - Many-celled
 - Have chlorophyll
 - Have no cell walls
 - Ingest food
- Which characteristic listed below is a characteristic of fungi? (1)
 - Have no cell walls
 - Do not respire
 - Do not reproduce
 - Absorb food directly from other organisms or organic waste.
- Choose an example of discontinuous variation from the list below. (1)
 - Human blood groups
 - Human height
 - The number of seeds in pea pods
 - The sizes of leaves of one tree.
- What is meant by the term 'genotype'? (2)
- What is a congenital disease? (2)
- What association cannot be made with plants? (1)
 - Photosynthesis
 - Self-feeder
 - Producer
 - Heterotroph
- What association cannot be made with consumers in an ecosystem? (1)
 - Other-feeder
 - Heterotroph
 - Depend on plants
 - Photosynthesis
- Which of the following are reactants in photosynthesis? (1)
 - Carbon dioxide and glucose
 - Carbon dioxide and chlorophyll
 - Carbon dioxide and water
 - Sunlight and glucose
- Put the processes involved in human nutrition in the correct order: absorption – ingestion – egestion – digestion – assimilation (5)

- Label the structures A, B, C and D in the diagram of the human alimentary canal, shown below. (4)



- Write down the word equation for cellular respiration. (5)
- Complete this sentence: There is _____ between photosynthesis and cellular respiration. (1)
- Complete this sentence: Breathing is an _____ action that is controlled by the respiratory control centre at the base of the brain. (1)
- Fill in the missing words: Both diffusion and osmosis depend on the _____ movement of molecules _____ their concentration gradient. No additional _____ is needed to make these processes happen, because they are passive processes. (3)
- Name the two main types of vascular tissue in plants. (2)
- Choose the correct word. All arteries in the body contain oxygen-rich blood, except the _____. (1)
 - pulmonary arteries
 - aorta
 - renal arteries
 - coronary arteries

18. Which blood component plays a role in blood clotting?

(1)

19. The seed of a dicotyledonous plant:

- A is the result of single fertilisation
- B has two seed leaves
- C is called a grain
- D is called a gamete.

20. Fruits:

- A nourish embryo development
- B help with seed dispersal
- C attract pollinators
- D develop from endosperm.

(1)

21. List five functions of the placenta during pregnancy.

22. Explain what the difference is between the primary and secondary sex characteristics in humans.

23. Explain in detail what a non-infectious disease is.

(1)

24. Name three ways in which you can prevent others from getting an infectious disease when you have been infected.

25. Describe the lifecycle of the schistosoma parasite.

(1)

TOTAL: 65

Objectives

- State the applications of filtration, winnowing, magnetism and evaporation.

Introduction

In Form 1, you were introduced to a number of separation methods. These included filtration, evaporation, magnetism and decanting.

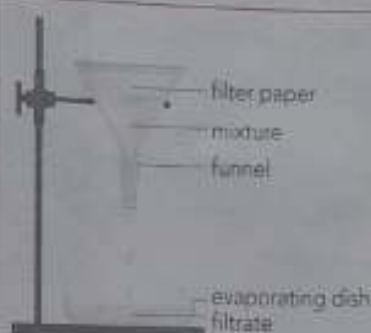
A number of these methods have practical applications in industry. In this unit, we will investigate some of these applications more closely.

Revision: Methods of separating mixtures

Table 7.1 shows some methods of separating mixtures that you learnt about in Form 1.

Table 7.1 Methods of separating mixtures

Filtration: A mixture consisting of a liquid and an insoluble solid is poured through filter paper so that the liquid runs through, and the solid substance stays behind on the filter paper.



Wet funnel with water and place a filter paper in the funnel. Press paper to the side of the funnel.

Figure 7.1 Separation by filtration

Evaporation: A mixture of a solvent and a solute is heated so that the solvent evaporates. The solute is left behind.



Figure 7.2 Separation by evaporation

Magnetism: A mixture of solids (in which at least one of the solids has magnetic properties) can be separated using a magnet.

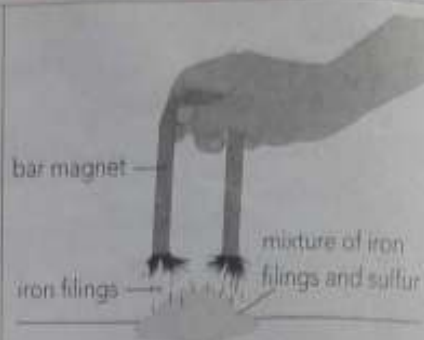


Figure 7.3 Separation by magnetism

Decanting: A mixture of two or more immiscible liquids can be separated by pouring off the less dense liquid (that is, the uppermost layer).

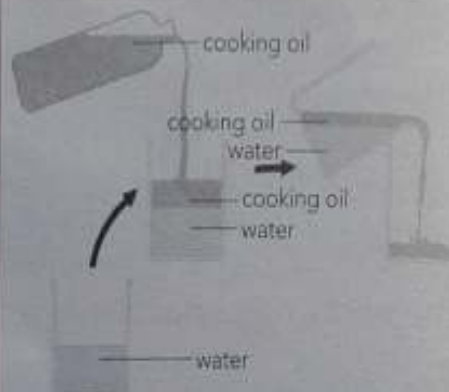


Figure 7.4 Separation by decanting

Sieving: A mixture of finer and coarser solids can be separated from each other by letting the finer solids fall through the sieve.



Figure 7.5 Separation by sieving

Winnowing: A mixture of grain and chaff can be separated by allowing the lighter chaff to be blown off by a stream of air.



Figure 7.6 Separation by winnowing

Exercise 7.1

1. Explain how you can separate paraffin from its mixture with water.
2. A toddler accidentally mixes salt and sand. Explain how you would obtain salt from the mixture.
3. Figure 7.7 shows the stages a learner followed when separating salt mixed with maize meal.
 - a) Arrange the stages in their correct order.
 - b) Describe what is happening at each stage.
 - c) Salt is a solute. What do you understand by the term 'solute'?
 - d) Maize meal does not dissolve in water. Which term is used to describe maize meal in water?

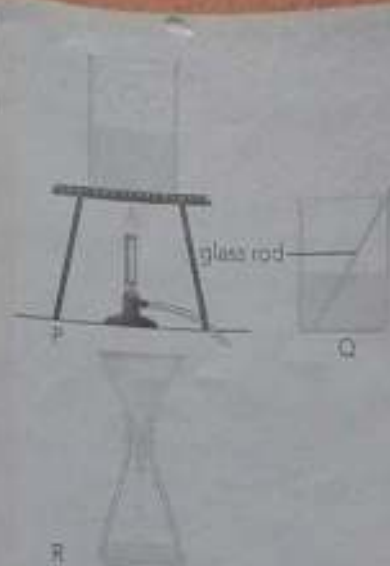


Figure 7.7 Stages of separation

Applications of some methods of separation

Filtration

Water treatment

There are undesirable substances found in water. Some of these impurities can be seen with the naked eye, but most of them can only be seen through a microscope. Dissolved substances are impurities of

water as well. These include gases and minerals in the soil. Harmful bacteria, which cause disease, are another impurity, and should be removed before the water is made available for drinking.

Experiment 7.1

Aim: To identify impurities in water

Materials: beakers, flasks, filter funnels, filter paper, measuring cylinders, bioviewers or microscopes, slides that show life in a pond, evaporating dishes, a burner, water samples from a dam, a well, a river, a pond, tap water

Procedure

1. Label your flasks as shown in Figure 7.8.
2. Measure 100 ml of each type of water and filter it as shown.
3. Compare the filter papers from each water source.
4. Try viewing the water from each source under a magnifying glass or microscope. Compare your observations.

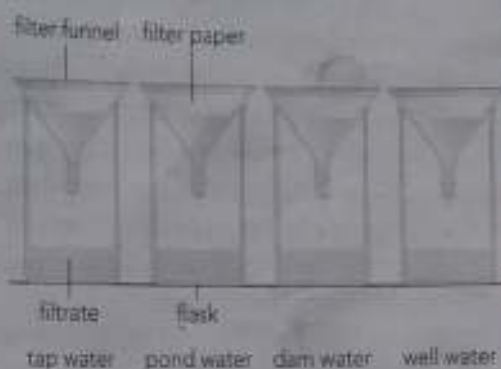


Figure 7.8 Identifying impurities in water

- Now view the slide, which shows life in a pond. Is it similar to any observation in any four of your water sources?
- Pour an equal amount of each filtrate in a beaker and evaporate the liquid until the beaker is dry. What remains at the bottom of each beaker?

Results

All suspended solids are removed by filtering the water. Micro-organisms are killed by boiling water. Dissolved salts remain at the bottom of the beaker after evaporation. All of these (suspended solids, micro-organisms and dissolved salts) are impurities found in water.

Questions

- Which water source had a lot of impurities left on the filter paper?
- Do the water sources have any micro-organisms?
- Besides boiling, how else can you destroy micro-organisms in water?
- Are there any residues left after evaporation?
- Where do you think such impurities came from?

Every town or city has a water treatment plant to treat water from its water source (such as a river, dam or lake). Before treatment, the water is raw. Large urban communities are supplied with piped

running water. The water is first treated to make it safe to drink. Figure 7.9 shows each of the main steps involved in water treatment.

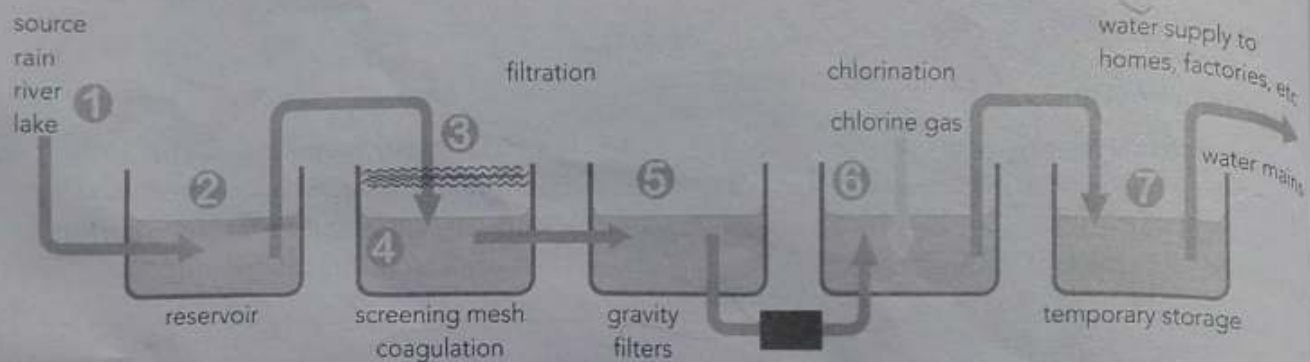


Figure 7.9 A modern water treatment plant

- Step 1 Water is pumped into a reservoir.
- Step 2 A filter screen (wire mesh) traps large, unwanted objects, such as leaves, grass, dead animals and stones.
- Step 3 Water is allowed to settle to allow removal of some of the solid particles. A chemical called a flocculant is added to the water. The flocculant speeds up the settling of clumps of solid particles at the bottom of the sedimentation tanks.
- Step 4 Air is blown through the mixture so that the flocculated clumps float to the top of the water. In this way, they can be skimmed off and discarded.
- Step 5 The remaining solid particles are removed by filtering the water through a sand filter. This makes the water clear, but it is still not safe for drinking. There are still some bacteria and other micro-organisms in the water. If needed, the water can be further filtered through finer sand or charcoal.
- Step 6 Chlorine is added to the mixture to kill any bacteria or other organisms that may still be present. This process is known as chlorination. If fluoride is added to the water supply, it is added immediately after this step. Fluoride is added to water sources to help prevent tooth decay.

- Step 7 The clean water is pumped into a storage reservoir, from where it can be piped to houses and businesses.

Activity 7.1

Aim: To purify water in the laboratory

Materials: 2 litre plastic bottle, flask, burner, sand, large and small stones, dirty water, a clean cloth

Procedure

1. Strain the water using a clean cloth to remove grass, leaves and other big objects.
2. Remove the bottom of your plastic bottle and set up a sand filter as shown below. Use clean sand and stones.
3. Filter the water using the sand filter.
4. Collect the clear water and boil it.

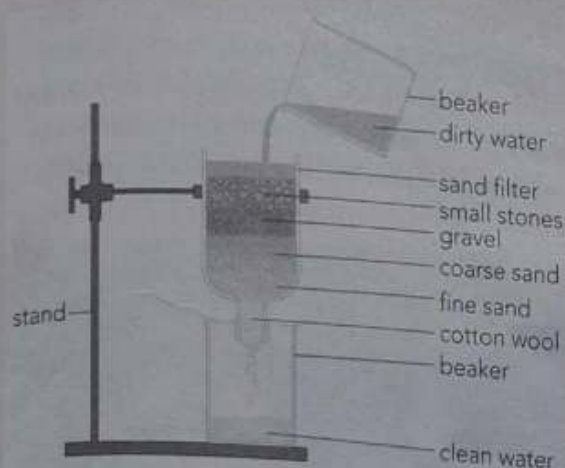


Figure 7.10 Using a sand filter in the laboratory

Winnowing

Grain separation

Winnowing by wind is a method of separating usable grain from chaff that has been used for hundreds of years. Any grain, such as rice or wheat, has lighter particles of dirt or plant material (chaff) attached to it as it is harvested. To remove the unwanted component, the mixture is thrown into the air where the wind (or a fan) carries the lighter particles away from the grain, which falls back down to the ground under the force of gravity.

In ancient societies, winnowing by hand was the only way to achieve a relatively pure component of grain. Today, threshers and winnowing machines typically do the same job on grain farms.



Figure 7.11 A traditional way of winnowing grains

Magnetism

Separation of metal objects from agricultural products

In Form 1, you learnt about the production of peanut butter and all the steps involved in processing the raw product to achieve the final product. One of the early steps in obtaining clean, raw and shelled peanuts was the removal of any metal objects or particles that may have been included in the harvest. Metal objects can be introduced by means of wind-blown litter, or by means of the tools that are used to work the land. For example, a piece of machinery could break off and end up in the soil. Either way, pieces of metal are not only unwanted in the final food product, they can also cause damage to the processing machinery.

Magnets are used to collect any remaining metal objects or particles from the agricultural product, usually at the point at which blowers and shakers are used to agitate the raw product. In this way, even small metal particles become dislodged from the mixture, and are therefore free to be attracted by a powerful magnet.

Separation of metal objects from recycling waste

In large-scale scrap metal plants, where valuable metal scraps must be collected and sorted from

other types of (less valuable) recyclable materials, an electromagnetic crane is often used to make the job easier. An electromagnetic crane has a large, flat magnetic disc instead of the mechanical pincers that a regular crane has. The operator can control the electromagnet so that large loads of scrap iron and steel can be attracted, moved and deposited in sorting piles.



Figure 7.12 More valuable scrap metal is separated from plastics and other waste by magnets at a scrapyard.

Evaporation

Cane sugar processing

Evaporation plays an important role in the extraction of solid sugar from sugar cane, as the sugar that processing plants seek is dissolved in the juice of the sugar cane.

At a sugar mill, sugar cane is crushed and shredded to extract the juice. The resulting mixture contains both the sugar-containing juice and the fibrous plant material. The solid plant material (called *bagasse*) must be removed by filtration before the juice is processed further.

Lime and other clarifiers are added to the liquid to remove particles of soil. The clear juice is piped into evaporators, which reduce the mixture to a thick, sugary syrup. This syrup is then separated into its sugar and molasses components. The sugar is further refined, while the molasses is used for animal feeds, the production of alcoholic beverages and other uses.



Figure 7.13 A sugar mill evaporator

Table salt processing

Evaporation is a necessity in the production of table salt from seawater. In countries that produce table salt from seawater, such as Namibia and South Africa, shallow ponds of seawater are left to dry in the sunshine until all the water has evaporated. The salt contains mostly sodium chloride, but also other mineral salts that were once dissolved in the ocean. Some chefs say that salt from seawater tastes better than salt that is mined on land, simply because of these other mineral salts.



Figure 7.14 Seawater salt production has not changed much since ancient times.

Exercise 7.2

1. Explain the role of gravity in the separation of grain from chaff and dirt by winnowing.
2. Explain how evaporation is involved in processing sugar from sugar cane.
3. Name two industrial applications of magnetism as a means of separating mixtures.

Summary

- The methods of separation that you learnt about in Term 1 include:
 - Filtration:** A mixture consisting of a liquid and an insoluble solid is poured through filter paper so that the liquid runs through, and the solid substance stays behind on the filter paper.
 - Evaporation:** A mixture of a solvent and a solute is heated so that the solvent evaporates. The solute is left behind.
 - Magnetism:** A mixture of solids (in which at least one of the solids has magnetic properties) can be separated using a magnet.
 - Decanting:** A mixture of two or more immiscible liquids can be separated by pouring off the less dense liquid (that is, the uppermost layer).
 - Sieving:** A mixture of finer and coarser solids can be separated from each other by letting the finer solids fall through the sieve.
 - Winnowing:** A mixture of grain and chaff can be separated by allowing the lighter chaff to be blown off by a stream of air. To remove the unwanted component, the mixture is thrown into the air where the wind (or a fan) carries the lighter particles away from the grain, which falls back down to the ground under the force of gravity.
- Water contains impurities that should be removed before you can drink it.
- Some of the impurities in water are invisible.
- Filtration and chemical treatment are the two methods used to purify water.
- Magnetism is used to separate metal scraps from other (recyclable) waste, or from raw agricultural products.
- Evaporation plays an important role in the extraction of solid sugar from sugar cane, as the sugar that processing plants seek is dissolved in the juice of the sugar cane. The clear juice is piped into evaporators, which reduce the mixture to a thick, sugary syrup.
- Evaporation is also a necessity in the production of table salt from seawater. Shallow ponds of seawater are left to dry in the sunshine until all the water has evaporated.

Glossary

baggasse – the dry plant material that remains once all the juice has been extracted from sugar cane

chlorination – process of adding chlorine to water to kill micro-organisms during purification of water

decanting – carefully pouring liquid from one vessel into another to separate two liquids with different densities

evaporation – the process of turning a liquid into a vapour, usually to obtain a solid that was dissolved in the liquid

filtration – the process of removing an insoluble solid from a liquid by pouring the mixture through filter paper

flocculant – substance added to water which makes solid particles gather together and settle at the bottom

magnetism – the ability of one substance to attract or repel another by means of a magnetic force

mixture – a combination of two or more substances

molasses – a dark, syrupy by-product obtained when refining sugar from sugar cane

sedimentation – the settling of small, insoluble particles during the purification of water

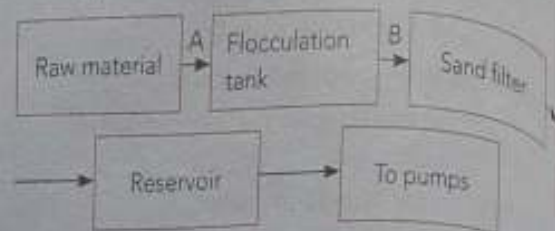
sieving – a method used to separate small things from large ones using a frame with a wire mesh

winnowing – a method used to separate grain from the rest of the plant by means of a stream of air, which blows off the lighter chaff

Revision questions

- Which is the correct sequence in water treatment?
 A Water source (reservoir) → chemical treatment → filtration → storage tanks
 B Chemical treatment → filtration → water source (reservoir) → storage tanks
 C Water source (reservoir) → filtration → chemical treatment → storage tanks
 D Storage tanks → water source (reservoir) → filtration → chemical treatment (1)
- Which type of impurities are removed using a sand filter in the treatment of water?
 A Micro-organisms
 B Dissolved impurities
 C Suspended substances
 D All of the above (1)
- The role of a flocculant in water purification is that of:
 A allowing suspended particles to remain suspended, so that removal is easier
 B causing large solid particles to clump together, so that removal is easier
 C filtering large particles before the water is settled in the sedimentation tanks
 D killing micro-organisms. (1)
- The name given to the remains of the sugar cane once the juice has been extracted from it, is:
 A flocculant
 B bagasse
 C molasses
 D refined sugar. (1)

- Magnetism can be used to separate recyclable material only if some of the material contains:
 A iron or copper
 B aluminium or copper
 C steel or aluminium
 D iron or steel.
- Below is a diagram which shows some stages in purification of water.



- What is the purpose of the flocculant at Stage A?
 - Stage B is the chemical treatment stage. Which chemical is added at this stage?
 - Water is said to be clean and safe if it is free from dissolved substances and _____.
- Below are some impurities that can be found in water:
 suspended solids
 dissolved salts
 micro-organisms

Explain how you would remove each of these impurities during purification of water.

(3 × 2 = 6)

TOTAL: 11

Objectives

- Determine the concentrations by colour densities of dissolved substances.
- Determine the concentration of a substance by varying the amount of solute in a given solvent.

Introduction

Some chemical substances have bright colours, so we can use this to determine their concentrations when dissolved in solution.

Although we can compare the colour densities of different concentrations of a substance, relying only on sight is not always precise enough in the laboratory. For this reason, chemists use a method of measuring concentration of coloured substances called **colorimetry**. It involves the use of a device called a **colorimeter**, which measures how much light of a certain wavelength a solution can absorb. The process involves preparing a reference solution, of which the concentration is known. A sample of the reference solution is used to calibrate the device, so that its results from the unknown samples can be compared to the initial (reference) reading.

In this unit, we will investigate the principles of this technique, but only rely on sight to determine solution concentrations.

Concentration

In Form 1, you learnt that chemical solutions are made by dissolving a substance (a solute) in a liquid (a solvent). The **concentration** of a chemical solution is the amount of solute that is dissolved in a solvent. Although we often think of a solute as being a solid substance, solutes can also be liquids or gases.

If we know the identity of the solute and the solvent in a solution, we can determine the

concentration of that solution. We can do this in different ways and using different units. For example, a solution of table salt (NaCl) in water can be expressed in units of mass of solute per volume of solvent, such as g/cm^3 or kg/l . However, it is sometimes more convenient to work with the number of particles of solute per volume of solvent – especially when we work with liquid and gas solutes.

Dilution

A **dilution** is a solution that is made by adding more solvent to a solution of known concentration. This will reduce the solution's concentration. We can see the effects of dilution in Figure 8.1.

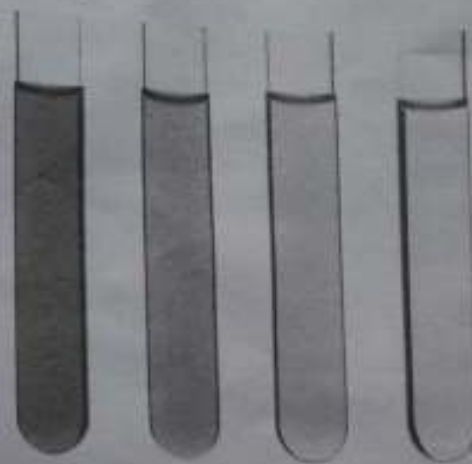


Figure 8.1 A series of dilutions

Higher concentrations of a substance will give solutions of darker or deeper colours. As solutions are diluted, their colour density decreases.

Experiment 8.1

Aim: To prepare a set of dilute solutions

Materials: four large beakers (A–D), dye, water

Procedure

1. Pour about 50 ml of water into beaker A.
2. Add five drops of dye to the water.
3. Stir the liquid using a glass rod until all the water is coloured. A solution of the dye has been made.
4. Pour about 5 ml of the prepared solution into beaker B.
5. Add water to beaker B until the water level rises to the 50 ml mark. Your solution has been diluted ten times.
6. Now compare the solutions in beakers A and B.
7. Repeat steps 4 and 5 another two times.

Results

Each time you dilute (or weaken) the solution ten times, the solution becomes paler, until the colour disappears, meaning there will be very few dye particles left in the 50 ml of water.

Questions

Discuss and answer these questions in groups.

1. Do you think it is possible to end up with a solution that has no colour?
2. Although you can see that the concentration of each successive solution is less than the solution before, can you determine the exact concentration of dilution 3 or dilution 4? Explain your answer.

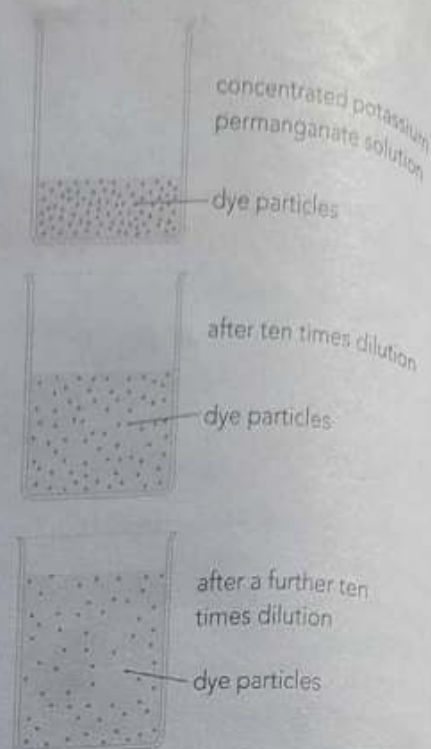


Figure 8.2 Diluting a dye

There are fewer and fewer dye particles with each dilution. However, even after a number of dilutions, some of the dye still remains. In Form 1, you learnt about the kinetic particle theory. Experiment 8.1 proves that matter is made up of particles, and that these particles have spaces between them.

Calculating dilutions

It is useful to use percentage values to represent dilutions. For example, a set of five concentrations of a solute in a solvent could represent a 100%, 80%, 60%, 40% and 20% concentration. To serve as a comparison, a reference sample that contains only the solvent is also included in the series. For liquid solutes, we use the equation:

$$\frac{\text{total volume of solution}}{\text{volume of solvent}} \times 100\%$$

Example 8.1

Complete Table 8.1 by calculating the volume of liquid solvent needed in each test tube, in order to create a set of sample solutions that represent 100%, 80%, 60%, 40% and 20% concentrations.

Table 8.1 Complete the table to create the concentrations in the first column

Concentration	Solute (ml)	Solvent (ml)	Total volume (ml)
100%	10	0	10
80%	a	b	10
60%	c	d	10
40%	e	f	10

Concentration	Solute (ml)	Solvent (ml)	Total volume (ml)
20%	2	8	10
0% (reference sample)	0	10	10

Concentration	Solute (ml)	Solvent (ml)	Total volume (ml)
100%	10	0	10
80%	8	2	10
60%	6	4	10
40%	4	6	10
20%	2	8	10
0% (reference sample)	0	10	10

Activity 8.1

For this activity, you will need water, a set of six large test tubes in a test tube rack, a small measuring cylinder, a pipette and some food colouring.

1. Copy and complete Table 8.2.

Table 8.2 Calculating dilutions

Concentration	Solute (ml)	Solvent (ml)	Total volume (ml)
100%			25
80%			25
60%			25
40%			25
20%			25
0% (reference sample)			25

2. Using your table as a guide, transfer the correct volumes of food colouring and water into each test tube in your set.
3. Your teacher will show you a number of solutions of different concentrations. For each solution of unknown concentration, compare it to your dilution series to determine the estimated concentration.

4. Discuss the following in groups:
 - a) Is it possible to determine exact concentrations by visually comparing solutions?
 - b) How could we improve the accuracy of our results?

Mole concept

Often, chemists work with samples that contain enormous numbers of atoms, so they use a special unit to describe very large numbers of atoms. Using a unit to describe a particular number of objects is not new, for example, one dozen represents twelve objects.

The unit chemists use is the **mole**. A mole is a measure of the number of particles (atoms or molecules) in a sample of substance. The number of particles (atoms or molecules) in one mole of anything is always the same number: 6.023×10^{23} . This number is called **Avogadro's constant**. $1 \text{ mole} = 6.023 \times 10^{23}$ particles. Of course, the mass of one mole of substance differs from substance to substance.

Example 8.2

One mole of oxygen gas (O_2) has a mass of 32 grams.



Figure 8.3 Oxygen gas



Figure 8.4 Copper metal

One mole of copper metal (Cu) has a mass of 63.55 grams.

We can use the number of particles in one mole of substance to determine concentrations of solutions more accurately than by comparing samples visually, as we shall see in the next experiment.

Experiment 8.2

Aim: To determine solution concentration from colour density

Materials: four large test tubes, test tube rack, measuring cylinder, pipette, potassium permanganate crystals, spatula, stirring rod, water

Procedure

1. One mole of potassium permanganate crystals has a mass of 158 g. Calculate the mass of 0.01 mole of potassium permanganate and the number of particles contained in that mass of potassium permanganate.
2. Put this amount of crystals in the first test tube (A) and add 20 ml water. Stir the solution in test tube A with the stirring rod until all the crystals have dissolved.
3. Copy and complete Table 8.3 as you complete the experiment.

Table 8.3 Copy and complete this table

Test tube	A	B	C	D
Mass of crystals (g)				
Number of moles				
Volume of solution (ml)				
Volume of water (ml)				
Total volume (ml)				
Concentration (mol/ml)				
Number of particles				

4. Use the pipette to measure 2 ml of the solution in test tube A, and transfer it to test tube B. Add 18 ml of water to test tube B.
5. Repeat Step 4 for test tubes C and D, each time taking 2 ml of solution from the previous test tube.

Results

The colour densities of the various solutions will range from a dark or deep purple to a light or very light pink-purple colour. Because a 1:10 dilution was created each time, each successive concentration is one-tenth of the previous concentration.

Questions

1. Describe the pattern formed in the numbers of moles in each dilution (the second row in the table).
2. The final dilution appears much lighter in colour than the others. However, it still contains a large number of particles of potassium permanganate. How many more times would you have to repeat the dilution before the number of particles in a solution is less than 1 000?

Summary

- Colorimetry is a method of measuring the concentration of a solution. It involves the use of a device called a colorimeter, which measures how much light of a certain wavelength a solution can absorb.
- The concentration of a chemical solution is the amount of solute that is dissolved in a solvent.
- If we know the identity of the solute and the solvent in a solution, we can determine the concentration of that solution.
- A dilution is a solution that is made by adding more solvent to a solution of known concentration.
- Higher concentrations of a substance will give solutions of darker or deeper colour.
- There are fewer and fewer solute particles with each dilution. However, even after a number of dilutions, some of the dye still remains.
- For liquid solutes, we use the equation:
$$\frac{\text{total volume of solution}}{\text{volume of solvent}} \times 100\%$$
 to determine the percentage concentration of the solution.
- A mole is a measure of the number of particles (atoms or molecules) in a sample of substance.
- The number is 6.023×10^{23} . It is called Avogadro's constant.

Glossary

Avogadro's constant – the number of particles (atoms or molecules) in one mole of any substance; 6.023×10^{23}

colorimetry – a method of determining concentration by measuring how much light of a certain wavelength a solution can absorb

concentration – the amount of solute that is dissolved in a solvent

dilution – a solution that is made by adding more solvent to a solution of known concentration

mole – a measure of the number of particles (atoms or molecules) in a sample of substance

Revision questions

- Colorimetry is a method of determining the concentration of a solution by:
 - measuring the mass of the solute
 - measuring the number of particles in the solution
 - performing a number of dilutions
 - measuring how much light of a certain wavelength a solution can absorb. (1)
- Avogadro's constant is:
 - the number of atoms in one gram of a substance
 - the number of atoms or molecules in one gram of a substance
 - the number of atoms or molecules in one mole of a substance
 - the number of moles in one gram of a substance. (1)
- A 50 ml solution contains 4.2 g of dissolved (solid) solute. What is the solution's concentration?
 - 0.084 g/ml
 - 0.084 mol/ml
 - 11.9 g/ml
 - 2.10 ml/g (2)
- A solution contains 24 ml of solute and 82 ml of solvent. What is the percentage concentration of the solution?
 - 17.1%
 - 29.3%
 - 22.6%
 - 58.6% (2)
- A learner prepares a set of 50 ml dilutions of sodium hydroxide solutions.
 - Copy and complete Table 8.4, to show how much solute and solvent there will be in each flask.

Table 8.4 Calculating the volume of solvent and solute

Concentration	Solute (ml)	Solvent (ml)	Total volume (ml)
100%			50
80%			50
60%			50
40%			50
20%			50
0% (reference sample)	0	50	50

- If the 100% solution contains one mole of sodium hydroxide, how many moles of sodium hydroxide are in the 40% solution? (1)
- A solution of silver nitrate has a concentration of 3 moles per litre.
 - Explain what a mole is. (2)
 - How many particles of silver nitrate are there in 1 litre of the solution? (2)
 - One mole of silver nitrate has a mass of 170 g. How many grams of silver nitrate are there in 500 ml of the solution? (3)
 - A 250 ml, 1:10 dilution of the solution is made. Prove by calculation that the new solution has a concentration of 0.3 moles per litre. (3)

TOTAL: 28

Unit
9

Acids and bases

Objectives

- Describe an acid-base reaction.

Introduction

In Form 1, you learnt about acids, bases (or alkalis) and neutral substances, and that the pH values of these substances can be determined by using litmus paper. In Form 2, you will learn about the reaction that takes place when an acidic and a basic substance react with one another.



Figure 9.1 Acids turn blue litmus paper red. Bases turn red litmus paper blue.

Reactions of acids with bases

When an acid and a base react, we call the reaction a **neutralisation reaction**, because both the acid and the base are effectively neutralised

when they come into contact with each other. As we would expect, the pH of the products is 7 (neutral). The term 'base' is used to describe all the compounds that can neutralise acids.

In an acid-base reaction, a salt is always produced. The type of salt that is formed depends on the acid used:

- Sulfuric acid forms a sulfate.
- Nitric acid forms a nitrate.
- Hydrochloric acid forms a chloride.

For example, a reaction between sodium hydroxide and sulfuric acid gives sodium sulfate, and a reaction between sodium hydroxide and hydrochloric acid gives sodium chloride. A neutralisation reaction also always produces water. Therefore, we can summarise neutralisation reactions using the following word equation:



We will investigate reactions of acids with the following different types of bases:

- metal oxides, such as copper oxide
- metal hydroxides, such as sodium hydroxide
- carbonates, such as calcium carbonate.

Experiment 9.1

Aim: To investigate the reactions of acids with metal oxides

Materials: beaker, conical flask, spatula, stirring rod, filter funnel and filter paper, burner, tripod, gauze, red and blue litmus paper, tongs, dilute sulfuric acid, 1 g of copper(II) oxide

Procedure

- Heat 20 cm³ dilute sulfuric acid in a beaker over a burner until it is almost boiling.
- Use a spatula to add small quantities of copper(II) oxide to the heated mixture. Stir the mixture after adding the copper(II) oxide each time. Use the tongs to lower pieces of red and blue litmus paper into the solution. Record the colour changes of the litmus paper, if any.

3. Heat the mixture for another minute or two, and then turn off the burner and let the mixture cool. Test the mixture using both red and blue litmus paper again, and record your observations.
4. Filter the cooled mixture.

Results

You should see a blue solution in the conical flask after filtering. If the solution is not clear, repeat the filtration.

Questions

1. Did the mixture change colour at any point of the experiment?
2. At which point do one of the pieces of litmus paper indicate that a change has occurred in the mixture's pH?
3. Write a word equation for this reaction.

Experiment 9.2

Aim: To investigate the reactions of acids with metal hydroxides

Materials: sodium hydroxide solution, dilute hydrochloric acid, dropper, universal indicator, test tube or small beaker, teaspoon or measuring cylinder

Procedure

1. Place about 10 ml (two teaspoons) of sodium hydroxide solution in the test tube or beaker.
2. Add a drop of universal indicator. Use the colour of the indicator to determine the pH of the sodium hydroxide.
3. Use the dropper to add hydrochloric acid to the sodium hydroxide. Swirl the beaker or test tube after every few drops.
4. When the indicator turns light blue, add one drop of acid at a time. Stop when the indicator turns green.

Results

The hydrochloric acid reacts with the sodium hydroxide solution to produce a specific set of products.

Questions

1. Explain the meaning of the green colour of the indicator.
2. Write a word equation for the reaction in the test tube.

Experiment 9.3

Aim: To investigate the reactions of acids with carbonates

Materials: test tubes, retort stand, spatula, sodium carbonate, nitric acid, hydrochloric acid, sulfuric acid, lime water, delivery tube, cork

Procedure

1. Place about 2 ml of dilute hydrochloric acid in a test tube.

2. Add a spatula of sodium carbonate (see Figure 9.2), fit the cork and delivery tube to the top of the test tube and submerge the other end of the delivery tube into lime water (in another test tube). Record your observations and test whether the gas evolved is carbon dioxide.
3. Repeat the experiment two more times – once using nitric acid and once using sulfuric acid. Copy the table below and record all your observations as you complete the experiment.

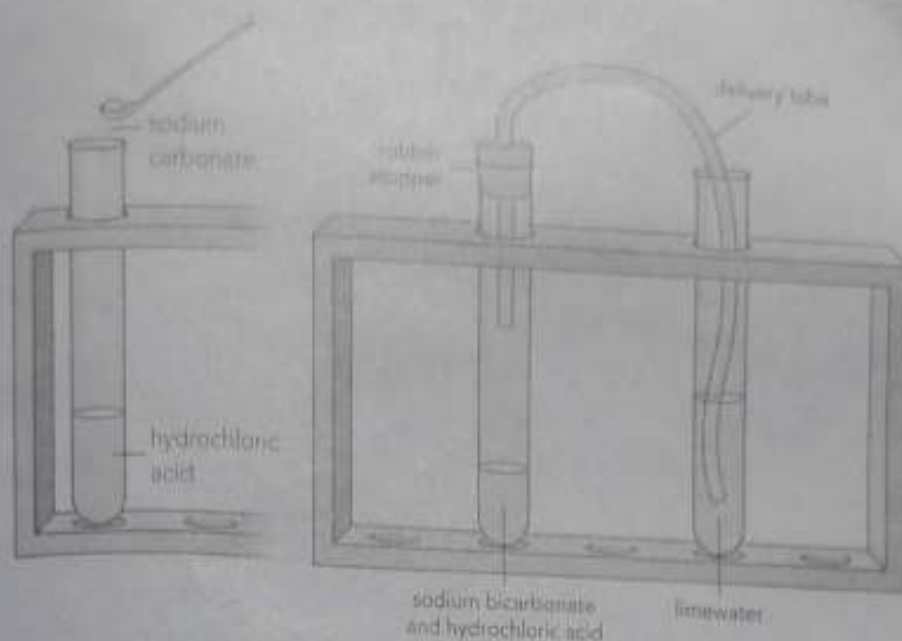


Figure 9.2 Investigating the reactions of acids with carbonates

Results

Record your results in Table 9.1.

Table 9.1 Investigating the reactions of acids with carbonates

Reaction of sodium carbonate and ...	Observations	Gas evolved (if any)
hydrochloric acid		
sulfuric acid		
nitric acid		

Questions

Write down the word equation for the reaction of sodium carbonate and:

1. hydrochloric acid
2. sulfuric acid
3. nitric acid.

A summary of the products produced in the three types of neutralisation reactions that we investigated are shown in Table 9.2.

Table 9.2 Three types of neutralisation reactions

Type of neutralisation reaction	Products formed
Acid + metal oxide	Salt + water
Acid + metal hydroxide	Salt + water
Acid + carbonate	Salt + water + carbon dioxide gas

Note: Because the reaction between an acid and a carbonate produces carbon dioxide gas as well as a salt and water, some scientists consider it not to be a true neutralisation reaction.

There are many everyday applications of acid-base reactions – examples of these can be seen in Table 9.3.

Table 9.3 Everyday applications of acid-base reactions

Bicarbonate of soda (NaHCO_3) is used for baking. When it is heated, it reacts with acidic ingredients in bread or cake dough and carbon dioxide gas is released. This makes the bread or cake rise.



Figure 9.3 Bread rises because of an acid-base reaction.

Excess stomach acid causes heartburn and indigestion. Antacids are used to neutralise excess stomach acid. Different antacids contain different bases. Examples include CaCO_3 , MgCO_3 , Mg(OH)_2 and Al(OH)_3 .



Figure 9.4 Antacid tablets neutralise excess stomach acid.

When an insect bites or stings you, it injects a small amount of a chemical into your skin. The acid of a bee sting can be neutralised using a base such as bicarbonate of soda. The alkaline sting of a wasp can be treated with a weak acid such as vinegar.



Figure 9.5 Take the sting out of insect bites with an acid-base reaction.

Toothpaste often contains particles of calcium carbonate or aluminium hydroxide. These particles are slightly rough and when you brush your teeth, they clean plaque (a sticky leftover from food and bacteria) from your teeth.



Figure 9.6 Toothpaste neutralises acids in your mouth to protect your teeth.

Some crops cannot grow if the soil is too acidic. Farmers use agricultural lime, which contains calcium hydroxide and calcium carbonate, to lower the acidity in the soil.



Figure 9.7 Agricultural lime neutralises acidic soil, thereby allowing sensitive crops to flourish.

The pH of swimming pools has to be controlled carefully. Pool acid is used to lower the pH if necessary. Soda ash is used to increase the pH.

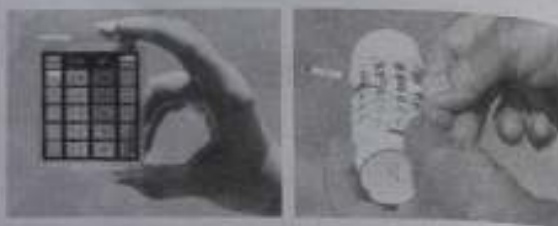


Figure 9.8 Swimming pool water requires a constant balance between acid and base.

Exercise 9.1

1. Which products are produced when an acid and a metal hydroxide react in a neutralisation reaction?
2. Which gas is evolved when a dilute acid reacts with a carbonate?
3. Name three applications of neutralisation reactions that we typically use in the home.
4. A learner is stung by a bee on his way home from school. Suggest what he can do to relieve the pain of the sting the minute he gets home.

Summary

- The reaction between an acid and a base is called a neutralisation reaction.
- The term 'base' is used to describe all the compounds that can neutralise acids.
- Bases react with acids to produce a salt and water.
- Neutralisation reactions are summarized by the following word equation:
$$\text{acid} + \text{base} \longrightarrow \text{salt} + \text{water}$$
- The type of salt that is formed depends on the acid used:
 - Sulfuric acid forms a sulfate.
 - Nitric acid forms a nitrate.
 - Hydrochloric acid forms a chloride.
- The products of a neutralisation reaction have a pH of 7 (i.e. they are neutral).
- The different types of bases that we investigated were:
 - metal oxides, such as copper oxide
 - metal hydroxides, such as sodium hydroxide
 - carbonates, such as calcium carbonate.
- Practical applications of neutralisation reactions include the use of:
 - bicarbonate of soda in baking
 - antacids to neutralise stomach acids
 - household acids and bases to treat insect stings
 - toothpaste to neutralise acids in the mouth
 - agricultural lime to neutralise acidic soils
 - chemicals to maintain the correct pH level in swimming pools.

Glossary

acid – a chemical compound that neutralises a base

base – a chemical compound that neutralises an acid

neutralisation reaction – the type of reaction that occurs when an acid and a base react to produce a salt and water

Revision questions

1. A neutralisation reaction produces the following products:
A An acid and a base
B Water and a base
C An acid and a salt
D Water and a salt (1)
2. When hydrochloric acid and magnesium oxide react, the products that are formed are:
A magnesium chloride and water
B hydrogen chloride and water
C magnesium sulfate and water
D hydrogen oxide and water. (1)
3. The type of salt that is produced when sulfuric acid reacts with a base is called:
A sodium sulfate
B a nitrate
C a sulfate
D a chloride. (1)
4. The gas that is always produced when an acid reacts with a carbonate is:
A carbon monoxide gas
B carbon dioxide gas
C water vapour gas
D hydrogen gas. (1)
5. Which salt is formed when zinc oxide and nitric acid react?
A Zinc dioxide
B Zinc sulfate
C Nitrous oxide
D Zinc nitrate (1)
6. Fill in the correct words or terms: A neutralisation reaction occurs when an acid reacts with a _____ to form a salt and _____. The pH of the products is always _____. (2)
7. Write down the word equation for each reaction.
a) hydrochloric acid and sodium carbonate (2)
b) sulfuric acid and magnesium oxide (2)
8. A farmer finds that the pH level of her soil is slightly too acidic for the types of crops that she wants to grow. Which chemical should the farmer combine with the soil to raise the soil's pH level: calcium carbonate or ash? Give a reason for your answer. (2)

TOTAL: 15

Industrial processes

Objectives

- Outline the manufacture of soap.

Introduction

We use soaps in various forms every day to clean ourselves, our clothes and our homes. In fact, humans have been making and using soaps and detergents since the earliest times. Archaeologists have discovered evidence of soap-making in ancient Babylon (around 2800 BC) and ancient Egypt (1500 BC).

In this unit, we will investigate the chemical processes involved in making soap, and learn how to make our own soap.



Figure 10.1 A long time ago, people would make soap bars with mechanical presses, such as these.

What is soap?

In the previous unit, you learnt about acid-base reactions, and some applications of these types of reactions. Human skin has a pH of around 5.5, so it is slightly acidic. This helps to protect the skin from germs, but also ensures that the correct moisture level is retained. Regular soap has a pH level that is slightly higher than that of

skin, so that dirt and germs can be washed away without affecting the skin's natural moisture level. Therefore, most soaps contain alkalis, such as lye, to raise the pH level. Indeed, regular soaps are sodium salts that are made by reacting sodium hydroxide with fats or oils. The first step in preparing soap is to start with plant or animal fats and oils. When a fat or oil is boiled in an alkali such as sodium hydroxide, soap is produced according to this word equation:



Saponification is the process of combining a fat and an alkali to form a soap and glycerol. It is therefore the name given to the chemical process of making soap. Although glycerol is a by-product in the soap-making process, it has many industrial applications. For example, it is used in skin moisturiser, sweetened cool drinks, various hair care products and as a precursor to explosives.

How does soap work?

Soap is an effective cleaning agent, because it can dissolve in water and remove dirt and oils from our bodies, clothes and other objects at the same time. A molecule of soap has one hydrophilic end and one hydrophobic end. **Hydrophilic** means water-loving, and **hydrophobic** means water-hating.

When we wash something with soap and water, the hydrophilic end of the soap is attracted to the water molecules. The hydrophobic end mixes with the oils. The oily particles from the dirty object detach themselves from the object and attach themselves to the soap. They then dissolve in the water.

Exercise 10.1

1. Explain the term 'saponification'.
2. Which two ingredients are needed to make soap?
3. Explain how soap is able to dissolve in water, yet also remove oily dirt from skin and clothes.

How is soap made?

Industrial soap manufacturers use a clean, purified form of fat – often vegetable-based oils, such as palm kernel oil, coconut oil or olive oil. In the past, animal fats were used, but these contained a number of impurities.

The alkali determines how soluble the soap is in water, so soaps made with sodium hydroxide tend to be more water-soluble, and are therefore called **soft soap**. Alternatively, when potassium hydroxide is used as the alkali, the soap product has slightly different characteristics. Liquid soaps are generally made from potassium hydroxide.



Figure 10.2 Small-scale soap makers produce smaller batches of product, but it takes them relatively longer to produce one batch compared to large-scale producers.

Small-scale soap manufacturers use the batch process, which follows this basic procedure for making soaps:

Step 1: Boiling

The alkali and fat are boiled in a large **kettle** (steel tank) until the two components react to form soap and glycerol.

Step 2: Separation

Salt is added to the mixture to cause the soap product to rise to the top of the kettle, and allow the glycerol to settle at the bottom. In this way,

the glycerol is simply tapped off from the bottom of the kettle.

Step 3: Refinement

Not all of the fat in the mixture reacts the first time round, so a strong alkali is added to the mixture, and the mixture is brought to the boil again to react the last of the fat.

Step 4: Pitching

The process of **pitching** the soap involves boiling the soap again, but this time with water. The soap separates into two layers. The bottom layer, called the **nigre**, contains most of the water, but also any impurities that were present in the starting materials. The top layer, called the **neat soap**, is a liquid layer that contains all the purified soap and some of the water. The neat soap is poured into moulds and allowed to cool.



Figure 10.3 Large-scale (industrial) soap makers can produce products continually because of the equipment that they use.

Large-scale soap manufacturers use a more efficient and quicker procedure to produce soap. This is called the **continuous process**, because the fat is added and the product is removed in a continuous cycle.

Step 1: Splitting

Liquid fat or oil is pumped into one end of a large stainless steel column, called a **hydroliser**. On the other end, super-heated water is allowed into the column. When the two components meet, the heated water causes the natural fat to split into glycerol and fatty acids. These two components are pumped out of separate outlets, while more fat and water are pumped into the column. The fatty acids go through a distillation process for further refinement.

Step 2: Mixing
This is where the alkali is introduced, as well as any additives, colourants or fragrances.

Step 3: Cooling
The soap mixture is poured into moulds and allowed to cool. Eventually, the cooled slab of soap is cut into smaller pieces, which are then packaged and distributed to retail outlets.

Activity 10.1

You will need the following materials for this activity: 340 g sodium hydroxide pellets, 600 cm³ ice-cold water or an ice-and-water mix, 2.5 kg lard or vegetable fat, a 1.5 litre ovenware china bowl, a 5 litre plastic bowl in a water bath or cast iron pot, plastic rubbish bag, cardboard box about the size of a shoebox, handheld blender, colouring and scents, a sharp knife

1. Your teacher will supervise this part of the experiment as sodium hydroxide is very corrosive and can cause serious damage if it is splashed into the eyes. Use safety goggles to protect your eyes and wear plastic gloves. Have a solution of vinegar in water nearby to neutralise any splashes. Put the ice water in the ovenware bowl and while stirring continually, slowly add the sodium hydroxide. Try not to splash any of the solution. Wash off any spills with lots of water.
2. The solution will become very hot. Leave it to cool until it reaches approximately 30 °C and the outside of the bowl does not feel warm to the touch. This solution is called lye.
3. Place the 5 litre plastic bowl inside a pot or another bowl that contains boiling water. Melt the fat in the plastic bowl. Do not use an aluminium bowl, or any other type of metal bowl.
4. When the fat has melted, take the plastic bowl out of the hot water and let it cool down to approximately 35 °C so that you can hold the bowl without it feeling hot.
5. Line a flat cardboard box with a plastic rubbish bag.

6. Pour a thin stream of lye into the fat and stir continually. Then stir vigorously or use a hand-held blender. Do not stop stirring! You may need to stir for at least 10 minutes. You will know that you have the right consistency when drops from the stirrer do not sink back into the mixture. They should stay at the top.
7. At this stage, you can add colours, scents or textures to the mixture.
8. Pour this mixture carefully into the cardboard box. Leave it for a day and then score it with a knife. Cut it into blocks after another four days. Keep it for a month before using it.



Figure 10.4 Homemade soap

Project

Your teacher will arrange a field trip to a soap-manufacturing factory. Someone there will give a talk on how that factory produces its products, which starting materials they use, how they refine the soap to make it attractive to customers, and where they sell their product.

In groups, create a small business plan for a soap-manufacturing factory of your own. Include the following:

1. the chemistry involved in soap-making (5)
2. the additives, such as colouring and scents, that you will add to your soap (4)
3. the packaging and marketing that will be used (3)
4. suggestions for the types of markets where this soap can be sold. (Think about the type of people you wish to sell to, and where they would go to buy soap.) (3)

TOTAL: 15

Summary

- Saponification is the process of combining a fat and an alkali to form a soap and glycerol, according to the word equation: fat + alkali $\longrightarrow$ soap + glycerol.
- A molecule of soap has one hydrophilic end, which is attracted to water molecules, and one hydrophobic end, which is attracted to dirt and oil.
- Usually, sodium hydroxide or potassium hydroxide is used as the alkali in soap-making, and natural vegetable-based fats and oils are used for the fat.
- Small-scale soap manufacturers use the batch process, which involves boiling, separating, refining and pitching.
- Large-scale soap manufacturers use the continuous process, which involves splitting, mixing and cooling.

Glossary

hydroliser – a stainless steel tank used in industrial soap-making that allows for the continuous addition of starting products and removal of glycerol and fatty acids

hydrophilic – water-loving

hydrophobic – water-hating

kettle – stainless steel tank, used in soap-making, in which fats and alkalis are boiled (reacted) on a small scale

neat soap – the pure, liquid layer of soap that is obtained after the soap mixture undergoes pitching

nigre – layer of impure soap by-products (mostly water)

pitching – a process in small-scale soap manufacturing whereby the relatively pure soap is boiled with water to purify it even more

saponification – the process of combining a fat and an alkali to form a soap and glycerol

soft soap – soap that is made using sodium hydroxide as the alkali, and hence is more soluble in water than soaps made from other alkalis

Revision questions

1. The soap-making process produces a valuable by-product. What is this by-product?
A Fatty acids
B Soft soap
C Nigre
D Glycerol (1)
2. Which alkali produces a soap that is more water-soluble than others?
A Potassium hydroxide
B Sodium hydroxide
C Glycerol
D Calcium carbonate (1)
3. The part of the batch process of soap-making that separates the final product from the nigre is called:
A pitching
B mixing
C refining
D splitting. (1)
4. The key piece of equipment that is used in industrial soap manufacture is called the:
A kettle
B hydroliser
C mould
D nigre. (1)

5. Match the term with its correct meaning. Not all the meanings provided in the second column will be used.

Term	Meaning
a) Nigre	A: A process in industrial soap manufacturing in which liquid fat or oil is split into glycerol and fatty acids by means of super-heated water in a hydroliser
b) Hydroliser	B: Water-hating
c) Soft soap	C: A process in small-scale soap manufacturing whereby the relatively pure soap is boiled with water to purify it even more
d) Splitting	D: Stainless steel tank used in soap-making in which fats and alkalis are boiled (reacted) on a small scale
e) Glycerol	E: Industrial soap-making apparatus that allows for the continuous addition of starting products and removal of glycerol and fatty acids
f) Hydrophobic	F: The by-product of soap-making that is used in many other industrial products
	G: The step in which the alkali is introduced, as well as any additives, colourants or fragrances (scents), in the industrial preparation of soap
	H: Soap that is made using sodium hydroxide as the alkali
	I: Water-loving
	J: A layer of impure soap by-products

6. Outline the four key steps involved in soap production. Briefly explain what happens in each step.

(6)

(6)

TOTAL: 16

Unit
11

Oxidation and reduction

Objectives

- Write simple word equations.
- Define oxidation and reduction in terms of oxygen.
- Distinguish between physical and chemical changes.

Introduction

In the activities we have carried out so far in our high school science careers, we have encountered many changes. Some substances were dissolved in water and we saw some changes. Other substances were heated and again changes were observed.

Some of these changes could be reversed, for example, liquid water was heated to produce steam, and steam was cooled back to liquid water.

However, not all changes can be easily reversed. There are two types of changes, **physical changes** and **chemical changes**. Only one type is easily reversible.

Physical and chemical changes

Physical changes

A physical change is reversible and no new substance is formed. For example, when you

dissolve salt in water to get a salt solution, you can reverse this process by evaporation and you will obtain salt crystals.

All changes of state that we observed in previous units are physical changes and are reversible. This means we can change certain conditions to reverse the states. For example, when a liquid evaporates, you have added energy and forced a physical change to take place. There is a change in the state of matter (from a liquid to a gas).



Figure 11.1 A physical change takes place when a liquid evaporates.

Experiment 11.1

Aim: To observe physical changes

Materials: beaker or boiling tin, burner, tripod stand, wire gauze, ice

Procedure

1. Put the ice in a beaker with a little water and heat it over a burner. What happens to the ice?
2. Continue heating until no ice is seen. Do you think the temperature in the beaker is still the same?

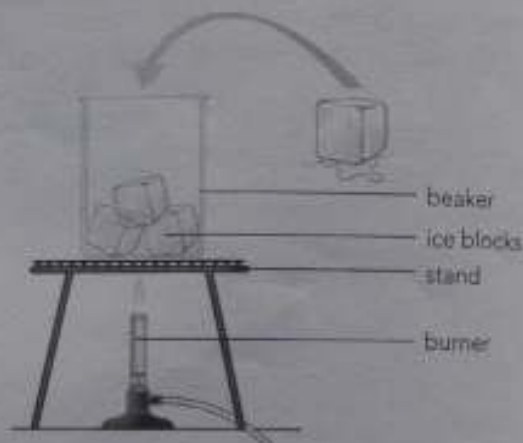


Figure 11.2 What happens when ice is heated?

3. Heat the contents of the beaker further and see what happens next. What does water turn to when heated to boiling point?

Results

The ice melts to a liquid and then boils away to water vapour.

Questions

1. Once the ice has melted and then evaporated, is it possible to reverse all these processes and produce ice once more? Give a reason for your answer.
2. Copy the diagram of the apparatus shown in Figure 11.2. Label the diagram.

If you step on a can and crush it, you have forced a physical change because the shape of the can has changed. You can use forces such as motion, temperature and pressure to cause physical changes.

Chemical changes

Chemical changes take place on a much smaller scale than physical changes. While some experiments show obvious chemical changes (such as a change in colour when something burns, or the release of heat or light), most chemical changes are not easily noticeable. For example, when iron rusts, you will only notice the change after a long time.



Figure 11.3 A chemical change takes place when iron rusts.

Experiment 11.2

Aim: To observe chemical changes

Materials: two test tubes, two beakers, two measuring cylinders, litmus paper, a pipette, spatula, stirring rod, dilute sulfuric acid, concentrated sulfuric acid, copper oxide, iodine solution, starch solution, water

Procedure

1. Copy and complete Table 11.1 as you perform the various experiments.
2. Your teacher will pour about 5 ml of concentrated sulfuric acid into a small beaker, and add it (drop by drop) to 10 ml of water in another beaker. Feel the outside of the second beaker with your hands.
3. Put a little dilute sulfuric acid into a test tube. Note the colour of litmus paper and drop it into the test tube with sulfuric acid. What happens to the litmus paper?
4. Using a spatula, place some copper oxide into a beaker. Add about 5 ml of dilute sulfuric acid, stirring the mixture with a glass rod. What do you observe? Throw away the contents and rinse the containers thoroughly.
5. Place a small lump of calcium carbonate into a test tube. Add about 5 ml of dilute sulfuric acid. What do you observe?
6. Put about 30 ml of starch solution into a test tube. Add 5 drops of iodine solution, drop-by-drop, to the starch solution. What changes do you observe?

Results

Record your observations in Table 11.1.

Table 11.1 Observing chemical changes

Substances mixed	Observation
1. Concentrated sulfuric acid and water	
2. Dilute sulfuric acid and litmus paper	
3. Copper oxide and dilute sulfuric acid	
4. Calcium carbonate and dilute sulfuric acid	
5. Starch solution and iodine solution	

Question

Can you reverse any of the changes that occurred in this series of experiments?

We will carry out further experiments to establish some characteristics of chemical changes.

Experiment 11.3

Aim: To demonstrate that a chemical change takes place when magnesium ribbon is burnt.

Materials: magnesium ribbon, sandpaper, tongs, burner, piece of paper

Procedure

1. Clean the ribbon with sand paper.
2. Examine a piece of magnesium ribbon and note its colour. Feel the ribbon.
3. Hold the ribbon using tongs and set it alight over a Bunsen flame.
Caution: Do not look directly at the flame as it is very bright and may cause harm to your eyesight.
4. Allow the ashes to drop onto a piece of paper. Note the colour of the ashes. Feel them.

Results

Magnesium ribbon is a shiny, silver metal. It burns with a bright white flame and leaves some white ashes.

Questions

1. What was the appearance of the ribbon before heating?
2. Describe what you observe when the magnesium burns.
3. Describe the appearance of the ashes.
4. Can you reverse this reaction somehow?



Figure 11.4 Burning magnesium ribbon

In Experiment 11.3, a chemical change occurred when magnesium ribbon was burnt. A new substance was formed, called magnesium oxide. A lot of heat was given off when magnesium oxide

was formed. This change is not reversible. You cannot get magnesium ribbon back from the ashes. A chemical change has occurred.

Experiment 11.4

Aim: To demonstrate that a chemical change takes place when iron and sulfur react when heated

Materials: boiling tube or strong test tube, clamp and stand, burner, magnet, piece of paper, spatula, iron filings, sulfur

Procedure

1. Using a spatula, mix equal amounts of iron and sulfur thoroughly. Can you separate this mixture?
2. Pour the mixture in a boiling tube and clamp the tube to a stand.
3. Heat the mixture until it starts to glow.
4. As soon as it starts to glow, remove it from the flame, turn the Bunsen burner off and wait for it to cool.

Results

A substance with different characteristics to that of either iron or sulfur forms in this reaction.

Questions

1. Were you able to separate iron from the mixture of iron and sulfur?
2. What was the colour of iron in the mixture?
3. Were you able to see sulfur powder after heating the mixture?
4. Was the magnet able to attract the iron after heating?
5. What is the colour of the new substance?

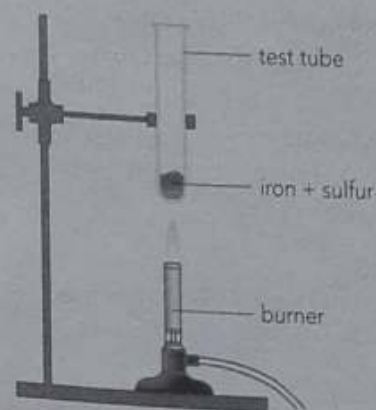


Figure 11.5 Forming a new substance from iron and sulfur

The solid substance in Experiment 11.4 looks different from the mixture of iron and sulfur. You cannot see the iron filings or powdered sulfur anymore. Even the magnet cannot attract the iron filings in the heated substance. A new substance must have been formed. A chemical change occurred when iron and sulfur were heated together.



Figure 11.6 Iron sulfide

Chemical changes involving fuels

Melting a sugar cube is a physical change because the substance is still sugar. However, burning a sugar cube results in a chemical change. The fire activates a reaction between the sugar and the oxygen in the air.

This is true for all fuels when they are burnt in oxygen. Fuels also release carbon dioxide gas when they are burnt in oxygen, as you will see in the next experiment.

Experiment 11.5

Aim: To demonstrate that carbon dioxide is produced when fuels burn.

Materials: jam jar with lid, metal spatula, lime water or bicarbonate indicator, maize meal, sugar

Procedure

1. Pour about 10 ml of lime water into a gas jar.
2. Burn maize meal on a metal spatula in a gas jar. Close the jar as soon as burning starts.
3. When burning stops, shake the lime water gently.
4. Repeat the procedure for sugar. Melt the sugar first before it burns.

Results

A gas is liberated when either sugar or maize meal is burnt in oxygen.

Questions

1. What happens to the lime water in each case?
2. Do all food substances produce the same amount of carbon dioxide?

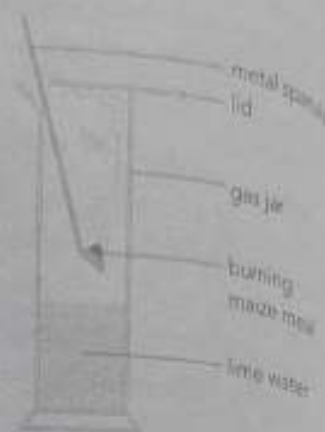
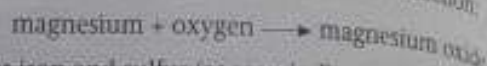


Figure 11.7 Burning maize meal

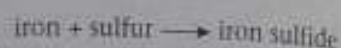
Writing word equations

Chemical changes are irreversible. New substances, which are totally different from the original substances, are formed. Chemical reactions are summarised by word equations. For example, magnesium burns in air (oxygen) to form magnesium oxide, as you saw in Experiment 11.3.

We can write a word equation for this reaction:



For iron and sulfur (as seen in Experiment 11.4), we have this word equation:



Experiment 11.6

Aim: To investigate the reaction between copper sulfate and iron

Materials: test tubes, iron nails, copper sulfate solution, sand paper

Procedure

1. Clean the iron nail with sand paper.
2. Half-fill a test tube with copper sulfate solution.
3. Add an iron nail and leave the set up until the next lesson.

Results

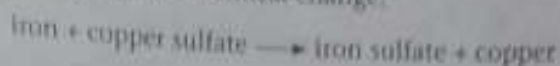
The appearance of the iron nail changes over time as it reacts with the copper sulfate solution.

Questions

1. What do you observe on the nail after some time?
2. What has happened to the colour of the copper sulfate solution?
3. Suggest what the brown substance on the iron nail could be.
4. Did a chemical change occur?

New substances were formed, which are different from those you started with. The deep blue colour of copper sulfate solution changes to light blue. Some brown particles of copper were pulled from the copper sulfate solution and were deposited

onto the iron nail. A chemical change took place. The change is not reversible. We can write a word equation for this chemical change:



Experiment 11.7

Aim: To investigate the reaction between copper sulfate and sodium carbonate

Materials: test tubes, dropper, copper sulfate solution, sodium carbonate solution

Procedure

1. Quarter-fill a test tube with copper sulfate solution.
2. Using a dropper, add a few drops of sodium carbonate solution to the copper sulfate solution.

Results

The appearance of the copper sulfate solution changes as it reacts with the sodium carbonate solution.

Questions

1. Did you observe any colour changes?
2. Does a new substance form?
3. What method of separation can you use to separate the products?

We can write a word equation for this reaction:
copper sulfate + sodium carbonate $\longrightarrow$ copper carbonate + sodium sulfate

Solid copper carbonate has a bright green colour.

Table 11.2 gives a summary of the characteristics of chemical and physical changes.

Table 11.2 Characteristics of physical and chemical changes

Physical change	Chemical change
Mixtures formed	Compounds formed
No new substance is formed	Always produces a new substance
Generally reversible	Generally not easily reversible
Properties of substances do not change	Properties of new substances completely different from those of reactants

Oxidation and reduction

Reactions with oxygen are usually called **oxidation** reactions and reactions in which oxygen is lost are called **reduction** reactions. The most common oxidation and reduction reactions are burning, rusting and respiration.

Long ago, chemists did not have much knowledge about oxidation and reduction. They used these terms to describe reactions that involved oxygen (in burning, rusting and respiration) and hydrogen (in water formation) only. Nowadays, **redox** processes (reduction and oxidation) are understood to include reactions that involve **electron transfer**. You will learn more about this in Form 3.

Oxidation and reduction in terms of oxygen

A substance is oxidised when it gains oxygen.

Sulfur + oxygen gas $\rightarrow$ sulfur dioxide gas	Sulfur is oxidised to sulfur dioxide by gaining oxygen. A substance is reduced when it loses oxygen.
Lead oxide + hydrogen gas $\rightarrow$ water + lead	Lead oxide is reduced by losing oxygen. Hydrogen is oxidised to water by gaining oxygen.

In redox reactions, the reduction and oxidation processes usually occur at the same time: one substance is oxidised and another substance is reduced, as seen in the second example where oxygen is added to hydrogen (oxidation) to

form water whilst oxygen is removed from lead (reduction) to form lead.

Exercise 11.1

- Say whether the underlined reactant in each reaction is oxidised or reduced.
 - magnesium + oxygen gas $\rightarrow$ magnesium oxide
 - hydrogen peroxide + carbon $\rightarrow$ carbon monoxide + hydrogen gas
 - calcium + oxygen gas $\rightarrow$ calcium oxide
- Say which reactant undergoes oxidation and which undergoes reduction in each of the following reactions:
 - iron + water $\rightarrow$ iron oxide + hydrogen gas
 - sulfur dioxide + sodium $\rightarrow$ sodium oxide + sulfur

Summary

- Physical changes are easily reversible.
- No new substance is formed in a physical change.
- All changes of state are physical changes and are reversible. You can change certain conditions to reverse the states.
- Chemical changes are not easily reversible.
- A new substance is formed in a chemical change.
- While some experiments show obvious chemical changes (such as a change in colour), most chemical changes are invisible.
- When fuels are burnt in oxygen, a chemical change takes place. Fuels also release carbon dioxide gas when they are burnt in oxygen.
- Word equations summarise chemical reactions which have taken place.
- Reactions with oxygen are usually called oxidation reactions and reactions in which oxygen is lost are called reduction reactions. The most common oxidation and reduction reactions are burning, rusting and respiration.

Glossary

chemical change - non-reversible process that takes place when substances combine or break up to form new different substances

electron transfer - movement of electrons from one atom to another

oxidation - reaction involving oxygen

physical change - reversible process which takes place when substances mix without forming

new substances

redox - chemical reaction in which at least one reactant is oxidised and another reduced

reduction - reaction that involves the loss of oxygen from at least one reactant

reversible - capable of going back to an original form

Revision questions

1. In a chemical change:
 - A the change cannot be seen
 - B a new substance is formed
 - C the reactants keep their properties
 - D one product is formed.
2. Which type of change can be reversed? (2)
 - A Chemical change
 - B Physical change
 - C Both physical and chemical changes
 - D Neither physical nor chemical changes
3. Which products are formed when iron and sulfur are heated? (2)
 - A Iron sulfide
 - B Magnesium oxide
 - C Iron and sulfide (no reaction occurs)
 - D Iron sulfide and sulfur dioxide
4. When sodium carbonate is added to a solution of copper sulfate, the copper sulfate solution changes colour from: (2)
 - A blue to green
 - B blue to clear
 - C clear to blue
 - D clear to green.
5. When magnesium metal burns in oxygen gas, magnesium oxide is formed. What kind of reaction is this? (2)
 - A Reduction reaction
 - B Physical change
 - C Oxidation reaction
 - D None of the above
6. For each of the following, say whether the change is physical or chemical: (2)
 - a) A solid substance is dissolved in a liquid and gas bubbles form.
 - b) Two clear liquids are mixed and they form a cloudy blue solution.
 - c) A glass window shatters.
 - d) A piece of paper burns.
 - e) Water freezes at 0°C .
 - f) Hydrogen gas is liberated when hydrochloric acid reacts with a metal.
 - g) A slice of apple is left on a plate and eventually turns brown.
 - h) You cut a piece of paper into small pieces.

7. Sulfur burns in oxygen to form sulfur dioxide gas. (1)
 - a) What is the colour of sulfur?
 - b) Is this a physical change or a chemical change?
8. State two properties of chemical changes. (1)
9. Complete the sentences: (2)
 - a) When elements combine in a chemical change, they form _____.
 - b) When a fuel is burnt in oxygen, _____ gas is formed.
 - c) Reduction is a reaction that involves the _____ of oxygen from at least one reactant.
10. Complete Table 11.3, which tabulates the differences between physical and chemical changes. (1)

Table 11.3 Differences between physical and chemical changes

	Physical change	Chemical change
Is a mixture or a compound formed?		
Is a new substance formed?		
Is the reaction reversible?		
What happens to the properties of the reactants?		
Is there a difference in mass between the products and reactants?		

(10)

TOTAL: 35

Objectives

- Define complete and incomplete combustion of fuels.
- List the products of complete and incomplete combustion of fuels.
- Describe the effects of burning fuels.

Introduction

Fuels play an important role in our homes, industries and agriculture. Examples of fuels include those that are used to make our vehicles and aeroplanes move, those we use to light and heat our homes, and those that are used to run machinery in factories. This unit looks at the different types of fuel, how they burn and what the effects of their combustion are.



Figure 12.1 Fuels burn to give off energy.

Complete and incomplete combustion

When substances burn in air, they combine chemically with oxygen. Fuels, such as petrol and diesel, burn in the presence of oxygen to form carbon monoxide and carbon dioxide gases. Carbon monoxide is a poisonous, colourless, odourless and tasteless gas.

Elements basically burn in oxygen to form oxides:

- Magnesium + oxygen gas $\rightarrow$ magnesium oxide
- Sulfur + oxygen gas $\rightarrow$ sulfur dioxide

Activity 12.1

For this activity, you will need two methylated spirit burners, some matches and a pair of pliers.

1. Adjust the wick lengths of the two spirit burners so that the wick length of one burner is less than 5 mm, and that of the other is at least 1 cm.
2. Light both spirit burners.
3. Observe height, colour and intensity of the flames produced by both burners.
4. Record any other observations.

In Activity 12.1, which burner produced the brighter flame? Flames are brighter and hotter the more oxygen there is. The gas we use in Bunsen burners contains carbon and hydrogen. These burn to release carbon dioxide and the oxide of hydrogen, which is water. The air hole on the burner controls the amount of oxygen that mixes with the gas. So, we say that when fuels burn, the carbon is oxidised to carbon dioxide and the hydrogen is oxidised to water.

Complete combustion occurs when a fuel burns in plenty of oxygen to produce carbon dioxide and water. More heat is given off in complete combustion than in incomplete combustion. **Incomplete combustion** occurs when there is insufficient oxygen to complete

...reduction. Carbon monoxide and soot are the products of incomplete combustion, and only a small amount of energy is produced. Carbon

monoxide can cause suffocation. Carbon that has not been burnt is called soot (a black substance). Soot is a pollutant as it causes smog.

Experiment 12.1

Aim: To demonstrate that water is a product of complete combustion

Materials: burner, burner stand, evaporating dish, beaker, spatula, sugar, cold water, cobalt chloride paper

Procedure

1. Put a few spatulas of sugar into an evaporating dish and heat it over a flame. When it turns brown, light the sugar to burn it. Remove the burner as soon as the sugar starts burning.
2. Hold a clean beaker, half-filled with cold water, slightly above the burning sugar.
3. Do you see any liquid forming at the bottom of the beaker? Test this liquid using cobalt chloride paper. Record your observation.

Results
When fuels burn, they produce water.

Question
What do fuels produce when they burn? Discuss this question in groups and consult your teacher for help.

Experiment 12.2

Aim: To demonstrate incomplete combustion

Materials: two beakers, Bunsen burner, thermometer, clock, burner stand, water

Procedure

1. Pour equal amounts of water into the two beakers and place them on stands.
2. Adjust the sleeve on the Bunsen burner so that the air hole is closed. Record the initial temperature of the water. Heat the water until its temperature has increased by 15°C . Record the time that this takes.
3. Note the colour of the flame.
4. Examine the underside of the beaker by wiping it with your finger.
5. Record the temperature of the water in the second beaker. Heat the water with the Bunsen burner with the air hole wide open. Record the time it takes for the temperature of the water to rise 15°C .
6. Repeat steps 3 and 4.

Results

You should have seen a luminous flame when the air hole was closed. With the air hole in this position, the small amount of heat that was produced meant that it took much longer for the water temperature to rise by 15°C . You should also have noticed a substance on the underside of the beaker. When the air hole was closed, the gas could not mix with air before it burnt. The result was incomplete combustion.

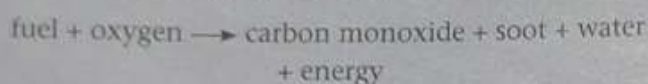
Questions

1. Which demonstration produced the most heat – when the air hole was open or when it was closed?
2. What did you observe when you wiped the underside of the beakers with your finger?
3. Were the flames the same colour when the air hole was open and when it was closed?

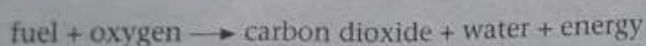
Word equations for complete and incomplete combustion

As you saw in Experiment 12.2, when the air hole is closed on the burner, the gas cannot mix with the air before it burns. The flame is not very hot. The temperature of the water after 5 minutes of heating is low.

We can write an equation for incomplete combustion of the gas (fuel) as follows:



With an open air hole, the gas mixes with the oxygen before it burns. The flame is very hot and clean. There is complete combustion. We can write an equation for complete combustion as follows:



More heat energy is released when carbon is completely burnt.

The effects of burning fuels

With the growing demand for fuel, there is a danger that some sources of energy will run out soon. These **non-renewable** resources include fossil fuels. However, some fuel sources are **renewable**, including the sun, wind, waves, geothermal energy, hydroelectric power and biomass. Maize, sunflowers and sugar cane are cash crops that supply both food and energy. We say that they have biomass (this refers to the amount of a living material something contains).

Deforestation

Deforestation refers to the uncontrolled cutting down of trees. Deforestation upsets the balance of nature. Plant and animal habitats are destroyed by deforestation, and wind and rain cause soil

erosion in areas where deforestation has occurred. The eroded soil causes siltation in rivers and lakes. Siltation is a type of water pollution. It is caused by particles of soil, sand and clay that enter rivers and lakes. Siltation is made worse by some farming practices, construction sites, deforestation, and almost any type of human activity. If deforestation is not controlled, some parts of the country could become deserts. To combat this problem, we need to plant trees where they have been cut down.



Figure 12.2 Large tracts of land are destroyed over time due to deforestation. Not only is the resource (wood) depleted, but the habitats of many plant and animal species are destroyed too.

Effects of the by-products of burning coal and petroleum

Burning hydrocarbons such as fossil fuels releases tonnes of carbon dioxide into the atmosphere. Increased carbon dioxide in the atmosphere is one of the causes of **global warming**. Carbon dioxide absorbs heat that is reflected from the Earth (which should escape from the Earth's atmosphere), and the Earth is thus warmed more than usual. Other waste gases in the atmosphere that are created by burning fuels contribute to global warming. These gases are also hazardous to our health.

Pollution

Oil spills occur when accidents happen during the transportation of unrefined fossil fuels. This results in pollution of water and consequent problems with bird and marine life. For example, the feathers of birds become clogged, and aquatic animals die as a result of a lack of oxygen.



Figure 12.3 Oil spills spell disaster for marine life.

Incomplete combustion produces carbon monoxide and soot. Carbon monoxide is a poisonous gas. It lowers the blood's ability to carry oxygen around the body. This can cause a person to suffocate and die. Soot blackens buildings, pollutes the air and contributes to the smog in towns and cities.

The complete and incomplete combustion of some fuels can also result in emissions of sulfur dioxide and various nitrogen oxides. These combine with water vapour in the atmosphere to form **acid rain**. Acid rain pollutes water systems and damages buildings and vegetation.



Figure 12.4 A forest damaged by acid rain.



Figure 12.5 Air pollution has direct – and harmful – consequences for human health.

Project

In groups, discuss how you could design and implement an environmental awareness campaign. You will need to prepare a short presentation and write a report to hand in.

In your report, you will have to:

1. describe the issues that you feel need more public awareness
2. state how our community is contributing to various forms of pollution
3. think of ways in which your community can make changes to avoid polluting the air and environment
4. suggest alternatives for fuel (for heat and cooking) that would be better to use for the environment.

Marks will be awarded according to the following:

1. Written report:
 - a) Structure (3)
 - b) Content/conclusion (15)
 - c) Neatness (3)
 - d) List of resources consulted (3)
2. Oral presentation (6)

TOTAL: 30

Summary

- Fuels, like petrol and diesel, burn in the presence of oxygen to form carbon monoxide and carbon dioxide gases.
- Complete combustion occurs when a fuel burns in the presence of plenty of oxygen to produce carbon dioxide and water. More heat is given off in complete combustion than in incomplete combustion.
- Incomplete combustion occurs when a fuel burns in insufficient oxygen. Carbon monoxide, soot and water are the products of incomplete combustion, and only a small amount of energy is produced.
- We can write an equation for incomplete combustion of the fuel as follows:
$$\text{fuel} + \text{oxygen} \rightarrow \text{carbon monoxide} + \text{soot} + \text{water} + \text{energy}$$
- We can write an equation for complete combustion as follows:
$$\text{fuel} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy}$$
- More heat energy is released when carbon is completely burnt.
- Non-renewable resources, such as fossil fuels, will run out one day. We are dependent on these non-renewable resources, and therefore need to find alternative sources of energy for our future fuel consumption.
- Renewable resources include the sun, wind, waves, geothermal energy, hydroelectric power and biomass.
- Deforestation refers to the uncontrolled cutting down of trees. Deforestation upsets the balance of nature. Plant and animal habitats are destroyed by deforestation, and wind and rain cause soil erosion in areas where deforestation has occurred.
- Oil spills cause great damage to the environment and the organisms that live there.
- Burning hydrocarbons, such as fossil fuels, releases tonnes of carbon dioxide into the atmosphere. This contributes to global warming.
- Soot blackens buildings, it pollutes the air and contributes to the smog in towns and cities.
- Sulfur dioxide and various nitrogen oxides (which are sometimes produced during fuel combustion) combine with water vapour in the atmosphere to form acid rain.
- Acid rain pollutes water systems and damages buildings and vegetation.

Glossary

acid rain – a form of precipitation that is more acidic than usual, and therefore has harmful effects on the natural and built environments

complete combustion – when a fuel burns in plenty of oxygen to produce carbon dioxide and water

deforestation – uncontrolled cutting down of trees

incomplete combustion – when a fuel burns in insufficient oxygen to give off carbon monoxide, water and soot

global warming – gradual heating of the Earth that is caused by elevated levels of carbon dioxide gas in the atmosphere

non-renewable – resources that cannot be regenerated or produced again

renewable resources – that can be regenerated or produced again

Revision questions

- What are the products of complete combustion of a fuel?
- Carbon dioxide and water
 - Soot and water
 - Carbon monoxide and soot
 - Carbon monoxide and water
- Which gas is used up during burning?
- Nitrogen
 - Oxygen
 - Carbon dioxide
 - Hydrogen
- What are the products of burning petrol in a limited supply of air?
- Carbon monoxide, carbon dioxide and oxygen
 - Carbon dioxide, water vapour and heat
 - Carbon monoxide, water vapour and soot
 - Carbon dioxide, water vapour and oxygen
- Which of the following is a non-renewable source of energy?
- Wood
 - Sun
 - Water
 - Coal
- Answer the following questions about combustion.
- What is complete combustion? (2)
 - Write a word equation for the complete combustion of coke. (3)
 - Name a dangerous gas that is produced when fuels undergo incomplete combustion. (1)

6. Figure 12.6 shows the findings for an experiment on fuels.

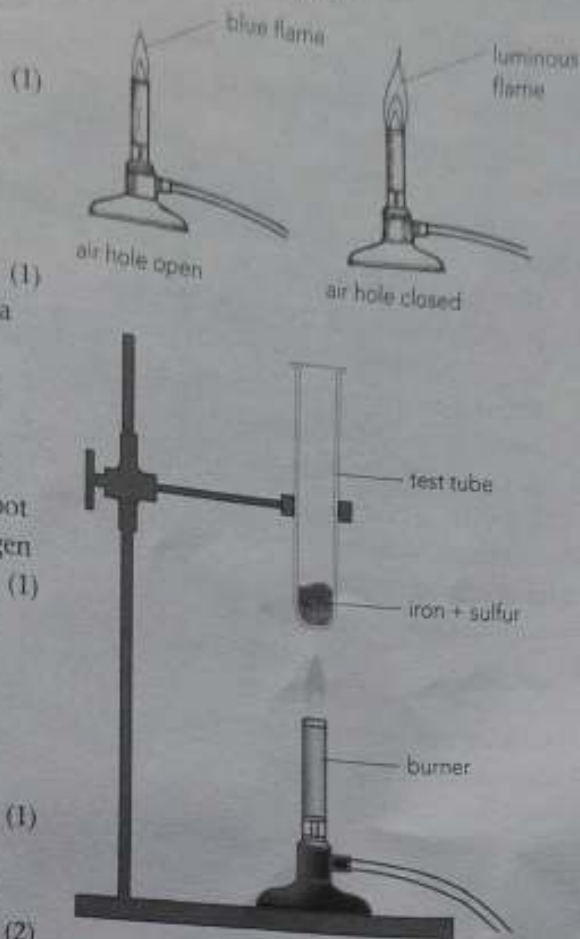


Figure 12.6 Observations for the experiment on fuels

- Name the type of combustion that is described by the burner on the right. (1)
 - Explain why there is a difference between the colours of the two flames. (2)
 - Which burner produces more energy: the burner on the left or the burner on the right? (1)
7. Explain how incomplete combustion contributes to air pollution. (2)

TOTAL: 16

End-of-topic revision test

- Micro-organisms in water must be killed to make the water safe to drink. In which two ways can this be achieved?
 A Filtration or boiling
 B Boiling or chlorination
 C Filtration or sedimentation
 D Boiling or evaporation (1)
- The role of a flocculant in water purification is that of:
 A killing micro-organisms
 B allowing suspended particles to remain suspended, so that removal is easier
 C filtering large particles before the water is settled in the sedimentation tanks
 D causing large solid particles to clump together, so that removal is easier (1)
- A solution contains 24 ml of solute and 82 ml of solvent. What is the percentage concentration of the solution?
 A 17.1% B 29.3%
 C 22.6% D 58.6% (1)
- When a sodium hydroxide solution is added to dilute hydrochloric acid in a beaker, which of the following does NOT happen?
 A The solution changes from alkaline to neutral
 B The acid is neutralised.
 C The solution changes from acidic to neutral.
 D The volume of water increases. (1)
- The type of salt that is produced when nitric acid reacts with a base is called:
 A nitrogen sulfate B a nitrate
 C a sulfate D a chloride (1)
- The process of saponification can be described by the following word equation:
 A fat + alkali $\longrightarrow$ soap + glycerol
 B fuel + oxygen $\longrightarrow$ carbon dioxide + water
 C soap + fat $\longrightarrow$ glycerol + alkali
 D fuel + oxygen $\longrightarrow$ alkali + water (1)
- Which of the following characteristics is NOT true for physical changes?
 A The properties of the substances remain the same.
 B The reaction is reversible.
 C No new substance is formed.
 D The mass of the product is different to that of the reactants. (1)
- Which of the following processes is an example of a physical change?
 A Photosynthesis
 B Rusting
 C Evaporation
 D Combustion (1)
- Which of the following changes is NOT an example of a chemical change?
 A Vinegar combines with baking soda to form carbon dioxide gas.
 B A piece of iodine undergoes sublimation.
 C A candle burns to give light.
 D Bread rises in an oven. (1)
- What are the products of incomplete combustion of a fuel?
 A Carbon dioxide and water
 B Soot and water
 C Carbon monoxide and soot
 D Carbon monoxide and water (1)
- Sugar cane is processed at a sugar mill to extract sugar. Draw a flow diagram that shows the steps involved in order, using the key words provided below. For each step, explain briefly what happens.
 evaporation; refining; shredding; separation; clarification; filtration (12)
- A solution of sodium hydroxide has a concentration of 2 moles per litre.
 a) Explain the term 'concentration'. (1)
 b) How many particles of sodium hydroxide are there in one litre of the solution? (2)
 c) One mole of sodium hydroxide has a mass of 40 g. How many grams of sodium hydroxide are there in 250 ml of the solution? (2)

- d) A 200 ml, 1:10 dilution of the solution is made. Prove by calculation that the new solution has a concentration of 0.2 moles per litre. (3)

13. A learner finds two chemical substances in a laboratory storeroom. One of the substances (X) has a pH of 2.5 when dissolved in water. The other substance (Y) is a white powder. When they are mixed together, X and Y react to give off a gas product.

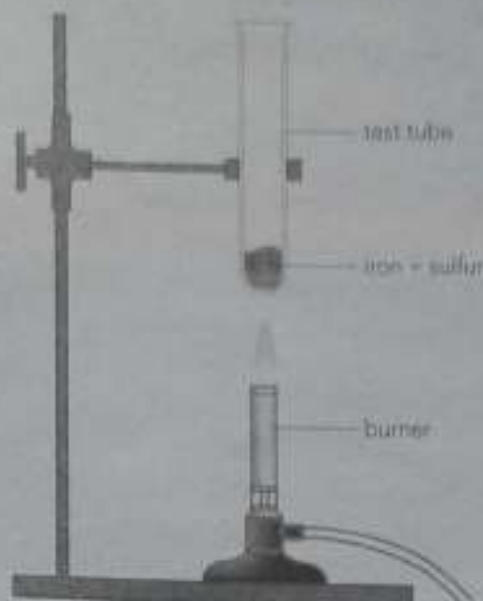
- Which is a possible identity for substance X: Calcium oxide, hydrochloric acid or magnesium sulfate? (1)
- Y is a carbonate. Identify the gas that is formed in the reaction between X and Y. (1)
- Apart from the gas product, name two other products formed in reaction between X and Y. (2)
- State what the pH values of all the products of the reaction between X and Y will be. (1)

14. The industrial preparation of soap involves three main steps. Identify which three of the following processes or steps are part of this type of preparation, and briefly describe what happens in each step.

Separation	Pitching
Cooling	Mixing
Boiling	Splitting

(6)

15. Iron and sulfur powder are heated in an experiment as shown below.



- Will any gas be liberated as the reaction proceeds? (1)
 - Is this an example of a physical or chemical change? (1)
 - Will the product(s) have the same properties as the reactants? (1)
 - What colour will the product be? (1)
16. You are given a burner, a burner stand, an evaporating dish, a beaker, a spatula, sugar, cold water and cobalt chloride paper. Describe an experiment that you could perform using these apparatus that demonstrates that water is a product of complete combustion. State what the expected results would be. (5)

TOTAL: 50

Unit
13

Data presentation

Objectives

- Construct a straight line graph from appropriate data.
- Interpret straight line graphs.

Introduction

Different **graphs** allow a scientist to present different types of **data** or **information** in a way that is easy for people who are unfamiliar with the data, to understand. By now, you should have learnt how to record data as well as how to represent data using a bar graph.

Activity 13.1

The average rainfall in millimetres (mm) over a number of months is represented in Table 13.1.

Table 13.1: Average rainfall from January to June, in millimetres

Month	Average rainfall (in mm)
January	40
February	50
March	45
April	35
May	20
June	10

Draw a bar graph to represent this data. Place the month on the x-axis and the rainfall in millimetres on the y-axis.

Line graphs

A line graph, like a bar graph, shows one set of data in relation to another and is also plotted on a set of axes.

Drawing a straight line graph

As mentioned, a **straight line graph** shows one set of data in relation to another. We construct the axes in the same way as we would for a bar graph, choosing an appropriate scale. When deciding which axis should show which set of data, the measured data (the data that has been collected) will always be presented on the vertical (y) axis.

Next, you will make dots in line with the corresponding data. This is called **plotting** the points of data.

Finally, you will join the dots using a straight line between adjacent dots.

Straight line graphs

In physics, we often need to know how two variables influence each other. A lot of experiments are done to investigate such relationships. The following section introduces variables and discusses how variables are investigated to produce line graphs, with particular emphasis on the straight line graph.

Investigating the relationship between voltage and current in a series circuit

Consider the series circuit in Figure 13.1 on the next page.

when we connect cell 1 to the bulb, it is likely to only burn dimly. When we move the flying lead to include cell number 2, the bulb will burn brighter. The brightness increases when we include cells 3

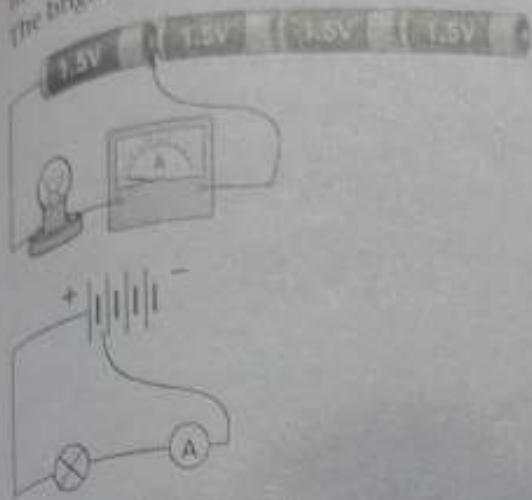


Figure 13.1 A series circuit

and 4. There is clearly a relationship between the brightness of the bulb and the number of cells. In physics, the number of cells is one variable and the brightness is the other variable. Notice that the brightness of the bulb is dependent on the number of cells in the circuit. The number of cells is the independent variable while the brightness is the dependent variable.

We can also represent each cell by its voltage, which is 1.5 V. A voltage drives a current through the circuit and the size of current determines the brightness of the cell. Instead of labelling the variables 'number of cells' and 'brightness of bulb', we can investigate the relationship between the voltage of the battery and the current it pushes through the circuit by completing Experiment 13.1.

Experiment 13.1

Aim: To investigate the relationship between voltage and current in a series circuit

Materials: 4 cells, a resistor, an ammeter, connecting wire

See Figure 13.2 where the bulb has been replaced by a nichrome wire.

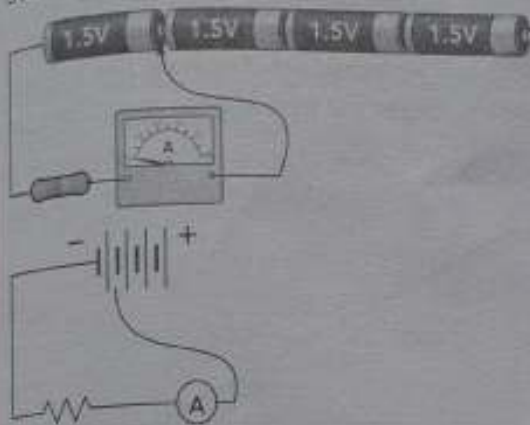


Figure 13.2 Setup of experiment

Procedure:

1. Connect a circuit as shown in Figure 13.2.
2. Record the current for zero voltage, which is zero.
3. Include one cell and record the current flowing by copying and completing the table below.

No. of cells	0	1	2	3	4
Voltage (V)					
Current (A)					

4. Then include two cells and record the current flowing.
5. Continue by increasing by one cell at a time and recording the corresponding currents until all four cells are connected.
6. Plot a graph of voltage versus current.

Results:

Voltage (V)	0	1.5	3.0	4.5	6
Current (A)	0	0.2	0.4	0.6	0.8

A plot of these results yields a straight line graph as shown in Figure 13.3.

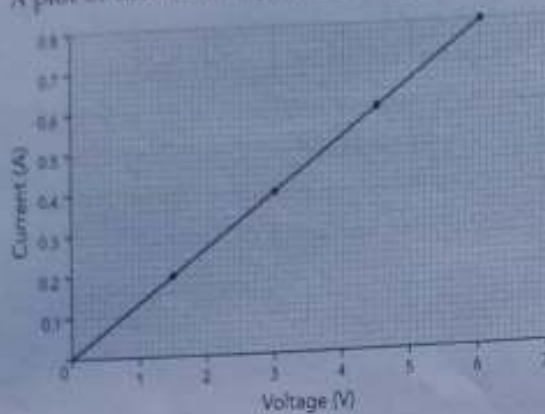


Figure 13.3 Current versus voltage graph

Interpreting a straight line graph

With such a straight line, we can conclude that the voltage is directly proportional to the current flowing. If the voltage increases, current also increases proportionately. The gradient of such a graph often has a physical meaning. In Experiment 13.1, the gradient gives a measure of the resistance of the bulb. In physics, there are other pairs of variables that will also give straight line graphs. These graphs are all interpreted the same way.

Inverse proportionality

Sometimes the relationship between variables is such that when one variable increases, the other

decreases. The product of these variables will always be a constant. Suppose variables P and Q are inversely proportional. A straight line graph of these variables will look like the one shown in Figure 13.4.

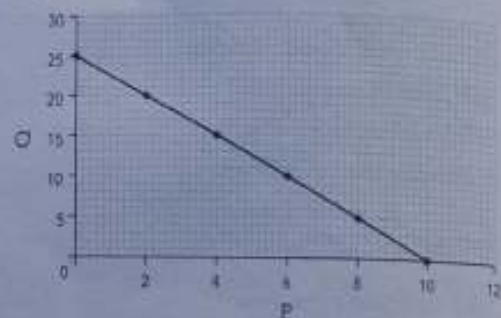


Figure 13.4 A graph showing the inversely proportional relationship between P and Q

Activity 13.2

An experiment was done to investigate the relationship between the distance travelled and the time taken for a cart to move along a linear track. The following data was collected and analysed.

Table 13.2 The distance travelled by the cart

Time (s)	0	2	4	6	8	10
Total distance (m)	0	5	10	15	20	25

1. Identify the variables being investigated.
2. Plot a graph of distance travelled against time.
3. What is the relationship between the variables?
4. What does the gradient of this graph tell us about the cart?

Summary

- Using graphs is a way of presenting information in a way that makes the information easy to understand and compare.
- Straight line graphs show one set of information in relation to another in the form of a line between two axes.
- A straight line graph is best used to show progression of information.

Glossary

data – information collected using specific methods for a specific purpose

graph – a way of presenting data or information in the form of diagrams

information – re-organised or processed data that can be used to make decisions

plotting – making points on a graph to represent collected data

straight line graph – a diagram where data is presented in a line form.

Revision questions

1. For the following sets of information, state whether a bar graph or line graph would be better to represent the information.
 - a) The number of different types of insects found in the school garden (1)
 - b) The speed of a car, measured every 10 minutes (1)
 - c) Test scores for a Science test for ten learners (1)
 - d) The number of learners from different countries in an International School (1)
 - e) The temperature of water that is being boiled over a five-minute period (1)
2. If a learner walks at a steady speed, then the more time he takes, the greater the distance he walks. Table 13.3 gives corresponding times and distances of the learner walking at a steady speed.

Table 13.3 Time taken to for the learner to walk 50 m

Time (s)	0	1	2	3	4	5
Total distance (m)	0	10	20	30	40	50

Draw a straight line graph to present the information.

(5)
TOTAL: 10

Unit
14

Measurements

Objectives

- Convert units.
- Measure mass of a liquid.
- Measure the volume of an irregular object.
- Determine the thickness, volume and mass of small objects.
- Calculate density.

Introduction

Taking measurements is an integral part of all scientific investigations. Accuracy in taking these measurements is vital in order to make sure that the findings of investigations and experiments are correct. In order to make measurements understandable to anyone who reads the information, we use *Système International* (SI) units of measure, such as metre for distance and kilogram for mass. You were introduced to these units in Form 1.

Table 14.1 shows the quantity being measured, the scientific (SI) unit for that quantity and the most commonly used sub-unit.

Table 14.1 SI units for common quantities

Quantity	SI unit	Commonly used unit	Other units
Length	metre (m)	centimetre (cm) = $\frac{1}{100}$ m	millimetre (mm) = $\frac{1}{1000}$ m
Mass	kilogram (kg)	gram (g) = $\frac{1}{1000}$ kg	milligram (mg) = $\frac{1}{1000}$ g
Time	second (s)	hour	day, week, month, year
Temperature	degrees Celsius ($^{\circ}\text{C}$)		
Volume of solids	cubic metre (m^3)	cubic centimetre (cm^3)	cubic millimetre (mm^3)
Volume of liquids	litre (L)	millilitre (ml) = $\frac{1}{1000}$ L	

Multiples of SI units

In learning and doing science in the classroom, we often measure small lengths, small masses and small volumes. Most of the times, therefore, we need to use multiples of SI units. For instance, you have measured length in millimetres and centimetres before. These are multiples of the metre. The question is, how do we get millimetres from a metre or centimetres from a metre?

To indicate the multiples of SI units, we use specific prefixes such as kilo-, centi- and milli-. These can be seen in Table 14.2.

Table 14.2 Prefixes for the most common multiples of SI units

Prefix	Abbreviation	Factor relative to the base unit
kilo	k	1 000
hecto	h	100
deca	da	10
deci	d	0.1
centi	c	0.01
milli	m	0.001

Using this information, to convert 1 metre to kilometres, you would divide the value in the base unit, metre, by the factor of the multiple, 1 000 (for kilo) to get to the value.

Example 14.1

Convert 4 500 metres to kilometres.
 $4\,500\text{ m} \div 1\,000 = 4.5\text{ km}$

Exercise 14.1

Convert the following values to the stated unit of measure.

1. 1 m to centimetres
2. 5 000 grams to kilograms
3. 5 cm to millimetres
4. 10 kg to milligrams
5. 500 litres to kilolitres

Time conversion

Unlike the other SI units, time works in multiples of 60 rather than 10. There are 60 seconds in a minute and 60 minutes in an hour. However, when it comes to fractions of a second, like milliseconds, the convention in Table 14.2 is used. In other words, there are 1 000 milliseconds in 1 second.

Activity 14.1

1. Using a stopwatch, measure how long it takes a classmate to run 400 m.
(Note: the stopwatch measures time in seconds as well as centi-seconds.)
2. Convert the time taken to run the distance to minutes.

What is mass?

Mass is a scientific measure of the amount of matter an object is made up of. No matter where you are at given moment in time, mass is constant. So, whether you're walking to the shop to stock up on food or bouncing around on the moon, your mass is the same.

Some key points:

- Mass can never be zero. What we mean by this is that everything in the universe has mass. If it didn't it simply wouldn't exist.

- Mass is not related to gravity. However, when gravity acts on mass, we end up with weight. Weight is a force.
- The SI unit of mass is the kilogram. Smaller masses are measured in sub-units of the kilogram, which is the gram and milligram.

Mass is measured using a balance. In science it is common to hear people call a balance a scale. A balance works by comparing a known mass with an unknown mass.

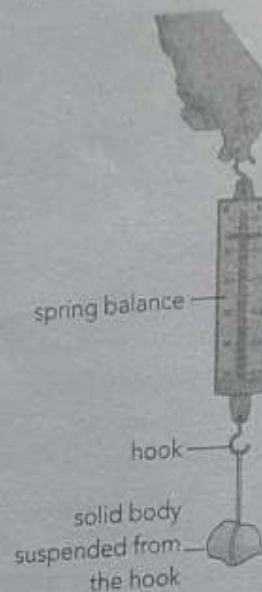


Figure 14.1 A spring scale



Figure 14.2 A balance scale

Weight is a form of measurement that is dependent on gravity and, unlike mass, your weight can vary depending on where you are in the universe.

Formula: Weight (W) = Mass (m) multiplied by gravitational acceleration (g)
($W = mg$)

Experiment 14.1

Aim: To measure the mass of a liquid

Materials: a beaker, water, a balance

Procedure:

1. Place the empty beaker onto the balance and measure its mass.
2. Take the beaker off the balance and fill it with water.
3. Place the full beaker back onto the balance and take the measurement again.
4. Subtract your first reading from your second reading.

Result:

The difference between the two readings is the mass of the water. This method can be used to measure the mass of any liquid.

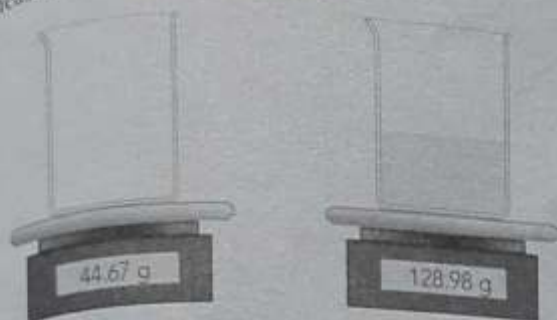
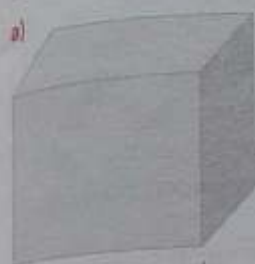


Figure 14.3 Measuring the mass of a liquid

Measuring the volume of a regular and an irregular object

It is easy to find the volumes of regular objects such as the ones shown below. For regular objects, a ruler or measuring tape can be used to obtain its dimensions, and a simple calculation can be used to determine its volume.



$$\text{Volume} = l \times w \times h$$



$$\text{Volume} = \pi r^2 h$$

(where r is radius and h the height.)

Figure 14.4 a) a cube b) a cylinder

Table 14.3 Calculations of regular objects

Shape	Calculation
Cube	side $\times$ side $\times$ side
Rectangular prism	height $\times$ breadth $\times$ length
Cylinder	$\pi \times \text{radius}^2 \times \text{height}$

Exercise 14.2

Calculate the volume of the following objects.

1. A book with a height of 5 cm, a length of 30 cm and a breadth of 15 cm
2. A cooldrink can with a radius of 5 cm and a height of 20 cm
3. A die with a side length of 2 cm

The unit of volume is the cubic metre (m^3), cubic centimetre (cm^3) or cubic millimetre (mm^3). The volume of a liquid is measured using a measuring cylinder, burette or pipette. We measure the volume of liquids in litres (L) or millilitres (ml).

Table 14.4 Conversion of liquid volumes

Volume of a liquid	Converted value
1 litre (1 L)	1 000 millilitres (1 000 ml)
1 millilitre (1 ml)	1 cubic centimetre (1 cm ³)
1 000 litres (1 000 L)	1 cubic metre (1 m ³)

Irregular objects are objects such as a stones, a queen's crown and other objects. Go to your

library and find out who Archimedes was and what he was famous for.

You can measure the volume of an irregular object by the amount of space the object occupies. The volume of irregular objects can be measured by calculating the amount of water displaced, for example, a stone immersed in a measuring cylinder.

Experiment 14.2

Aim: To measure the volume of an irregular object

Materials: measuring cylinder, water, irregular object (a stone or toy), string

Procedure:

1. Tie the string around the object.
2. Place the object into the measuring cylinder, with the string hanging outside the beaker.
3. Pour enough water into the beaker to submerge (cover) the object. Measure the volume of the water.
4. Remove the object from the water carefully and measure the volume of water again.
5. Subtract your second reading from your first.
6. Convert your answer to cubic centimetres.

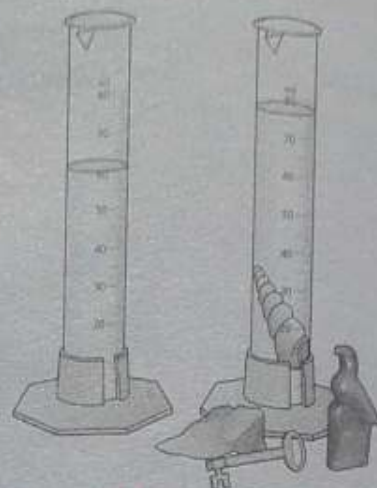


Figure 14.5 Measuring the volume of an irregular solid

Determining the thickness, volume and mass of small objects

Experiment 14.3

Aim: To determine the volume of a single ball bearing (or a bead)

Materials: 50 ball bearings or beads, a measuring cylinder, water

Procedure:

1. Pour some water to about a third of the measuring cylinder as in Figure 14.6a).
2. Record the volume of this water.
3. Place the 50 ball bearings (or beads) into the water in the measuring cylinder without splashing the water. See Figure 14.6b).
4. Record the new volume (ball bearings + water).
5. Calculate the volume of the 50 ball bearings (subtract the volume of water from that of water plus ball bearings).

6. Calculate the volume of one ball bearing.
7. Record your results with their units.

Results:

Volume of water alone = ____

Volume of water plus 50 ball bearings = ____

Volume of 50 ball bearings alone = ____ - ____

= ____

∴ Volume of 1 ball bearing = ____ ÷ ____

= ____

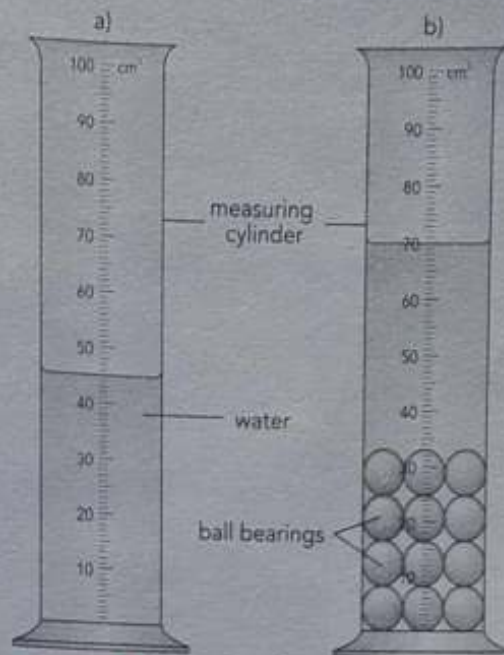


Figure 14.6 Measuring the volume of ball bearings

Experiment 14.4

Aim: To determine the thickness of a sheet of paper

Materials: a thick book (your science text book will do), a ruler

Procedure:

1. Separate the cover of the book from the rest of the pages as shown in Fig. 14.7.
2. Press together the rest of the pages and then measure the thickness of the book.
3. Find out how many sheets of paper you measured. Remember that a sheet has two pages.

Results:

Total number of sheets = ____

Thickness of ____ sheets = ____

∴ Thickness of one sheet = ____ ÷ ____

= ____

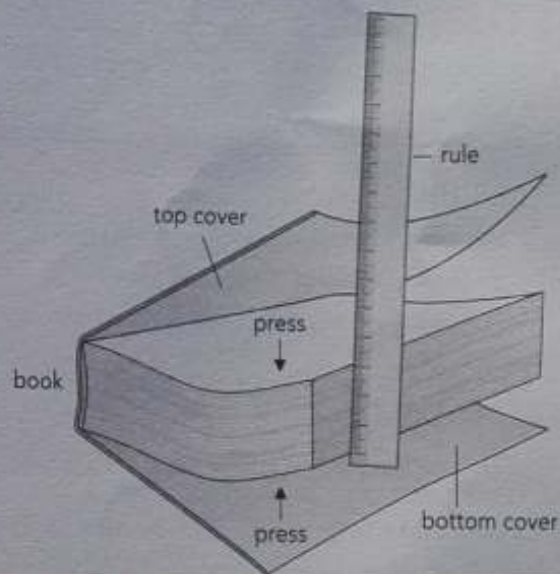
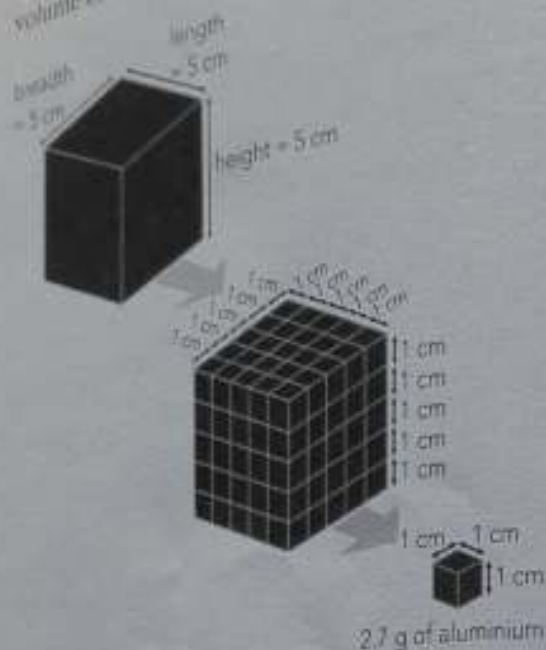


Figure 14.7 Measuring the thickness of sheets of paper

Density

The density of a substance is the amount of matter (mass in g) of that particular substance that can be packed into one cubic centimetre (1 cm^3) of volume of the substance. The unit of density is

grams per cubic centimetre (g/cm^3). For example, aluminium has a density of 2.7 g/cm^3 . This means that in every cubic centimetre, there is 2.7 g of aluminium.



$$\begin{aligned}\text{Density} &= \frac{\text{mass of substance}}{\text{volume of substance}} \\ &= \frac{\text{grams}}{\text{cubic centimetre}} \\ &= \text{g/cm}^3\end{aligned}$$

Figure 14.5 Aluminium has a density of 2.7 g/cm^3 . This means that 2.7 g of aluminium is present in every cubic centimetre.

Example 14.2

1. Calculate the density of glass if 120 cm^3 of glass has a mass of 300 g .

$$\begin{aligned}\text{Density} &= \frac{\text{mass of substance}}{\text{volume of substance}} \\ &= \frac{300 \text{ g}}{120 \text{ cm}^3} \\ &= 2.5 \text{ g/cm}^3\end{aligned}$$

2. A cylinder of aluminium has a radius of 7 cm and a height of 20 cm . The mass of the cylinder is 8.316 kg . Calculate the density of aluminium.

$$\begin{aligned}\text{Density} &= \frac{\text{mass of substance}}{\text{volume of substance}} \\ &= \frac{8.316 \text{ g}}{\pi \times 7^2 \times 20 \text{ cm}^3} \\ &= 2.7 \text{ g/cm}^3\end{aligned}$$

* the value for π is 3.143 or $\frac{22}{7}$

3. What is the density of a cube of sugar weighing 11.2 g measuring 2 cm on a side?

Step 1: Find the mass and volume of the sugar cube.

$$\text{Mass} = 11.2 \text{ g}$$

$$\begin{aligned}\text{Volume of a cube} &= (\text{length of side})^3 \\ &= (2 \text{ cm})^3 \\ &= 8 \text{ cm}^3\end{aligned}$$

Step 2: Plug your variables into the density formula.

$$\begin{aligned}\text{density} &= \frac{\text{mass}}{\text{volume}} \\ &= \frac{11.2 \text{ g}}{8 \text{ cm}^3} \\ &= 1.4 \text{ g/cm}^3\end{aligned}$$

Answer: The sugar cube has a density of 1.4 g/cm^3 .

Summary

- SI units can be converted into various multiples of the base units.
- The mass of a liquid can be measured by determining the difference between the mass of the container of the liquid and the liquid itself.
- To determine the volume of an irregular object you can submerge it in water and measure how much water is displaced by it.
- The dimensions of a regular object can be used to determine its volume.
- The density of the object is how much mass is contained within its dimensions.

Glossary

density – the amount of matter within the volume of an object

displace – the change in position of an object

mass – the amount of matter within an object

volume – how much space an object occupies

weight – is a form of measurement that is dependent on gravity

Revision questions

- Convert:
 - 30 g to kg
 - 3 kg to g
 - 0.25 kg to g
 - 5 100 g to kg
 - 2.5 kg to g
 - 0.05 g to kg
 - 13.56 kg to g
 - 20.005 g to kg
- Describe how you would measure the mass of water in a glass. (2)
- A 200-page book has a thickness of 20 mm when covers have been excluded. What is the thickness of each leaf making up the book? (note – one leaf has 2 pages) (2)
- A block of plastic has a mass of 60 g and a volume of 40 cm^3 . Calculate the density of the plastic. (2)
- A rectangular wooden block is 5 cm long, 3 cm high and 2.5 cm wide. The mass of the block is 26 g. Calculate the density of the wooden block. (2)
- The mass of 175 cm^3 of yoghurt is 322 g. Calculate the density of the yoghurt. (2)
- A rectangular block of iron has a density of 750 kg/m^3 and measures $40 \text{ cm} \times 30 \text{ cm} \times 20 \text{ cm}$. Calculate the mass of the iron in kg. (3)
- When a lead sinker was lowered into a measuring cylinder with water, the water level rose from 35 ml to 42 ml. The mass of the lead sinker was 79 g. Calculate the density of the lead sinker. (6)

- A learner is given a bottle containing 100 very small identical ball bearings and asked to determine the volume of each bearing. The learner decides to use the displacement method to measure the volume. His results are shown in Figure 14.9. Calculate the volume of each bearing. (3)

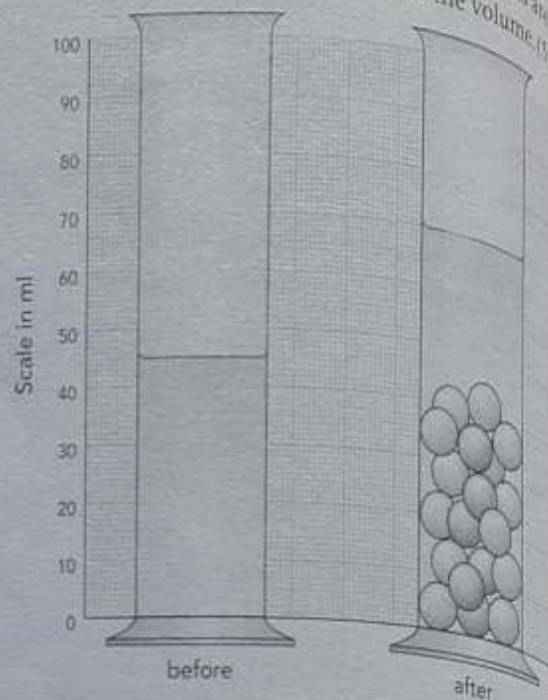


Figure 14.9

TOTAL: 32

Forces

Objectives

- Calculate the resultant force of a pair of inline or co-linear forces.
- Define moment of a force.
- Calculate moment of a force.
- State the principle of moments.
- Apply the principle of moments in simple calculations.
- Define friction.
- Measure friction.
- State the applications of frictional force.

Introduction

A force can be applied on an object in the form of a push, a pull, a compression or stretching. A push or a pull will result in a change in the acceleration or change in direction of the body, whilst compression or stretching will result in a change of shape. The study of the application of forces allows us to predict these changes in objects. When describing a force, the following four points need to be noted:

- what is exerting the force
- on what is the force being exerted
- the direction in which the force is being exerted
- the magnitude (size) of the force.

A force is represented using arrows. The magnitude (size) of the force is labeled on the arrow as shown in Figure 15.1. The direction of the force



Figure 15.1 A force pushing a box that can slide across the floor.

is represented by the direction of the arrowhead. The magnitude of forces is measured in units called newton (N) and can be measured using a spring balance or force meter (also called a newton meter).



Figure 15.2 A spring balance

Resultant force

A force is a vector quantity. A vector is a physical quantity that has a magnitude and direction. Because forces are vector quantities, they cannot just be added in any way. Forces that act in the same direction can be added together to produce a new force called the **resultant** or **net force**. A resultant or net force is defined as a single force that has the same effect as two or more forces that act on an object.

Inline or co-linear forces

Inline forces, like the name implies, are forces acting on the same line. Forces pointing to the right are positive whilst forces pointing to the left are negative. These forces can either be balanced or unbalanced.

Balanced forces have equal magnitudes, but act in opposite directions. They cancel each other out and the overall effect of the force is zero. When forces are balanced, there is no change in the object's state of motion, so the object will either remain stationary (if it is stationary), or continue moving at a constant speed (if it is already moving at a constant speed).

Unbalanced forces do not completely cancel each other out, so they have an overall effect on an object and there is a non-zero resultant force. When forces are unbalanced, they cause a change in the object's motion.

Figure 15.3 shows how to add vectors when they act in a straight line.

- The force to the right is $50\text{ N} + 100\text{ N} = 150\text{ N}$.
- The force to the left is 50 N .
- The resultant force is $150\text{ N} - 50\text{ N} = 100\text{ N}$ to the right.

The vectors in Figure 15.3 act in the same straight line, but they do not all act in the same direction – two act to the right, while one acts to the left. One way to work out the directions of vectors is to use a sign convention. In this example, we say that forces acting to the right are positive (+) while forces acting to the left are negative (-). So the force of 50 N to the left should be regarded as -50 N . Then we add the forces like this:

$$\text{resultant force} = 50\text{ N} + 100\text{ N} + (-50\text{ N}) = +100\text{ N}$$

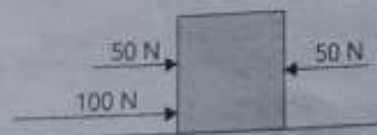


Figure 15.3 The resultant of these three forces is 100 N to the right.

Because the final answer is positive, we know that the resultant force acts to the right. If the answer had been negative, we would have known that it acted to the left.

Exercise 15.1

Determine the resultant force in the following

1. 6 N → 6 N ←
2. 1 N → 3 N ←
3. 8 N → 6 N ←
4. 10 N → 6 N → 3 N ←
5. 3 N → 5 N → 3 N ←
6. 4 N ↑ 7 N ↓

Moment of force

Sometimes, if we have a single force acting on an object, we get a turning effect. Consider a door that is opened or closed. You push (or pull) the door, yet instead of moving backwards or forwards the door swings open (or closed). This turning effect comes about because the door is fixed on one side to a hinge. This hinge acts as a pivot (fulcrum) around which the door turns. The door acts as a lever in this case. Levers and fulcrums (pivots) will be dealt with in detail in Unit 16.

Another example of the turning effect is shown in Figure 15.4 when we tighten a nut with a spanner. In tightening the nut, we apply a force on the spanner. One end of the spanner is fixed around the head of the nut. This acts as a pivot and the spanner acts as a lever.

To determine the magnitude of the moment, two values are needed: the magnitude of the force (measured in newton) and the perpendicular distance from the force to the pivot (measured in metres).



Figure 15.4 There is a turning effect when a single force acts at a distance from the pivot.

Calculating the moment of force

The formula to calculate **moment of a force** is as follows:

$$\text{moment of force} = \text{force} \times \text{perpendicular distance from force to pivot}$$

The unit of measurement for the moment of a force is newton metre, which can be abbreviated to N.m.

Example 15.1

A boy tries to loosen a nut with his hand, applying 100 N of force with a distance of 0.01 m between his fingers and the fulcrum (Figure 15.5a). The nut is too tight, so the boy fetches a spanner and applies the same force. However, this time the distance between the end of the spanner (where his hand is applying the force) and the fulcrum is 0.15 m (Figure 15.5b). This causes the nut to loosen. Why does this happen?

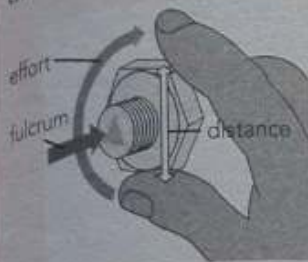


Figure 15.5 a) Using a hand to loosen the nut

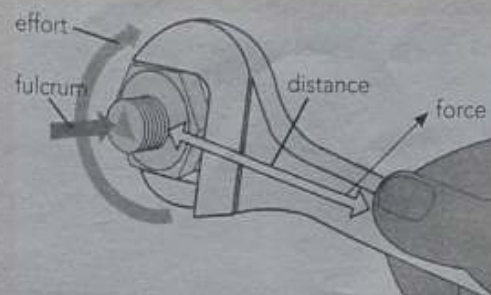


Figure 15.5 b) Using a spanner to loosen the nut

To find the answer, you can calculate the moment of each action.

Loosening the nut by hand: $100 \text{ N} \times 0.01 \text{ m} = 1 \text{ N.m}$

Using the spanner: $100 \text{ N} \times 0.15 \text{ m} = 15 \text{ N.m}$

Because the moment of force of using the spanner is larger, we are able to loosen the nut.

The principle of moments states that when an object is in equilibrium, the total sum of the anticlockwise moments about a turning point is equal to the total sum of the clockwise moments.

An example illustrating the principle of moments would be two children sitting on a see-saw without it moving. The forces exerted by the two children on the fulcrum (centre of the see-saw) are balanced forces, and cancel each other out.

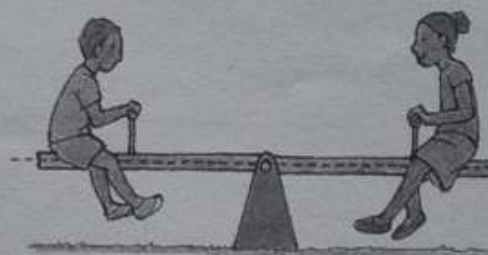


Figure 15.6 The forces exerted by the two children on the fulcrum (centre of the see-saw) are balanced forces, and cancel each other out.

Experiment 15.1

Aim: To investigate balancing

Materials: a metre ruler, a string, 2 different weights

Procedure:

1. Suspend a metre ruler at its middle point, either by passing a string through a hole or on a knife-edge (Figure 15.7a). If necessary, stick Plasticine on one end to make it balance exactly.

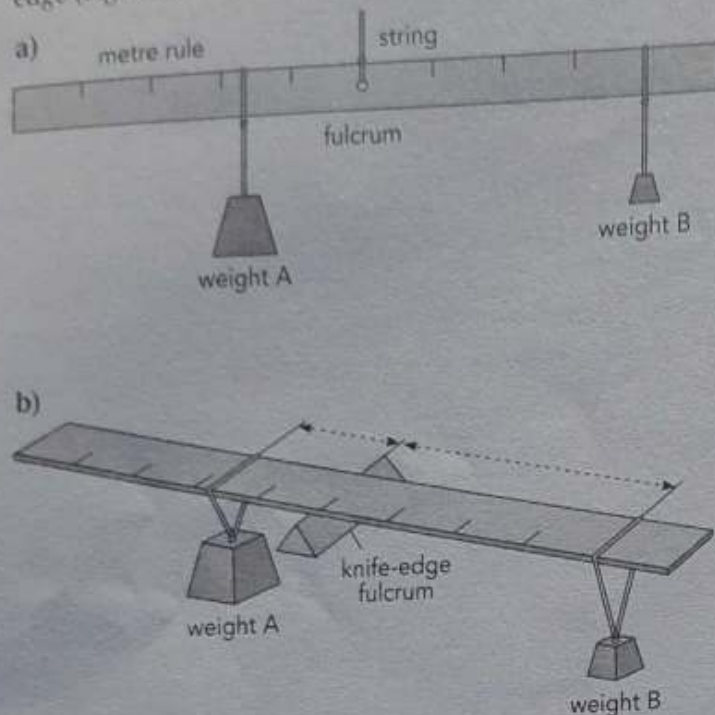


Figure 15.7 Studying moments on a lever

2. Tie loops of thread to several weights. Hang a 0.5 N weight A (the load) on the left-hand side of the ruler on the 34 cm mark (16 cm from the fulcrum). Place another weight B, 0.2 N (the effort), on the other side and move it until the ruler balances. Note the distances of the weights from the fulcrum, i.e. from the midpoint.
3. Record the results in a table, as shown in Table 15.1.
4. Repeat several times with (a) the same weights in a different position, and (b) different weights.

Table 15.1 Conversion of liquid volumes

Anticlockwise moments			Clockwise moments		
Weight A	Distance from fulcrum	Weight $\times$ distance	Weight B	Distance from fulcrum	Weight $\times$ distance
0.5 N	16 cm	8 N.cm	0.2 N	40 cm	8 N.cm

Friction

In simplistic terms, **friction** is a force that opposes motion. If there was no friction, a moving body would continue moving at that speed without any need for extra energy. In the case of a car, it would need to keep burning more fuel to keep it moving.

What is friction?

Before we answer this question, let us do Experiment 15.2 on the next page first.

Experiment 15.2

Aim: To demonstrate friction between surfaces

Materials: wooden block, a force meter, string

Procedure:

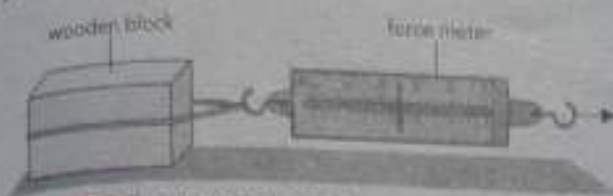


Figure 15.8 Experimental set-up

1. Tie a piece of string around the wooden block's length, with enough string left to tie to the force meter.
2. Hook the force meter to the string and use it to pull the wooden block as shown in Figure 15.8. Start with a small force and continue to increase the force, noting the readings on the scale of the force meter as you increase the pulling force. Record what happens.
3. Record the maximum force reached by the meter before the block starts moving.
4. Continue pulling the block so that it slides with minimum force. Record the reading on the force meter while the block is moving.

Observations:

1. You had to pull until a certain force was reached before the block starts moving.
2. A smaller force was needed to keep the wooden block sliding.

Discussion:

Why didn't the wooden block start moving as soon as you started pulling? It shows that as the block is pulled forwards, another force acting backwards occurs at the same time. This is the force of friction. It exists between two surfaces that are in contact. Figure 15.9 shows the wooden block, the pulling force and the force of friction. However, this opposing force has a maximum size for any two surfaces in contact such that when this size is exceeded, the object starts moving.

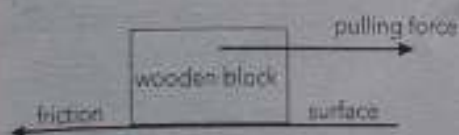


Figure 15.9 Forces acting on a wooden block

The maximum force that was recorded by the force meter just before the block started moving is called static friction. As soon as the block starts moving, a smaller force is required to keep the brick moving. The friction that now exists between the two surfaces is called kinetic friction or dynamic friction. Kinetic friction is smaller than static friction.

What causes friction?

To answer this question, we need to make some observations by doing Experiment 15.3.

Experiment 15.3

Aim: To find the relationship between the force of friction and the nature of the surfaces in contact.

Materials: 3 wooden blocks (1 with a smooth surface, 1 with a medium rough surface, 1 with a very rough surface), a flat cement surface, a string, a force meter

Procedure:

1. Set up the experiment as described in Experiment 15.2. Pull the wooden block on its smooth surface and record the static friction and kinetic friction.
2. Repeat step 1 with the medium rough surface and then again with the very rough surface.
3. Copy the table below and record your results.
4. Place a 1 kg mass on the wooden block and repeat the experiment. Record your results in the last column.

Observations:

Nature of surface of wooden block	Size of static friction	Size of kinetic friction	Size of static friction when extra load is placed on block
Smooth			
Medium rough			
Very rough			

1. The values in the table should show that the rougher the surface is, the larger the frictional force will be.
2. Increasing the mass of the block increases the force of friction between the two surfaces.

Surfaces may appear smooth but in fact they are quite rough and have protruding parts that we may not be able to see with the naked eye. When one surface sits upon another, the protruding parts interlock and make it difficult for one surface to slide over the other smoothly. This is what causes friction.

When is friction important?

The brakes of bicycles, cars and trains all have one surface rubbing against another trying to bring movement to a stop. Why do you think brakes should not be 'smooth'?

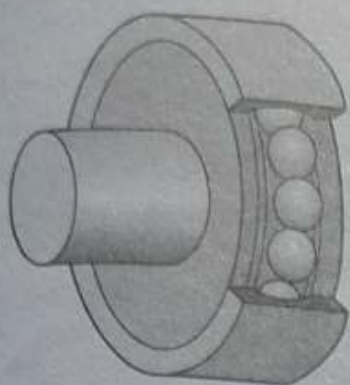
Bicycle and car tyres must have a good tread to improve their grip on the road. Even your shoes need good heels and soles. In all these cases, friction is a useful force that ensures we can control motion and remain safe.

When is friction a nuisance?

Moving parts in machines have one surface rubbing against another. This friction results in metal parts wearing away, which are often expensive to replace. Sometimes friction between two materials also generates a lot of heat, which may cause machines to stop functioning properly.

How to reduce friction

There are various ways to reduce friction. One way is to polish surfaces of moving parts of machines until they are smooth. Further examples are shown in Figures 15.10a) to d)



a) Put ball bearings between the two surfaces.



b) Use rollers.



c) Add wheels.



d) Put oil or grease between two moving parts of the machines.

Figure 15.10 Examples of how friction is reduced

Example 1

A man pulls a wooden chest along the floor. If the force of friction between the wooden chest and the floor is 50 N, what pulling force must he exert to just get the box moving?

Solution

In Experiment 15.2, the force on the wooden block kept increasing until we reached a maximum force when the block started to slide. This means friction between the floor and wooden box increased to a maximum equal to the pulling force before the block started moving. The maximum reading on the force meter equals the frictional force. Therefore the man has to exert 50 N before the box starts to move.

Example 2

The friction force between the floor and the bottom of a box is 10 N (see Figure 15.10). What is the resultant force if the box is pulled with a force of 15 N?

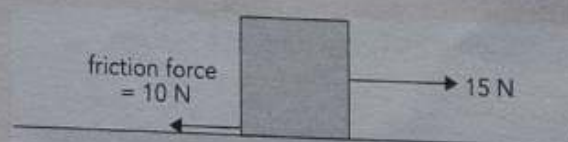


Figure 15.11 Forces acting on the box

Solution

$$\begin{aligned}\text{Resultant force} &= \text{pulling force} - \text{friction force} \\ &= 15 - 10 \\ &= 5 \text{ N}\end{aligned}$$

Factors influencing the magnitude of friction

- The nature of the surface: The rougher the surface, the greater friction it would cause.
- The mass of the object: The heavier the object, the more friction it would experience.

The effects of friction can be increased by increasing the roughness of the surface of an object, for example, the tread on car or bicycle

tyres or the patterns on the soles of shoes. Friction is often used to ensure safety, for example, the brakes on a bicycle or a car.

The effect of friction can be reduced by using a **lubricant**. Oil is a common lubricant that is used in car engines and bike gears to reduce friction effects. This is mostly used where components need to move often, but the force of friction would wear the surface away.

Summary

- A force can either be a push or a pull.
- When two or more forces act on an object on the same line, they will have a resultant force.
- Two or more forces acting on the same object on the same line can either be balanced or unbalanced.
- Unbalanced forces will change the speed, direction or shape of an object.
- A moment is the turning effect of a force.
- The moment of force is measured in newton metres and is calculated by multiplying the magnitude of the force by the perpendicular distance of the force to the fulcrum.
- Friction is caused by the nature of the surface of a material.
- Friction always acts in the opposite direction to the motion of an object.
- Friction can be increased by increasing the roughness of a surface.
- Friction can be decreased by using a lubricant on a surface.

Glossary

balanced forces – two or more forces that have equal magnitude but opposite directions; they cancel each other out

friction – a contact force between an object and another material; opposes the motion of the object

lubricant – a substance that can be applied to reduce the frictional force of a surface

moment of force – the turning effect of a force around a fulcrum

resultant (net) force – a single force that has the same effect as two or more forces that act on an object

unbalanced forces – two or more forces that are not equal in magnitude and therefore will have an effect on the object they are acting on

Revision questions

1. Calculate the resultant force in the following scenarios:

a) Two boys pull in opposite directions on a rope. The boy on the left pulls with a force of 5 N and the boy on the right with a force of 7 N. (1)

b) Figure 15.12 shows the forces acting on three different objects. Note: All the force vectors are not drawn to scale. Calculate the resultant force. (3)

2. Explain the difference between a balanced and an unbalanced force. (2)
3. State the principle of moment of a force. (2)
4. Define friction. (1)
5. Which direction does frictional force act in? (1)

TOTAL: 10

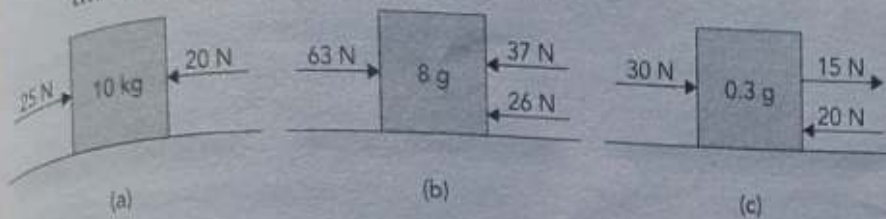


Figure 15.12

Objectives

- Define a machine.
- Construct a simple machine.

Introduction

To perform work requires **effort**. From the earliest days, people have looked for ways to reduce the effort needed to do their work, making it easier. As invention and technology advanced through the ages, people moved from using animals to perform work to using mechanisms called machines.



Figure 16.1 People use animals and machines to perform work for them.

Machines

A device that transmits or modifies a force is called a **machine**. In its most basic form, there are six simple types of machines: levers, inclined planes, screws, pulleys, wedges, and wheels and axles.

Levers

A lever is a machine that consists of a bar that **pivots** (turns) on a fixed support called a **fulcrum**, and transfers force. It changes the force applied on one end of the lever to an opposite force on the other side. Examples of levers are crowbars and see-saws.



Figure 16.2 A crowbar is an example of a lever.

Inclined planes

Also called ramps, an inclined plane is a flat surface, tilted at an angle so that one side is higher than the other. This is used to steadily raise or lower loads. It changes a linear (moving forward or backwards) force into a force going up or down. Examples of inclined planes are wheelchair ramps and portable ramps used to move cargo onto trucks.



Figure 16.3 Without a ramp, it is difficult for a person in a wheelchair to move up or down.

Screws

A screw is a machine that has ridges called threads along its sides. When it is pushed into another object and turned, the thread transfers the rotational (turning) force that makes the screw move forwards or backwards (linear).



Figure 16.4 Screws change turning movement into a forward movement.

Wheels and axle

A wheel and axle is a simple machine where a wheel is attached to a cylindrical bar called an axle. When the wheel is turned, it transfers force to the axle, which can be used to lift or lower loads. An example would be the mechanism used to lower a bucket into a well.



Figure 16.5 A well showing the wheel and axle system

Pulleys

A pulley is a wheel on an axle that supports a taut rope or cable. It allows a person to lift heavy loads by transferring force across the entire rope. Pulleys are often used on boats to lift and support sails as well as in mechanical cranes.

Wedges

Wedges are like portable, smaller inclined planes. Any force applied to its blunt end is transferred to its inclined surface. Examples are axes and door stops.

Any combination of simple machines is called a complex machine.



Figure 16.6 The axe has a wedge shape.

Exercise 16.1

State what type of simple machine the following devices are.

1. bench clamp
2. bicycle
3. wheelchair ramp
4. wheelbarrow
5. spade

Experiment 16.1

Aim: To build a simple lever

Materials: objects of different mass, ruler, sharpener or other small wedged object

Procedure:

Part A

1. Position the sharpener under the 10 cm mark on the ruler.
2. Place masses on the 2 cm mark on the ruler to balance it. This is your load.
3. Place masses on the ruler's 28 cm mark until the load moves up. This is the effort needed to lift the load.

Part B

1. Move the sharpener to the 20 cm mark on the ruler.
2. Add masses to the ruler's 28 cm mark (the effort) until the load moves up.

Questions:

1. In which of the two demonstrations was less mass required to lift the load?
2. What can we deduce from this?

Levers

As we have just seen, a lever is a stiff bar with a fixed turning point called a fulcrum or a pivot.



Figure 16.7 A lever is a stiff bar, with a fixed point.

Parts of levers

There are three parts to any lever: a fulcrum, load and effort.

- The fixed point at which the lever turns is called the fulcrum or pivot.
- The load is the object that is being moved by the lever.

- The effort is the force required to move the object.

The closer the load is to the fulcrum, the easier it is to move it.

Classes of levers

There are three classes of levers.

First class lever

In a first class lever, the fulcrum is between the load and the effort. Examples include a see-saw, a pair of scissors and a claw bar.



Figure 16.8 Children balancing on a see-saw is an example of a first class lever.

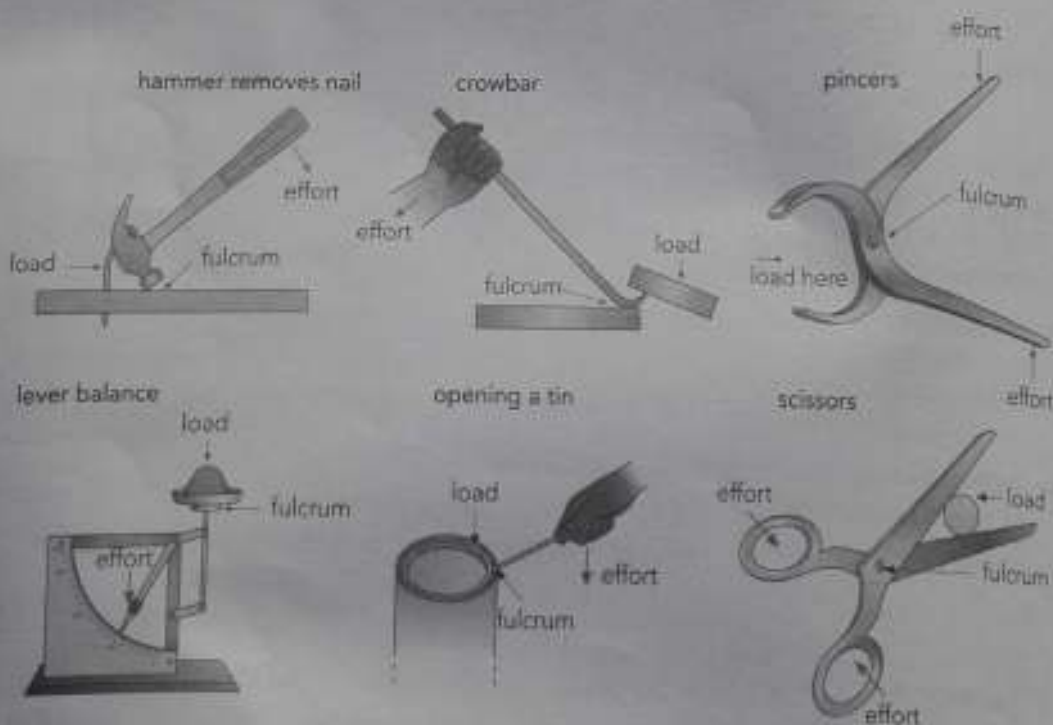


Figure 16.9 Examples of first-class levers. The fulcrum is between the effort and the load.

Second class levers
In second class levers, the load is between the fulcrum and effort. A wheelbarrow is a good example of a second class lever.



Figure 16.10 A wheelbarrow is an example of a second class lever.

Other examples include a nut cracker, a paper cutter and a bottle opener.

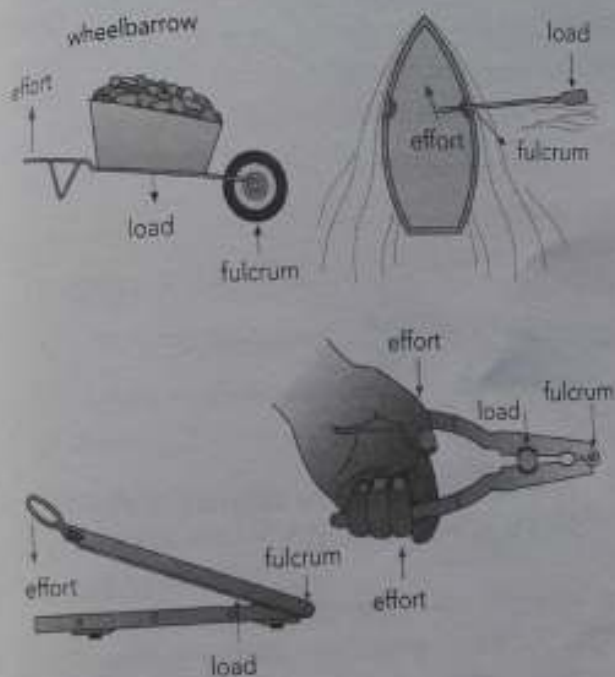


Figure 16.11 Examples of second class levers. The load is between the fulcrum and effort.

Third class levers

In third class levers, the effort is between the pivot or fulcrum and the load. Third class levers are useful if you want to apply a small force to objects and move them easily and carefully. Examples include a hoe, a fishing rod, a spade, a pair of tongs, a broom and tweezers.

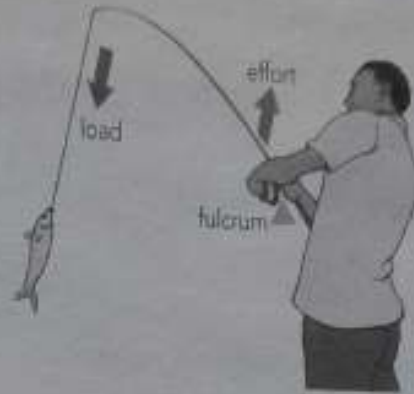


Figure 16.12 Fishing with a fishing rod is an example of a third class lever.



Figure 16.13 Examples of third class levers. The effort is between the fulcrum and load.

Summary

- Machines are used to make work easier by transmitting or modifying a force.
- There are six simple machines: levers, inclined planes, screws, pulleys, wedges, and wheels and axles.
- A complex machine is a combination of simple machines.
- A lever consists of three parts: a pivot (fulcrum), a load and an effort.
- In first class levers, the fulcrum is between the load and effort.
- In second class levers, the load is between the fulcrum and the effort.
- In third class levers, the effort is between the fulcrum and the load.

Glossary

complex machine – a combination of simple machines

effort – the force exerted on a lever in order to move the load

fulcrum (pivot) – the point at which a lever turns

load – an object lifted by a lever

machine – a device that transmits or modifies force

simple machine – levers, inclined planes, screws, pulleys, wedges, and wheels and axles

Revision questions

1. Give three examples of simple machines. (3)
2. What is the function of a machine? (2)
3. What is a complex machine? (1)
4. Draw and label the three classes of levers. (6)
5. How can you reduce the effort needed to lift a load using a lever? (2)

TOTAL: 14

Energy

Objectives

- State the law of conservation of energy.
- Define work and energy.
- State the SI unit of work and energy.
- Calculate the work done or energy used by forces.
- List sources of light energy.
- Show that light travels in a straight line.
- Explain the production and transmission of sound.
- Demonstrate the need for a medium in transmission of sound.

Introduction

Energy cannot be seen or touched, but it can be experienced all the time. Energy is what is stored up and used in all things around us. It allows for movement, life, light and heat. We know that energy comes in many forms, like kinetic,

chemical, electrical, light and sound energy. Energy is needed in order for a force to move an object. In this unit, we will explore the aspects of work and energy.

Experiment 17.1

Aim: To investigate energy in a pendulum.

Materials: retort stand, string, small heavy mass or stone

Procedure:

1. Tie the mass or stone to the retort stand using the string.
2. Pull the mass to one side of the retort stand, making sure that the string remains straight.
3. Let go of the mass and observe the movement.

Questions

1. What type of energy does the object possess at the top of the movement (at point A)?
2. What type of energy is this converted to when the pendulum swings down?
3. As the pendulum swings up, what energy does the object develop?

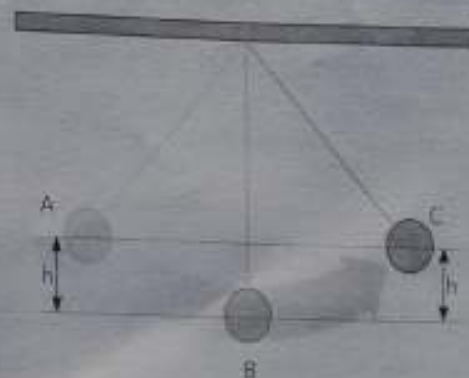


Figure 17.1 The motion of a pendulum, moving from point A to point C and back again.

Conservation of energy

Gravitational potential energy is the energy possessed by all objects that are in a position to move from a higher point to a lower point. This is because of the Earth's gravitational field, which pulls all things towards the centre of the Earth. The force involved is called the force of gravity.

As you saw in the pendulum experiment, energy is constantly converted between kinetic energy (at the bottom) and gravitational potential energy as the pendulum swings back up. To calculate how much gravitational potential energy is gained by an object, we can use the equation:

$$\text{gravitational potential energy (GPE)} = \text{weight (in newton)} \times \text{height raised (in metres)}$$

Example 17.1

A pendulum of mass 500 g is raised 25 cm. Calculate the gravitational potential energy it possesses at this point.

$$\begin{aligned}\text{GPE} &= \text{weight} \times \text{height raised} \\ &= \text{mass} \times \text{height raised} \\ &= 0.5 \times 9.8 \times 0.25 \\ &= 1.23 \text{ J}\end{aligned}$$

As the pendulum swings down, the 1.23 J is converted to kinetic energy. As it swings back up to 25 cm again, it will have 1.23 J of potential energy again.

The law of conservation of energy states that energy cannot be created or destroyed, but it can be transformed from one form to another.

In reality, when a pendulum swings, it eventually slows down to a standstill. This does not, however, disprove the law of conservation of energy. Energy is not only converted from potential to kinetic energy as the pendulum swings, but also into heat energy as a result of the friction between air and the mass of the pendulum.

Work and energy

When a force moves an object, we say work has been done on the object by the force. If an object does not move, regardless of the magnitude of the force applied, then no work has been done. Energy transferred is how much energy is required or used up to move an object. Work done is how much force is required or used to move an object.

We can see from the two definitions that work done and energy used is essentially the same. Energy is measured in joule, which is abbreviated to J. In order to calculate work done or energy used, we use the formula:

$$\text{Work done} = \text{force (N)} \times \text{distance (m)}$$

Exercise 17.1

Calculate the following.

1. The work done if a 5 N force pushes a box a distance of 30 m
2. The energy used to push a wheelbarrow 40 m, if you exert a 3 N force in the direction of motion
3. The force exerted if 500 J of energy is used to move an object 25 m

Experiment 17.2

Aim: To investigate the effect that the mass of an object has on the energy used

Materials: objects of different masses, string, force meter, ruler

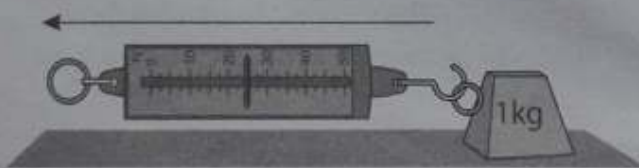


Figure 17.2 Set-up for Experiment 17.2

- procedure:**
1. Attach one of the objects to the force meter using the string.
 2. Mark the starting point and pull the object using the force meter.
 3. Take the reading on the force meter.
 4. Measure the distance that the object was moved.
 5. Calculate the energy used in pulling the object.
 6. Copy Table 17.1 and record your answers.

Table 17.1 Results for Experiment 17.2

	Object 1	Object 2	Object 3
Mass			
Distance (in m)			
Force used (in N)			
Energy used (in J)			

Questions:

1. What effect did the mass of the object have on the energy used?
2. What do you think the reason for this is?

Light energy

Light is a form of energy. The study of light energy is called optics. Natural sources of light include the sun, stars, fireflies and fire. Humans have invented other sources such as electric bulbs, candles and LEDs. Light energy enables humans and animals to see things around them. When objects such as the moon and all these other objects around us are illuminated, they reflect light into our eyes thus

enabling us to see. You are seeing these words right now because light reflected off them and into your eyes.

Light travels in straight lines

A ray of light is a very thin shaft of light. When a ray of light is considerably large, it is called a beam of light. Diagrammatically we represent a beam by several rays. A beam can be parallel, divergent or convergent as shown in Figure 17.3.

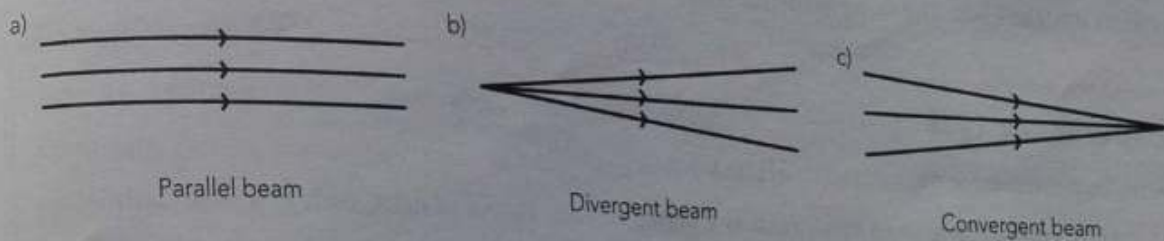


Figure 17.3 (a) a parallel beam; (b) a divergent beam; (c) a convergent beam

Experiment 17.3

Aim: To demonstrate that light travels in a straight line

Materials: three pieces of cardboard, cotton, a pin, a light source (a flashlight or lamp)

Procedure:

1. Place the pieces of cardboard on top of one another and punch a hole through the middle of all three.
2. Thread the cotton through the cardboard and pull it tight. This lines up all three pieces of cardboard.
3. Secure the cardboard to the table using wood, books or tape.
4. Remove the thread and place the light source behind one piece of cardboard.
5. You should see the light shining out through the other two pieces of cardboard.

Question:

What would happen if you moved one of the pieces of cardboard?

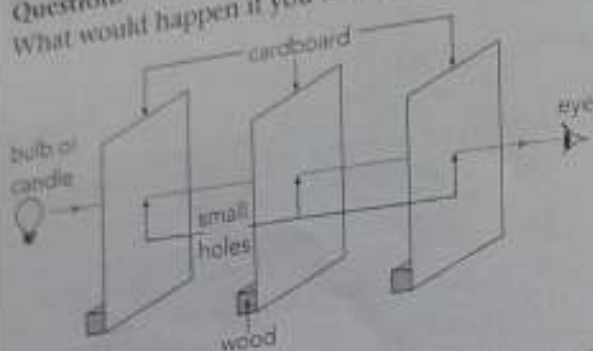


Figure 17.4 Testing the idea that light travels in straight lines

Formation of shadows

If light is made to pass through a narrow slit, a straight ray of light is produced. This ray has defined edges. Similarly, shafts of light passing through the canopy of trees in a jungle are straight. This leads to the conclusion that light travels in straight lines. The other evidence that

light travels in straight lines is the formation of shadows. If light is blocked by a figure with sharp straight edges, a sharp shadow of the object with well defined edges is cast onto a screen.

Experiment 17.4

Aim: To demonstrate the formation of shadows

Materials: compact source of light, such as a penlight, large source of light, such as a large fluorescent white screen or wall, a ball

Procedure:

Part A

1. In a dark room, arrange a compact light source so that its light falls on a white screen.
2. Hold the ball between the light source and the screen.

- A complete shadow (umbra) should appear on the screen. It is normally larger than the ball and its edges are sharp and clear.

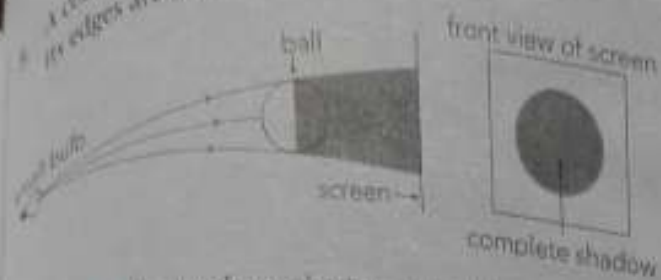


Figure 17.5 Shadow formed using a small bulb

Part B

- Repeat the experiment in Part A, this time using the large source of light.
- The complete shadow (umbra) should be surrounded by a partial shadow (penumbra).

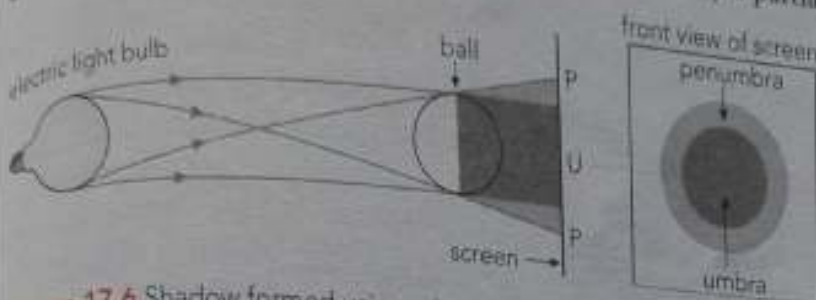


Figure 17.6 Shadow formed using a large bulb

Question:

If you move the light away from the ball, how does this alter the shadow cast?

Sound energy

Sound travels in waves and needs a medium through which to travel such as air, water, steel and so on. Sound cannot travel through a vacuum. So, if there is an explosion somewhere in space, you may be able to see a flash of light, but you

would not be able to hear the explosion.

The bell jar experiment is a common experiment used to demonstrate that sound needs a medium to travel.

Experiment 17.5

Aim: To demonstrate that sound needs a medium to travel

Materials: bell jar, electric bell, vacuum pump and hose

Procedure:

- Set up the apparatus and pump air into the bell jar.
- Ring the bell and observe the sound.
- Now start removing the air from the bell jar and observe the sound.

Observation:

At a particular vacuum, no more sound is heard from the bell, but we can see that the hammer continues hitting the gong and sound is produced. However, the sound cannot be heard because of the vacuum inside the jar.

Conclusion:

This demonstrates that the sound wave cannot travel through vacuum and sound needs a material medium for its travel.

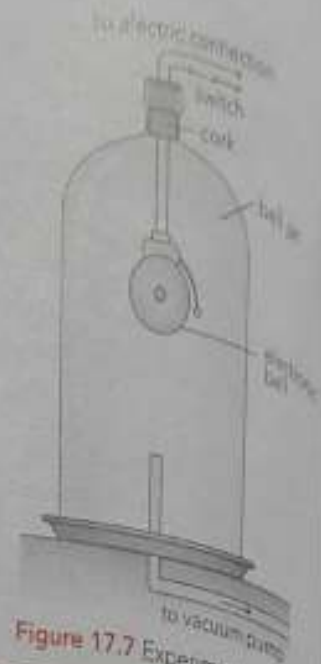


Figure 17.7 Experiment setup

How is sound produced?

Objects that vibrate produce sounds. An example is the strings on a musical instrument. We hear sound when vibrations that travel as sound waves reach our ears. When these molecules vibrate, they in turn cause the molecules around them to vibrate. The vibration will spread from molecule to molecule, causing the sound to travel.

The propagation of sound

Sound is a longitudinal wave. Longitudinal waves are propagated (transmitted) when particles of a medium vibrate back and forth in the same direction as the movement of the wave. For example, when a loudspeaker vibrates outwards, it brings air particles close to one another and a compression is formed. When the loudspeaker vibrates inwards, it pulls air particles apart to form a rarefaction. A series of compressions and rarefactions moves sound forwards (see Figure 17.8), but the particles of the medium only move backwards and forwards. This is why sound needs a medium in order to be propagated.



Fig 17.8: Compression and rarefaction produced by a vibrating tuning fork

Table 17.2 compares light and sound energy.

Table 17.2 Comparison between light and sound energy

Light energy	Sound energy
Can travel through a vacuum	Cannot travel through a vacuum
Travels in a straight line	Travels in waves radiating from the object
Produced by self-luminous objects	Produced by disturbing an object that can vibrate

Summary

- The law of the conservation of energy states that energy cannot be created or destroyed, rather it can only be transformed from one form to another.
- Energy used or work done is measured in joule (J).
- $\text{Work done} = \text{energy used} = \text{force} \times \text{distance moved}$
- Sources of light energy include the sun, fire and electric bulbs.
- Light travels in a straight line unless reflected by a reflective object, such as an angled mirror.
- Sound energy is produced through vibrations passing through different mediums or bodies. The kinetic energy of the vibrations is converted to sound energy.

Glossary

elastic – an object or medium that can be compressed

kinetic energy – energy that produces movement

penumbra – a partial shadow, with some light passing through it

potential energy – energy stored in an object by virtue of its position

self-luminous – an object that give off light energy

umbra – a complete shadow, with no light passing through it

vibration – the periodic movement of a rigid or elastic body

work – the amount of force required to move an object over a distance

Revision questions

1. State the law of conservation of energy. (2)
2. Explain what is meant by the term 'work done'. (2)
3. Calculate the energy needed to move a load of 50 N over 10 m. (3)
4. Calculate how far a cart would move if a force of 10 N is applied, using 5 000 J of energy. (3)
5. Name three sources of light energy. (3)
6. How is sound produced? (3)
7. A tuning fork, a violin string and a loudspeaker are producing sounds. This is because they are all in a state of: (1)
 - A compression
 - B rarefaction
 - C rotation
 - D tension
 - E vibration
8. Astronauts are in a spaceship orbiting the moon. They see an explosion on the surface of the moon. Why can they not hear the explosion? (1)
 - A Explosions do not occur in space.
 - B Sound cannot travel through a vacuum.
 - C Sound is reflected away from the spaceship.
 - D Sound travels too quickly in space to affect the ear drum.
 - E The spaceship would be moving at a supersonic speed.
9. Explain how shadows are produced. (2)

TOTAL: 20

Unit
18

Magnetism and electricity

Objectives

- Describe the properties of magnets.
- State the law of magnetism.
- Draw magnetic fields.
- Define current and voltage.
- State the SI units of current and voltage.
- Measure current and voltage.
- Determine electrical power.

Introduction

In ancient times, Chinese people and others observed that magnetite, an ore of iron, attracted other similar stones and had a tendency of pointing in a particular direction when hung freely and free to turn. This latter property was useful for navigation, hence the ore was later called lodestone (meaning 'leading' stone). It became the first compass.

Nowadays we no longer use lodestones as magnets and compasses but use manufactured magnets.

Magnets

Properties of magnets

The following activities and experiments will help to understand the laws and properties of magnets.

Experiment 18.1

Aim: To investigate which items are attracted to a magnet

Materials: bar magnet. Collect as many different items in your environment as possible. Items must be made from different materials.

Procedure:

Use the bar magnet to investigate which items are attracted to the magnet.

Results:

Write your results in a table such as the one shown below.

Attracted	Not attracted

Observations:

Very few substances are attracted to a magnet. The substances that are attracted are made from iron or steel. You are not likely to have items made from cobalt and nickel in your immediate environment, which are the other two elements attracted by magnets. Iron, cobalt and nickel are called *ferromagnetic*.

Experiment 18.2

Aim: To find where magnetism is concentrated in a bar magnet

Materials: bar magnet, iron filings, paper

Procedure:

1. Wrap the magnet around with cling wrap or ordinary paper to prevent iron filings clinging directly to the magnet as they are difficult to remove.
2. Place the bar magnet at the centre of the piece of paper.
3. Sprinkle iron filings generously along the length of the magnet.
4. Observe where, along the length of the magnet, the iron filings concentrate.

Results:

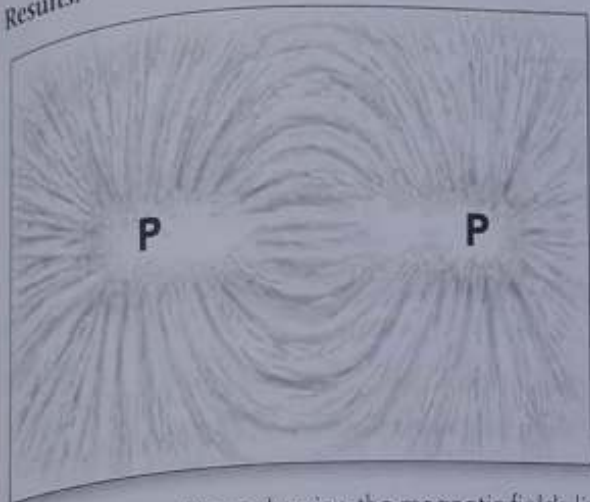


Figure 18.1 Iron filings showing the magnetic fields lines

Observations:

Iron filings are attracted mostly at two points that are near the ends of the bar magnet. These two points are where the magnetism is most concentrated and are called the poles of the magnet.

Experiment 18.3

Aim: To use a bar magnet as a direction finder

Materials: bar magnet; copper wire, cotton thread

Procedure:

1. Suspend the bar magnet as shown.
2. The other end of the cotton thread should not be tied to iron or steel (and not have iron or steel nearby) since this would influence the direction of the magnet.
3. Allow the magnet to swing freely until it comes to rest. Note the direction.
4. Disturb the magnet and again allow it to come to rest freely.

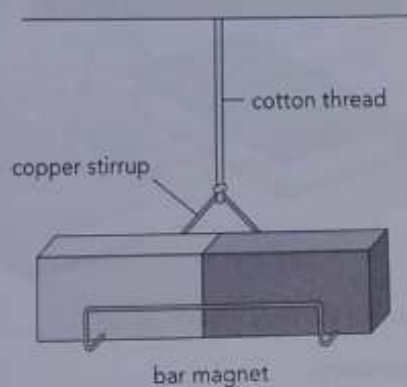


Figure 18.2 Experiment set-up

Results:

Each time the magnet comes to rest, it is found that it points in the same North-South direction. The pole that points in the northern direction is called the north-seeking pole or north pole. The other is the south pole.

Experiment 18.4

Aim: To find the behaviour of poles when they are next to each other

Materials: two bar magnets whose poles are known

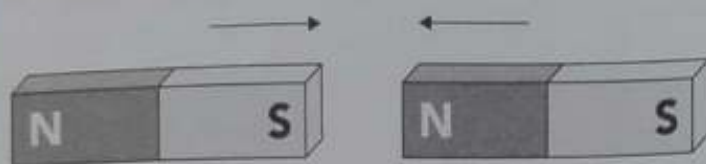


Figure 18.3 Two magnets showing attraction

Procedure:

1. Bring the magnets towards each other as shown in Figure 18.3.
2. Start with N-N, then S-S and finally N-S.

Observations:

1. The N-N poles pushed against each other. The same happened with S-S. N-S poles attracted each other.
2. Like poles repel each other whilst unlike poles attract.

Experiment 18.5a

Aim: To observe magnetic field patterns

Materials: two bar magnets, two books of the same thickness as the magnets, white A4 paper, fine iron filings.

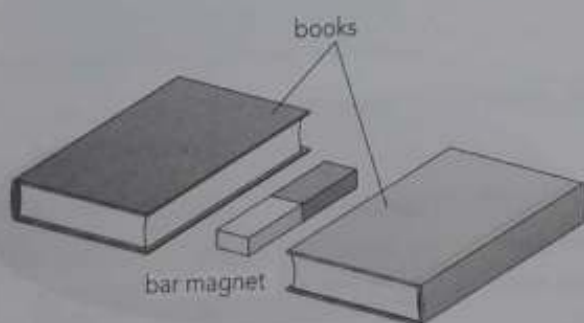


Figure 18.4 Experiment set-up

Procedure:

1. Position the two books and magnet as shown in Figure 18.4.

2. Cover the magnet with the A4 paper such that it rests on the two books.
3. Sprinkle iron filings in the area over and around the magnet and tap gently at the corners of the paper.

Observations:

As the iron filings are sprinkled and the tapping is done, the iron filings trace out a pattern as shown in Figure 18.5.

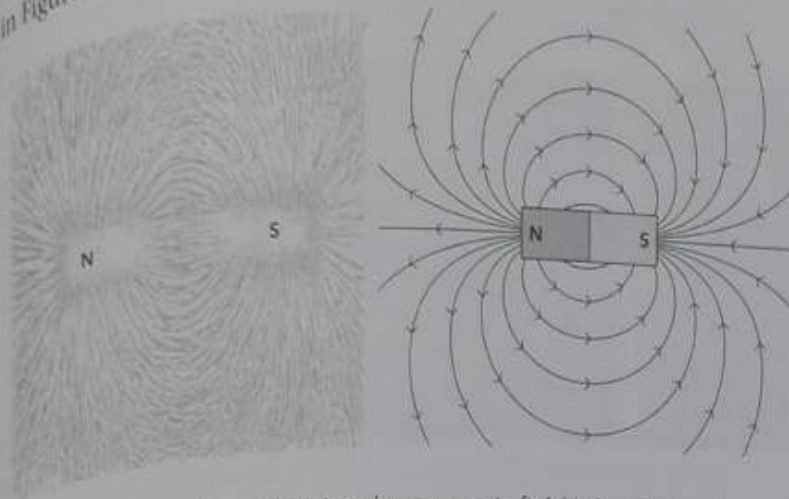


Figure 18.5 a) iron filings showing the magnetic field lines, b) how the results are represented in a diagram

Results:

1. Each line in the pattern is called a magnetic line of force.
2. Magnetic lines of force do not touch and do not cross.
3. Magnetic lines of force move from the north pole to the south pole always, therefore each line has a direction.
4. The area around the magnet where there are magnetic lines of force is called a **magnetic field**.
5. A magnetic substance placed inside a magnetic field will experience a magnetic force of attraction.

Experiment 18.5b

Aim: Alternative method to investigate magnetic field lines using a plotting compass

Materials: bar magnets, a plotting compass, A4 plain paper, iron filings, a pencil

Procedure:

1. Place a bar magnet in the middle of a sheet of paper and trace its outline with a pencil.
2. Place the plotting compass at the top of one end of the magnet. Put a dot on the edge of the compass in the northern direction as shown by the compass needle (see Figure 18.6).



Figure 18.6 Plotting magnetic field lines

End-of-topic revision test

1. How many millimetres in a metre?

- A 10
- B 100
- C 1 000
- D 10 000

(1)

2. When converting ml to cm^3 , what is the volume of 1 litre of liquid?

- A 1 cm^3
- B 10 cm^3
- C 100 cm^3
- D $1\,000 \text{ cm}^3$

(1)

3. The most appropriate graph to use when showing the cooling time for water after boiling is a:

- A bar graph
- B line graph
- C pie graph
- D tally.

(1)

4. Friction can be reduced by:

- A increasing the force pushing an object
- B applying lubricant to the object
- C heating the object
- D cutting grooves into the object, making the surface rougher.

(1)

5. A balanced force is:

- A two or more forces of equal magnitude, but opposite directions which cancel each other out as a result
- B two or more forces that are not equal in magnitude and therefore will have an effect on the object they are acting on
- C a single force which has the same effect as two or more forces that act on an object
- D the force exerted by a person sitting on a see-saw.

(1)

6. a) Explain how you would measure the volume of an irregular object.

(4)

b) Calculate the density of glass if

240 cm^3 of glass has a mass of 600 g. (3)

7. Draw a line graph to present the following information:

Test	Number of learners with marks above 80%
1	
2	30
3	35
4	45
5	35
6	20
	10

8. a) State the principle of moment of force. (4)

b) Define friction. (2)

c) Calculate the resultant force if two boys pull in opposite directions on a rope, the boy on the left with a force of 15 N and the boy on the right with a force of 17 N. (2)

d) Calculate the moment of force if a force of 400 N is applied on a lever, 0.3 m away from the fulcrum. (2)

9. a) Give three examples of simple machines. (3)

b) Define a second class lever. (2)

10. a) State the law of conservation of energy. (2)

b) Calculate the energy used to move a load of 40 N over a distance of 20 m. (3)

c) Explain how sound energy is produced. (2)

d) Name a source of light energy. (1)

11. a) Name three characteristics of magnets. (3)

b) Calculate the voltage required to produce a current of 8 A when connected to a 12 W appliance. (3)

c) State the law of magnetism. (2)

d) Define the term 'power'. (2)

TOTAL: 50

Practice examination

Marks: 140

Time: 60 minutes

Paper 1

Multiple choice

Each question has four possible answers marked A, B, C and D. Choose the letter of the correct answer and write it down.

1. Which of the following is an example of continuous variation in humans?
A Gender
B Foot length
C Blood type
D How earlobes are attached or not attached (1)
2. Genotype relates to variation that is:
A inherited
B due to the environment
C due to disease
D due to the environment and disease. (1)
3. The organelles inside plant cells that carry out photosynthesis are called:
A chlorophyll
B xylem
C phloem
D chloroplasts. (1)
4. Herbivores are:
A plants
B organisms living on decaying plant material
C organisms living on plants and animals
D heterotrophs. (1)
5. Humans are examples of:
A autotrophs
B heterotrophs
C self-feeders
D producers. (1)
6. The lungs are located in:
A the abdominal cavity
B the bronchial cavity
C the thoracic cavity
D the pulmonary cavity. (1)
7. During the respiratory process, exhalation is the:
A phase in which the diaphragm moves down
B inactive phase of breathing
C phase in which the ribcage moves outward
D active phase of breathing. (1)
8. Choose the description of transport that does not correspond to osmosis:
A Water from the soil moves into the roots of plants
B Water reabsorption in the colon
C Oxygen moving from the alveoli to the bloodstream
D Water reabsorption in the kidneys (1)
9. In plants, water is lost through:
A photosynthesis
B transpiration
C osmosis
D translocation. (1)
10. The products of photosynthesis are transported in:
A companion tubes
B xylem
C sieve tubes
D tracheid tubes. (1)
11. In dicotyledonous plant roots, the endodermis:
A is the outer layer of the root
B channels the flow of water from the cortex into the stele
C protects the tissue of the stele.
D Both B and C. (1)
12. In dicotyledonous plants, the young green stem:
A does not have a cuticle
B cannot photosynthesise
C is strengthened by collenchyma cells
D is strengthened by the epidermis. (1)
13. The bicuspid valve:
A has three folds
B has flaps that look like half-moons
C prevents blood from flowing back into the right ventricle
D prevents blood from flowing back into the left atrium. (1)

14. Arteries:
- A always carry blood to the heart
 - B always carry blood away from the heart
 - C supply the heart with oxygen-rich blood
 - D have valves, because of the low blood pressure inside them.
15. Which is the correct sequence in water treatment? (1)
- A Water source (reservoir) → filtration → chemical treatment → storage tanks
 - B Water source (reservoir) → chemical treatment → filtration → storage tanks
 - C Chemical treatment → filtration → water source (reservoir) → storage tanks
 - D Storage tanks → water source (reservoir) → filtration → chemical treatment (1)
16. A 40 ml solution contains 3.2 g of dissolved (solid) solute. What is the solution's concentration?
- A 8 g/ml
 - B 0.08 g/ml
 - C 12.5 g/ml
 - D 1.25 ml/g
17. A solution contains 25 ml of solute and 75 ml of solvent. What is the percentage concentration of the solution? (1)
- A 75%
 - B 25%
 - C 50%
 - D 22.5%
18. Which of the following reactions would not be used to produce magnesium chloride?
- A Magnesium carbonate and dilute hydrochloric acid
 - B Magnesium oxide and dilute hydrochloric acid
 - C Magnesium and dilute hydrochloric acid
 - D Magnesium hydroxide and dilute hydrochloric acid (1)
19. Which of the following products are always formed in an acid-base reaction?
- i) A salt
 - ii) Water
 - iii) Carbon dioxide gas
 - A All of the above
20. Which alkali produces a soap that is more water-soluble than others?
- A Potassium hydroxide
 - B Glycerol
 - C Sodium hydroxide
 - D Calcium carbonate
21. The part of the batch process of soap-making that separates the final product from the mixture is called:
- A mixing
 - B refining
 - C splitting
 - D pitching
22. Which of the following processes is an example of a physical change? (1)
- A Metals rusts
 - B Paper burns
 - C Water boils
 - D Fruit spoils
23. Which of the following is NOT a sign that a chemical change has occurred? (1)
- A A substance changes colour
 - B A substance gives off heat
 - C A substance gives off light
 - D A substance changes from one state to another
24. During a chemical change, a substance: (1)
- A loses electrons
 - B changes into another substance
 - C keeps its chemical properties
 - D always loses mass
25. Which type of radiation are greenhouse gases able to absorb? (1)
- A Infrared
 - B X-ray
 - C Solar
 - D Ultraviolet
26. Which of the following is a renewable source of energy? (1)
- A Wood
 - B Solar energy
 - C Crude oil
 - D Coal

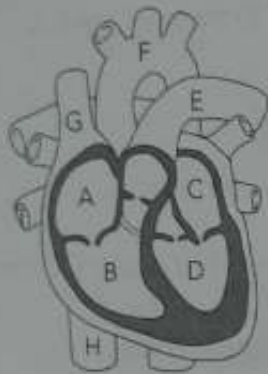
27. Which of the following gases is NOT a greenhouse gas?
 A Carbon dioxide
 B Methane
 C Water vapour
 D Nitrogen gas
28. A straight line graph refers to a:
 A diagram where data is presented in a line form (1)
 B diagram where data is presented in bar form
 C way of presenting diagrams in a table format
 D way of presenting photographs in table format
29. A moment of force describes the:
 A length of time over which a force is applied (1)
 B magnitude of a force
 C time when a force is applied
 D torque around a fulcrum
30. 150 ml is _____ cm^3 .
 A 150 (1)
 B 15
 C 1.5
 D 0.15
31. The volume of a block with dimensions 5 cm by 10 cm by 4 cm would be:
 A 400 cm^3 (1)
 B 2 cm^3
 C 200 cm^3
 D 20 cm^3
32. Pressure is measured in:
 A pascal (1)
 B kilogram
 C newton
 D metres squared.
33. Gravitational force is an example of a:
 A contact force (1)
 B field force
 C negative force
 D force field.
34. Sound energy is produced by:
 A vibrations
 B changes in pressure
 C rotation of objects
 D objects exerting force upon one another
35. The instrument used to measure force is called a:
 A mass balance (1)
 B measuring cylinder
 C spring balance
 D spectrometer.
36. Which of the following is NOT an example of how potential energy can be stored?
 A Gravitational potential energy (1)
 B Chemical potential energy
 C Elastic potential energy
 D Kinetic potential energy
37. A block of plastic has a mass of 120 g and a volume of 40 cm^3 . Calculate the density of the plastic.
 A 0.67 Pa (1)
 B 3 Pa
 C 40 Pa
 D 30 Pa
38. Work is done when:
 A forces are in balance (1)
 B effort is exerted
 C a force is applied to an object
 D a force causes an object to move.
39. Current is measured using a/n:
 A rheostat (1)
 B speedometer
 C voltmeter
 D ammeter.
40. Current describes the:
 A rate of flow of charged particles through a conductor (1)
 B amount of energy transferred to an electrical component
 C the amount of energy lost through an electrical component
 D work done by an electrical component (1)

PAPER 1 TOTAL: 40

Paper 2

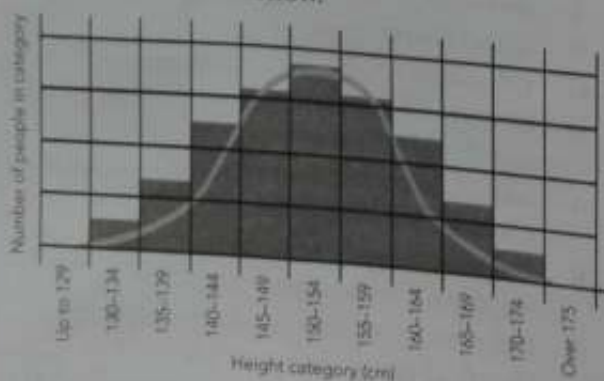
Section A

1. Study the diagram of the human heart and answer the questions that follow.



- a) Write down the letter representing the following:
- the right atrium
 - the left atrium
 - the right ventricle
 - the left ventricle
 - the aorta
- b) In which structures is there blood flow:
- to the lungs?
 - from the lungs?
- c) What is the main function of the heart in the body?
- d) Name the sac surrounding the heart and give its function.

2. Study the graph below and answer the questions that follow.



- a) Name and define the type of variation that is associated with the kind of distribution in this graph.
- b) What does the genotype of an individual refer to?

- c) Give two examples of the way in which humans benefit from variation.

3. Explain the difference between information and data.
4. What is meant by the term density?
5. Explain the three different classes of levers.
6. Explain four characteristics of magnets.
7. Explain and compare the difference between complete and incomplete combustion of fuels, and describe an experiment involving sugar that proves that a complete combustion experiment produces water and carbon dioxide.

Section B

Answer any TWO of the following questions.

1. Explain how the typical leaf is adapted for its role as the main photosynthetic organ in the plant.
2. Give an outline of the blood flow in the veins. Pay particular attention to the structure of veins and the pressure inside them.
3. Explain how the amnion that surrounds the human embryo is formed, and explain the role of the amnion and the amniotic fluid during pregnancy.
4. A small amount of marble chips (calcium carbonate) is dropped into a test tube that contains dilute hydrochloric acid.
- What would you observe as this reaction proceeds?
 - Describe a test that you would perform to prove which gas is evolved in the reaction.
 - Write a word equation for this reaction.
 - Once the reaction is complete, what would the pH of the products be?
 - Would you describe the reaction as a chemical change or a physical change? Give a reason for your answer.
 - Name two practical household applications of calcium carbonate in which it is used to neutralise an acid.
 - Name products that would form if calcium oxide was used instead of calcium carbonate.

6. Name and explain any five different simple machines. [10]
7. Explain the process to show the magnetic field around a bar magnet. [10]

7. Tabulate the differences between sound and light energy. [10]

PAPER 2 TOTAL: 60

Paper 3

Choose two of the following three questions to complete. In your exam, only two compulsory options will be provided.

1. You are provided with the following materials: two similar flasks with lids, bean seeds, two beakers with lime water.
- Make drawings to explain the set-up of an experiment to demonstrate that carbon dioxide is a product of cellular respiration. (4)
 - Explain the procedure that you will follow. (8)
 - Explain what lime water is used for in the experiment. (2)
 - Explain one way in which the experiment can go wrong. (2)
 - What were the results and conclusions of this experiment? (2)
- [20]
2. a) Access to clean drinking water is a basic human right. Explain how a typical water treatment plant treats water so that it is free from contaminants, such as bacteria and sand particles. Briefly describe the seven steps involved. (10)

- b) Soap is an effective cleaning agent, because it can remove dirt and oils from our bodies, clothes and other objects, and dissolve in water at the same time. Explain how soap works in chemical terms, and explain how this useful substance is produced industrially, by outlining the three basic steps. (10)

[20]

3. You are provided with the following: three pieces of cardboard, thread, a pin, a light source (i.e. a flashlight or lamp)

- Set up an experiment to show that light travels in a straight line. Your teacher will assess your experimental set-up. (8)
- Write the method you would use to conduct your experiment. (4)
- Explain how shadows are formed. (4)
- Differentiate between an umbra and a penumbra. (4)

[20]

PAPER 3 TOTAL: 40

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step ahead

Combined Science

FORM 2



Step Ahead Combined Science Form 2 is part of a new course especially developed to meet the needs of the 2015–2022 Zimbabwe Education Curriculum Framework.

The course seeks to develop learners who:

- are keenly aware of the values, ethos and attitudes necessary for the growth and development of our country
- realise that they already have natural talents and skills which they can build on, so that they are able to function beyond the classroom as effective citizens of Zimbabwe.

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- carefully drawn artwork and photographs
- varied activities and exercises designed to foster key skills, like critical thinking, problem-solving, leadership, ICT, communication and team-building.

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- useful background information for the teacher
- hints on how to teach concepts well
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